



COMSATS University Islamabad (CUI)

Software Design Description
(SDS DOCUMENT)
for

BrainBee

Version 1.0

By

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Bachelor of Science in Computer Science (2022-2026)

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Revision History

Name	Date	Reason for changes	Version

Application Evaluation History

Comments (by committee) *include the ones given at scope time both in doc and presentation	Action Taken
Animatios can be added to improve the learning process.	While we acknowledge that animations can enhance the learning experience, we have already defined a comprehensive and ambitious scope for our project, which includes multiple modules such as AI-generated flashcards, live classes, peer battles, performance tracking, and machine learning-based personalization. After thorough discussion with our supervisor, we collectively agreed to retain the current scope without additional features to ensure successful implementation, testing, and delivery within the available timeframe and resources. Expanding the scope at this stage may compromise the overall quality and stability of the system.

Supervised by
Supervisor's Name

Signature_____

1. Introduction

The BrainBee project aims to develop a mobile-based, AI-powered learning platform tailored for Pakistani students in grades 5–12. The system is designed to address gaps in accessibility, engagement, and personalized support in the education sector by leveraging artificial intelligence and gamification. The goal is to make high-quality education affordable, engaging, and effective for students in urban and rural regions alike.

1.1 Modules

This subsection lists the modules of BrainBee, along with their functionalities.

1.1.1 Student App

1.1.1.1 User Management

Manages user registration, profile updates, account deletion, grade selection, subject preferences, and premium upgrades.

1.1.1.2 Peer Competition Module

Enables students to participate in AI-generated quiz battles, track performance, maintain battle history, and view leaderboard positions.

1.1.1.3 Learning Module

Provides AI-generated summaries and explanations, interactive chatbot support, and personalized flashcards.

1.1.1.4 Assessment Module

Generates personalized quizzes, enables quiz and assignment submission, provides instant feedback, and maintains report cards and leaderboard metrics.

1.1.1.5 Achievement and Rewards Module

Tracks and displays badges, certificates, redeemable coins, and coin quests to gamify the learning experience.

1.1.1.6 Performance Analytics Module

Offers detailed performance reports across books, report cards, and leaderboard standings over time.

1.1.1.7 AI Powered Assistant Module

Allows students to ask questions and receive AI-generated answers with chat history maintenance.

1.1.2 Parent App

1.1.2.1 User Management

Manages parent user accounts including registration, profile updates, and deletion.

1.1.2.2 Monitor Children

Allows parents to add/remove children, monitor performance, view leaderboard rankings, and receive reports and notifications.

1.1.2.3 Goals Module

Enables parents to set, track, and reward academic goals for their children.

1.1.2.4 Communication Module

Facilitates chat with teachers and feedback submission to admins.

1.1.3 Teacher App

1.1.3.1 User Management

Enables account management for teachers, including registration, profile updates, and deletion.

1.1.3.2 Class Management

Allows creation and management of classes, students, study material, assignments, and quizzes.

1.1.3.3 Assessment Module

Provides performance monitoring and insights through ML algorithms, auto-grading of assignments, and personalized learning path generation.

2. Design Methodology and Software Process Model

This section outlines the chosen design methodology and software process model for the BrainBee educational application.

2.1 Design Methodology

BrainBee employs **Object-Oriented Programming (OOP)** as its design methodology due to the modular and scalable nature of the application. OOP provides a structured way to represent complex real-world entities such as students, parents, teachers, quizzes, rewards, and performance analytics through classes and objects. This approach ensures high code **reusability**,

maintainability, and **extensibility**, which is essential for a system that involves multiple interacting modules and diverse user types.

Encapsulation and abstraction enable developers to isolate features like assessments, goal tracking, and peer competition, making the application more manageable and easier to debug. OOP's use of inheritance and polymorphism also supports future feature expansions without major codebase refactoring. This design methodology aligns well with BrainBee's goal of building a flexible and robust platform that supports continuous improvement and evolving educational needs.

2.2 Software Process Model

BrainBee follows the **Agile Software Development** process model due to its iterative, collaborative, and adaptive nature. Agile is particularly well-suited for BrainBee because the project involves AI integration, dynamic feedback from students, teachers, and parents, and frequent improvements based on user behavior and evolving academic needs.

Agile allows the development team to work in **sprints**, delivering incremental updates with fully functional modules such as the Student Quiz Module, AI Chatbot, or Parent Monitoring Dashboard. Regular feedback from stakeholders ensures that any required changes are identified early and implemented quickly. This reduces risk, enhances product quality, and improves alignment with end-user expectations. Agile's emphasis on working software, adaptability, and close collaboration makes it ideal for an education-focused application aiming for continuous improvement and user engagement.

This figure illustrates the agile development process model and its stages.

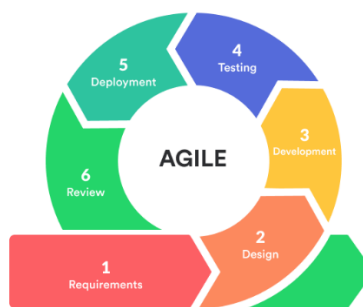


Figure 1 Agile Development Model

3. System Overview

This section provides a high-level summary of the overall system architecture and functionalities of **BrainBee**.

3.1 Functionality

BrainBee is an AI-powered mobile educational platform designed to enhance academic learning for students in grades 5 to 12 across Pakistan. It offers features such as user management for students, parents, and teachers; personalized learning content; AI-generated quizzes and explanations; gamified peer competition; performance analytics; and a reward system. The platform also provides tools for parental monitoring, goal setting, and teacher-based class and assignment management. Real-time feedback and AI tutoring ensure a customized, engaging, and efficient learning experience. The system promotes academic success through individualized learning paths, gamification, and collaborative support among users.

3.2 Context

Operating within the educational ecosystem of Pakistan, **BrainBee** addresses key challenges such as the lack of personalized instruction, rote learning, absence of student motivation, and limited parental and teacher engagement. The platform integrates artificial intelligence to personalize study material and assessments while encouraging healthy competition and progress tracking. By supporting students, parents, and teachers, BrainBee bridges the educational gap between urban elite and under-resourced communities, especially in Punjab. It aims to make quality education accessible, affordable, and interactive through AI and mobile technology.

3.3 Design

BrainBee utilizes a **multi-tiered architecture** that separates the system into distinct layers:

- **Presentation Layer** for user interfaces across mobile apps (Flutter-based)
- **Business Logic Layer** for managing core functionalities such as quizzes, performance tracking, and reward systems
- **Data Access Layer** for interacting with the MongoDB database securely

This design promotes **modularity**, **reusability**, and **scalability**, allowing different modules (e.g., student assessments, parent goals, teacher analytics) to work independently yet cohesively. RESTful APIs connect the layers to ensure seamless communication across different system

components and user apps. The AI components are integrated through Python-based machine learning models that assist in generating personalized content, performance predictions, and quiz recommendations.

3.4 Architectural Design

BrainBee is designed with a modular architecture to support maintainability, extensibility, and integration of new features. The system is divided into key modules:

- **User Management:** Handles registration, profile management, and user type switching
- **Assessment Module:** Manages AI-generated quizzes, assignments, feedback, and report cards
- **Learning Module:** Provides summaries, flashcards, and AI chatbot explanations
- **Peer Competition Module:** Enables quiz battles, leaderboard tracking, and win/loss history
- **Performance Analytics:** Tracks detailed academic data and AI-based progress insights
- **Parent & Teacher Modules:** Allow monitoring, goal setting, and personalized teaching interventions
- **Reward Module:** Handles badges, coins, certificates, and redemption tracking

This modular approach ensures that updates or changes in one part of the system do not affect others. The **multi-tiered architectural style** supports separation of concerns and eases testing, deployment, and scaling.

This figure illustrates the architecture style diagram showing the multi-tiered pattern that is followed by the **BrainBee** system.

Software Design Description for **BrainBee**

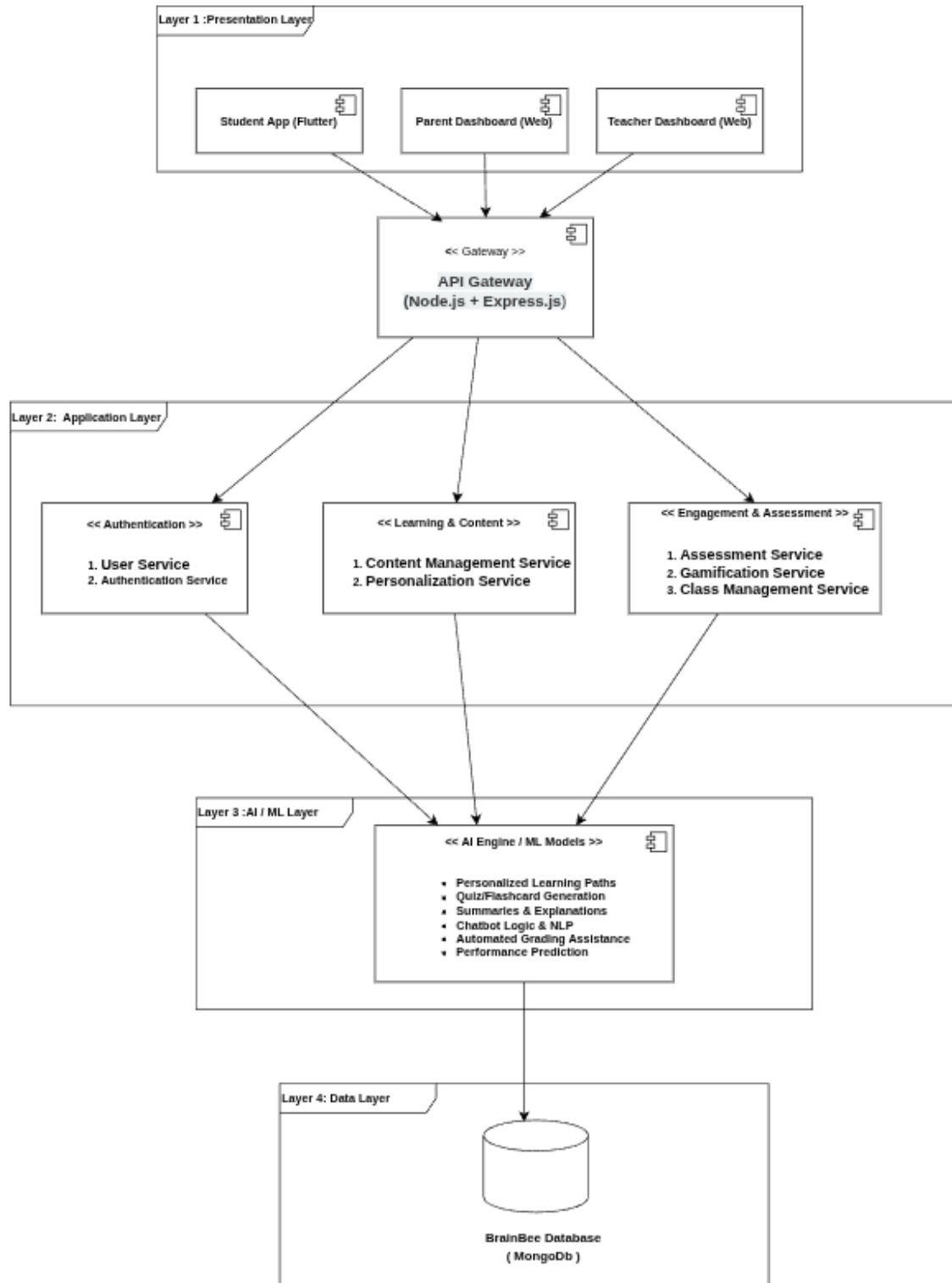


Figure 2 Architectural Style Diagram for BrainBee

Software Design Description for **BrainBee**

This figure illustrates the box and line diagram to represent the **BrainBee** system.

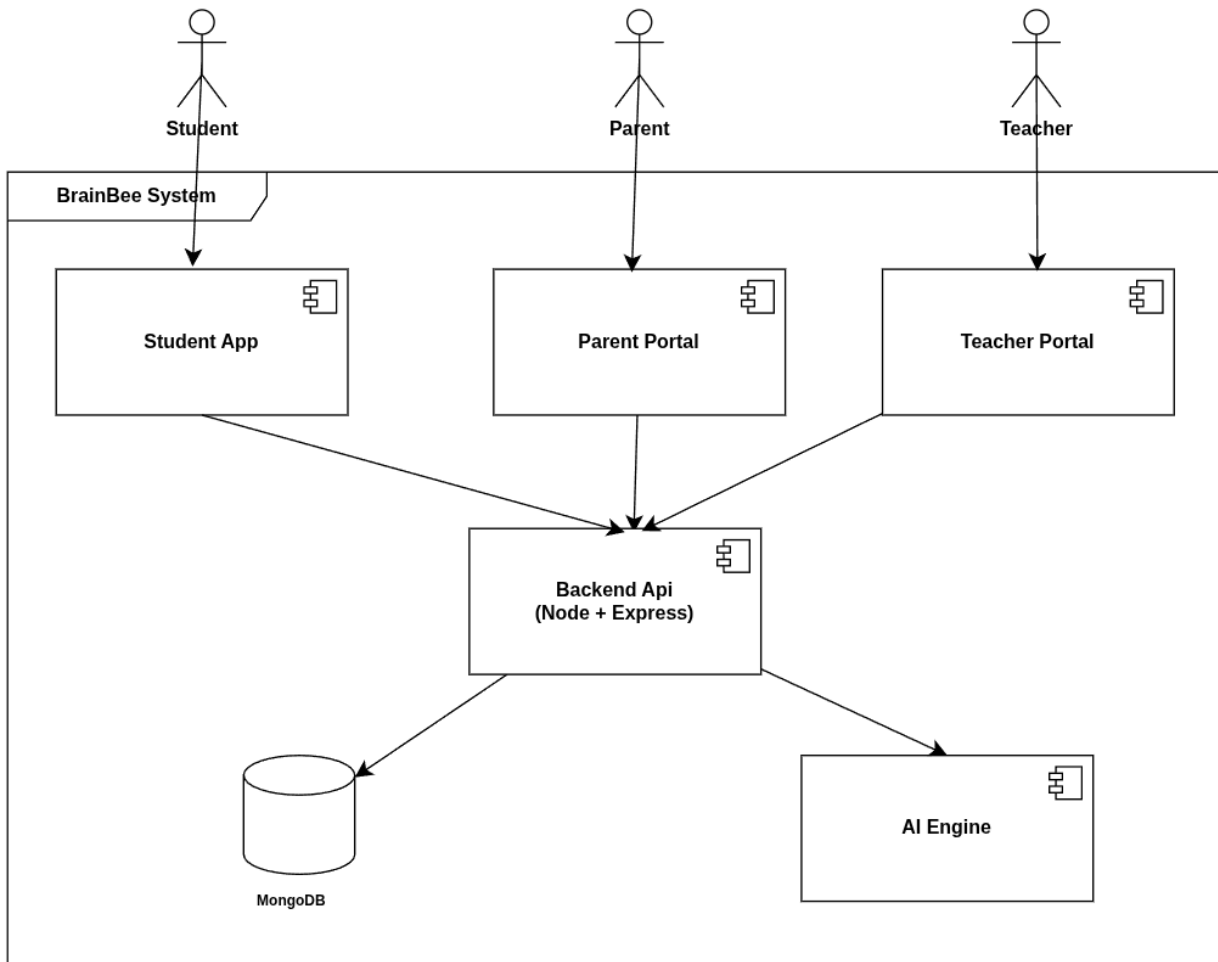


Figure 3 Box and Line Diagram for BrainBee

4. Design Models

This section details the design methodologies and models used to structure and implement the BrainBee system.

4.1 Activity Diagrams

This subsection presents activity diagrams illustrating the workflow of key processes within BrainBee

4.1.1 Activity Diagram for Module User Management

This figure illustrates the activity diagram detailing the flow of operations involved in the User Management module.

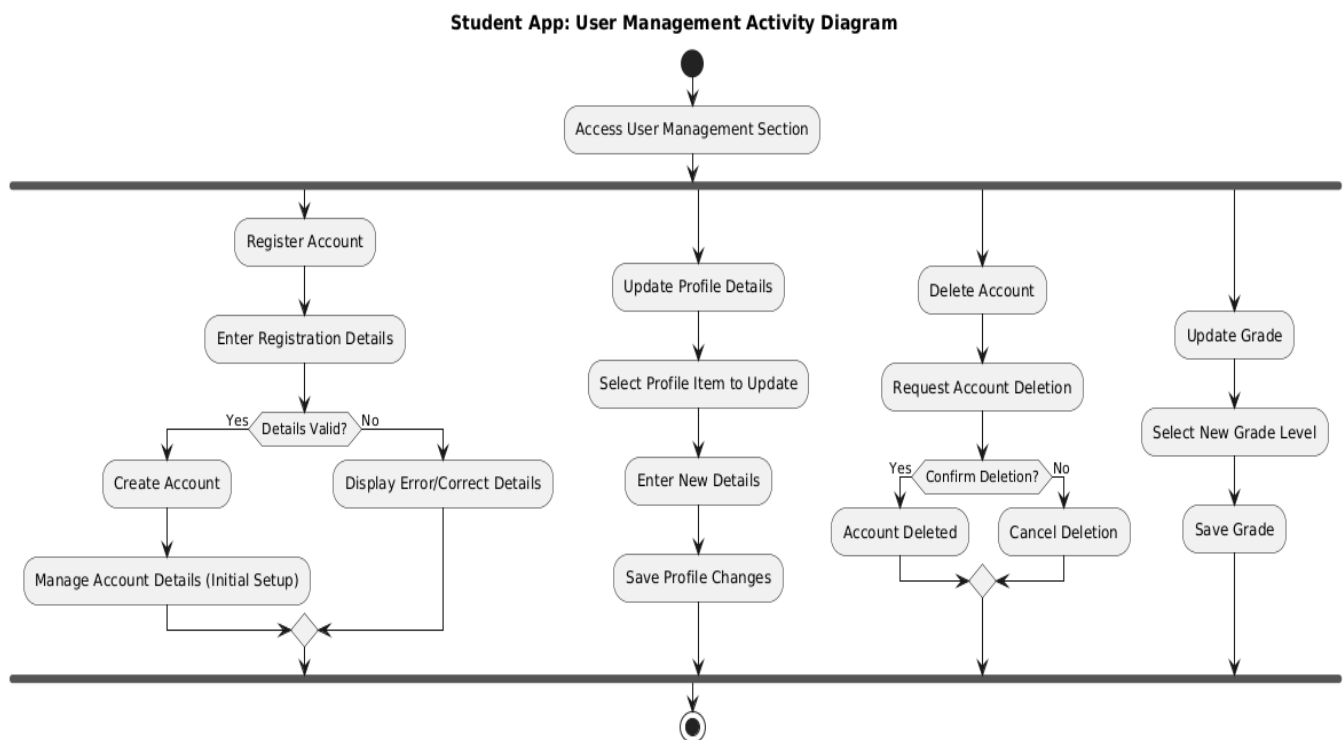


Figure 4 Activity Diagram for User Management

4.1.2 Activity Diagram for Module Peer Competition Module

This figure illustrates the activity diagram detailing the flow of operations involved in the Peer Competition Module



Figure 5 Activity Diagram for Peer Competition

4.1.3 Activity Diagram for Module Assessment Module

This figure illustrates the activity diagram detailing the flow of operations involved in the Assessment Module

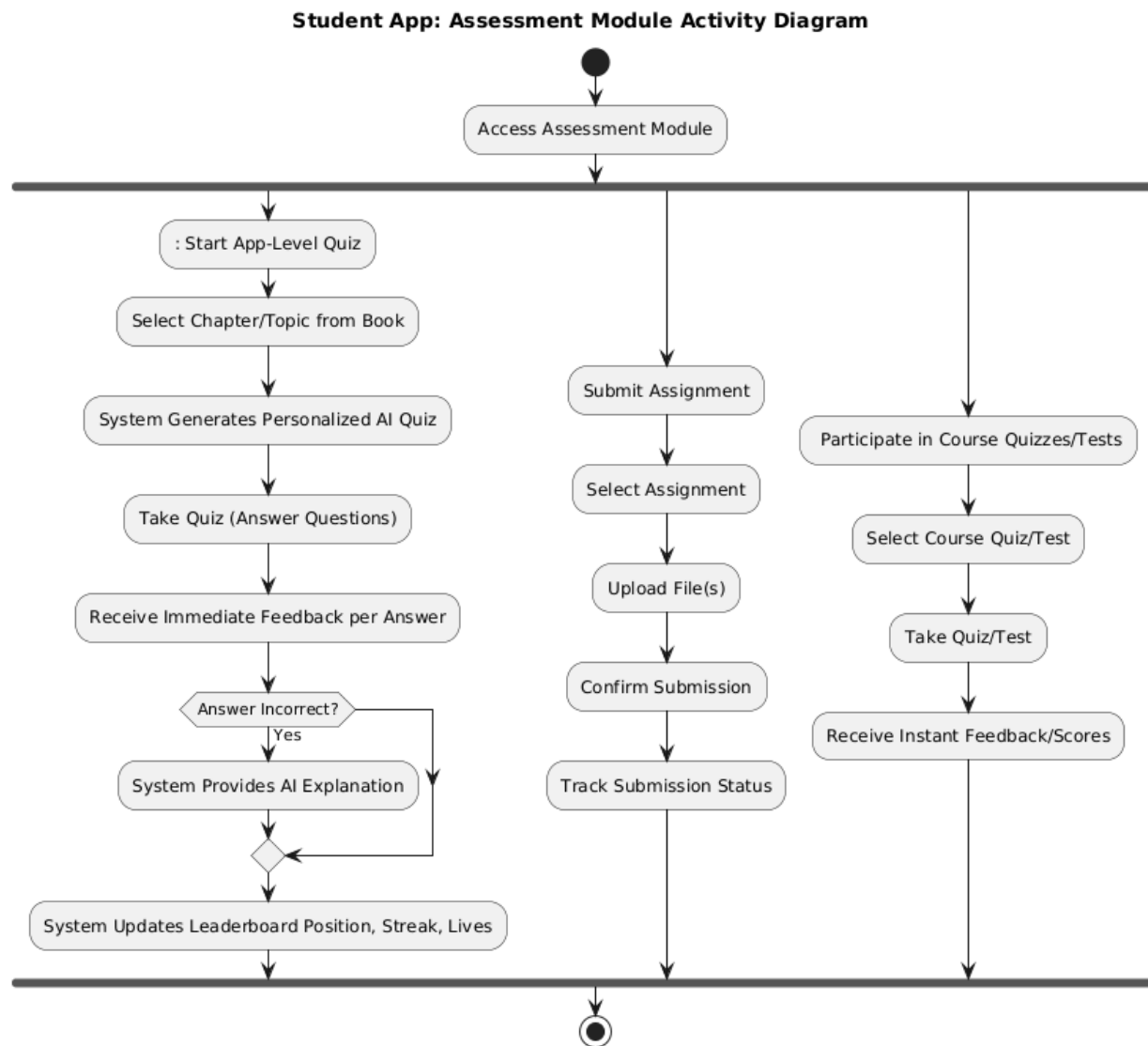


Figure 6 Activity Diagram for Assessment Module

4.1.4 Activity Diagram for Module AI Powered Assistant Module

This figure illustrates the activity diagram detailing the flow of operations involved in the AI Powered Assistant Module

Student App: AI Powered Assistant Module Activity Diagram

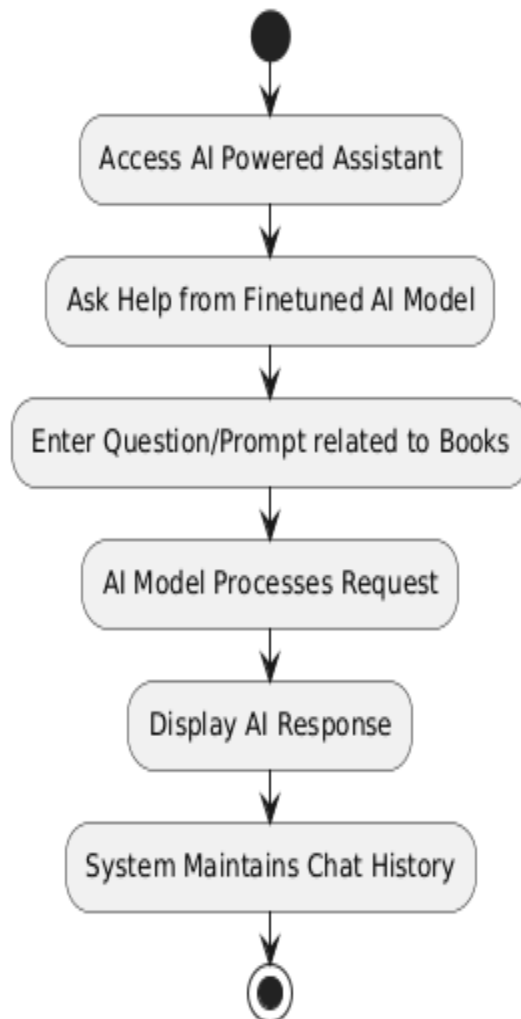


Figure 7 Activity Diagram for AI Powered Assistant Module

4.1.5 Activity Diagram for Module User Management

This figure illustrates the activity diagram detailing the flow of operations involved in the User Management module.

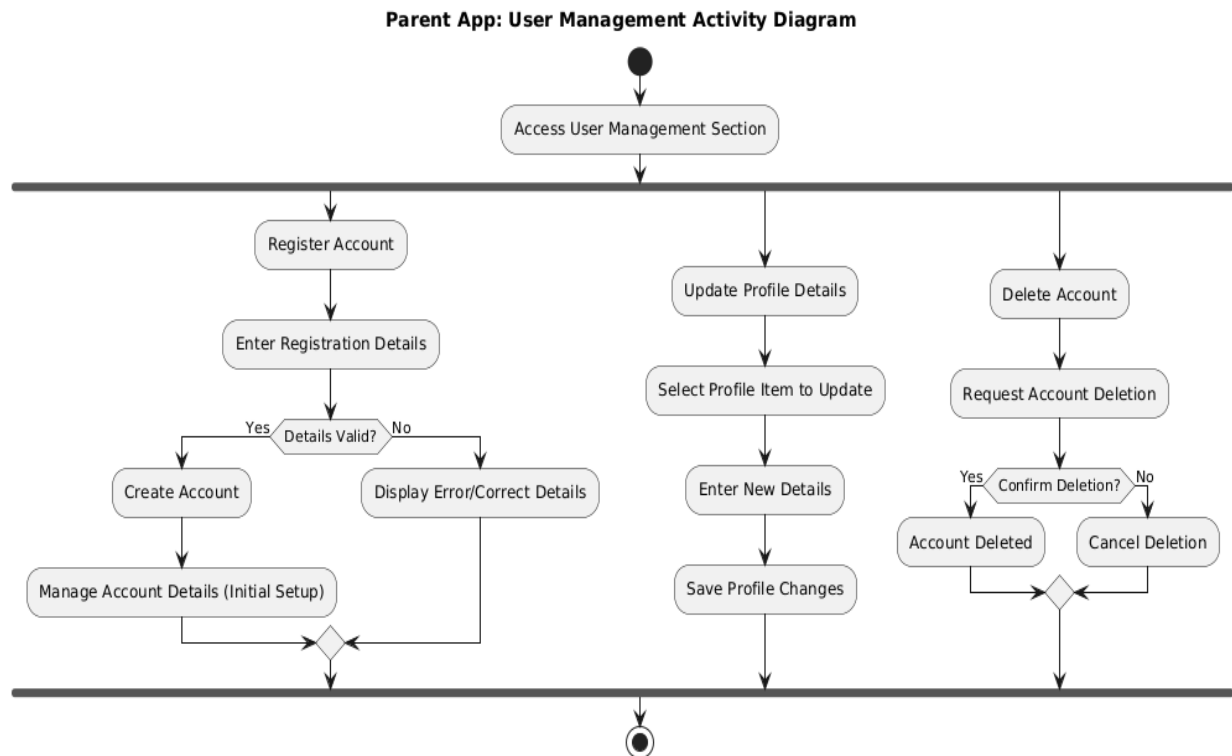


Figure 8 Activity Diagram for User Management

4.1.6 Activity Diagram for Module Monitor Children

This figure illustrates the activity diagram detailing the flow of operations involved in the Monitor Children

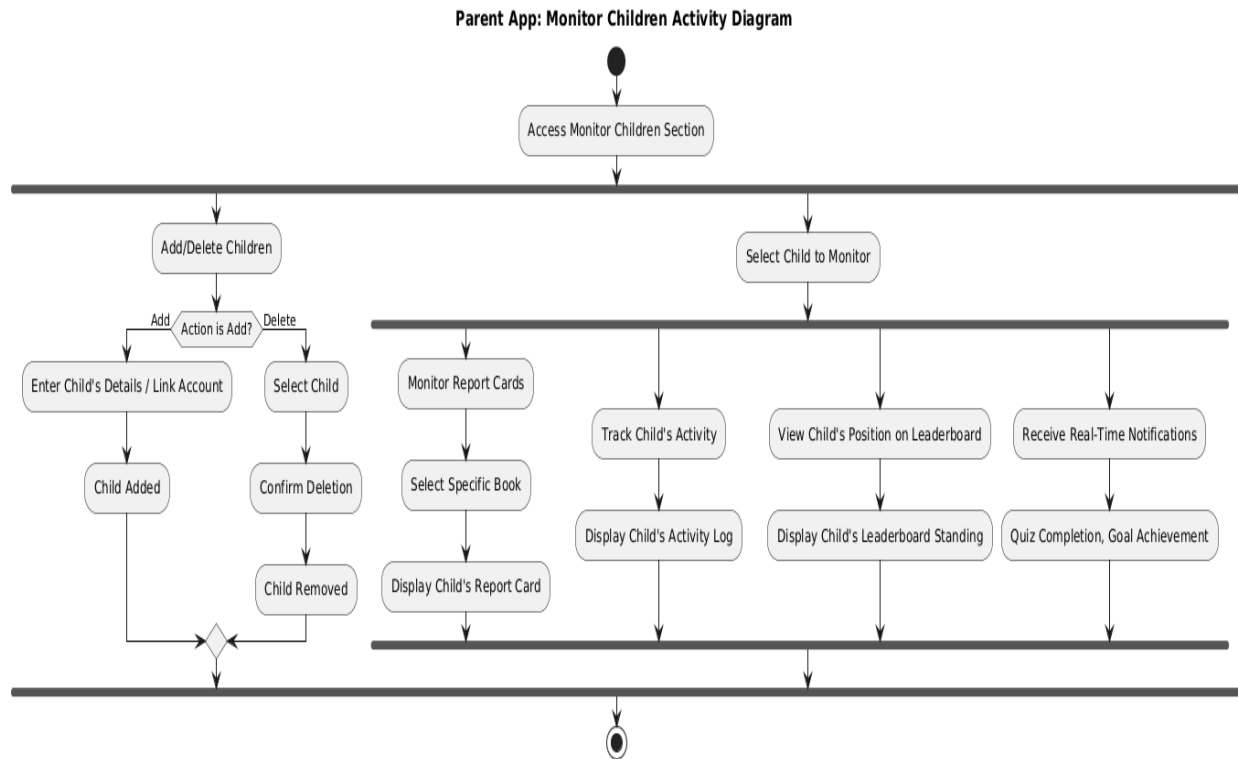


Figure 9 Activity Diagram for Monitor Children

4.1.7 Activity Diagram for Module Goals Module

This figure illustrates the activity diagram detailing the flow of operations involved in the Goals Module

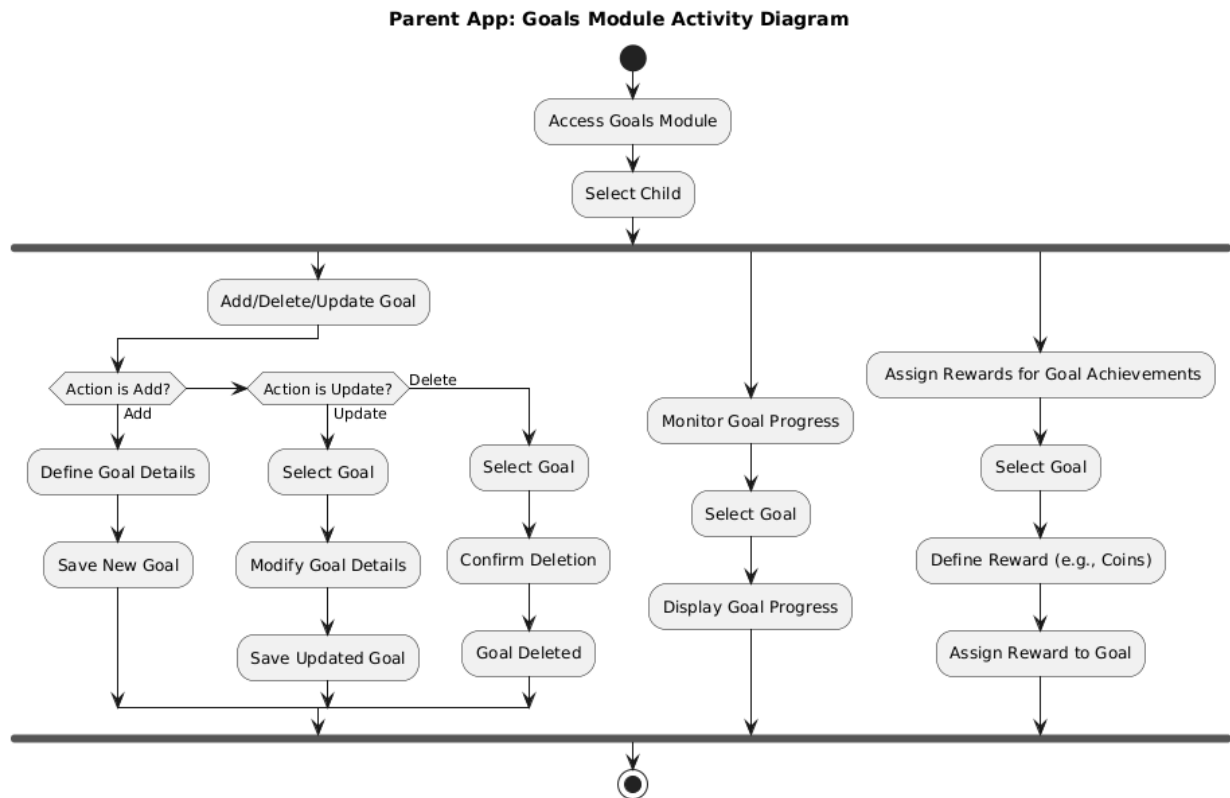


Figure 10 Activity Diagram for Goals Module

4.1.8 Activity Diagram for Module User Management

This figure illustrates the activity diagram detailing the flow of operations involved in the User Management module.

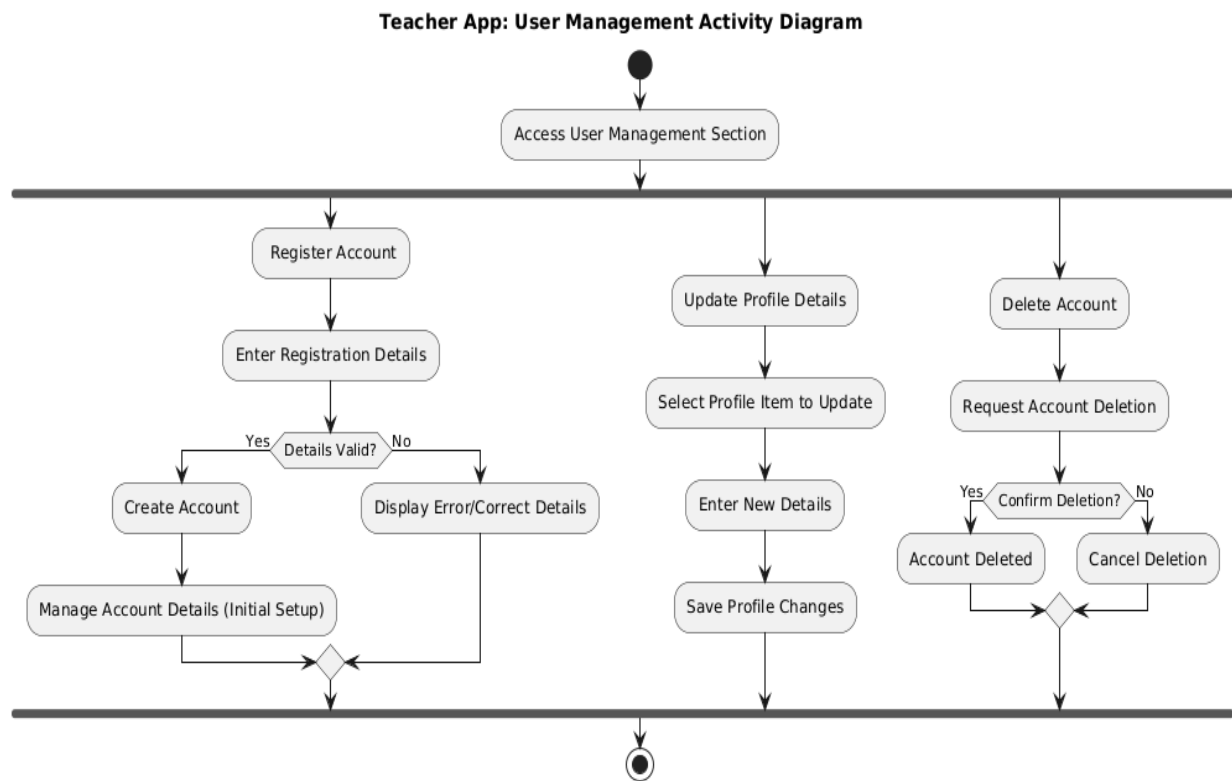


Figure 11 Activity Diagram for User Management

4.1.9 Activity Diagram for Module Class Management

This figure illustrates the activity diagram detailing the flow of operations involved in the Class Management

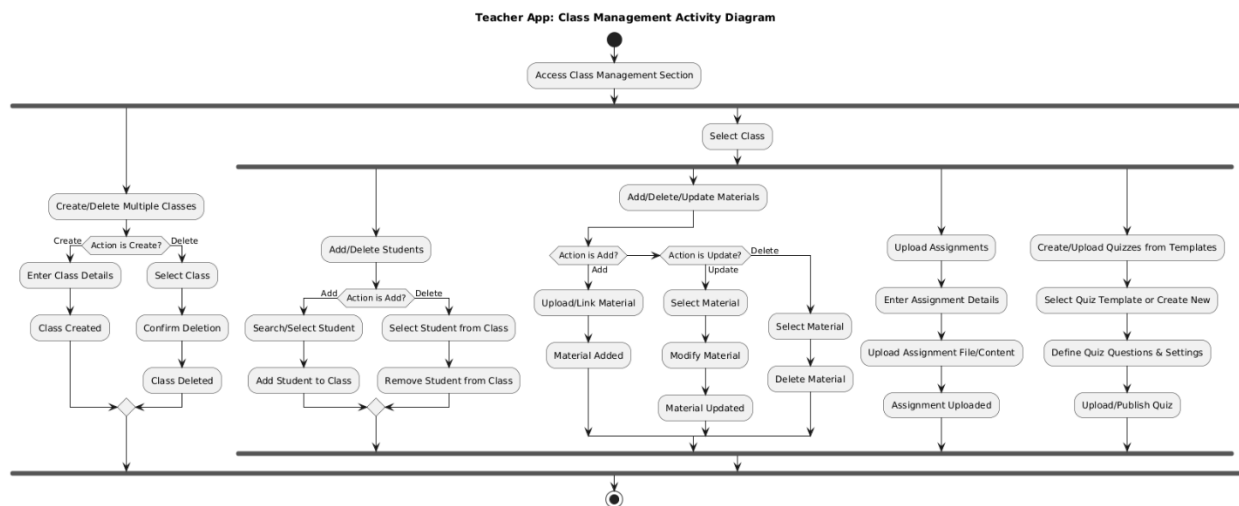


Figure 12 Activity Diagram for Class Management

4.1.10 Activity Diagram for Module Assessment Module

This figure illustrates the activity diagram detailing the flow of operations involved in the Assessment Module

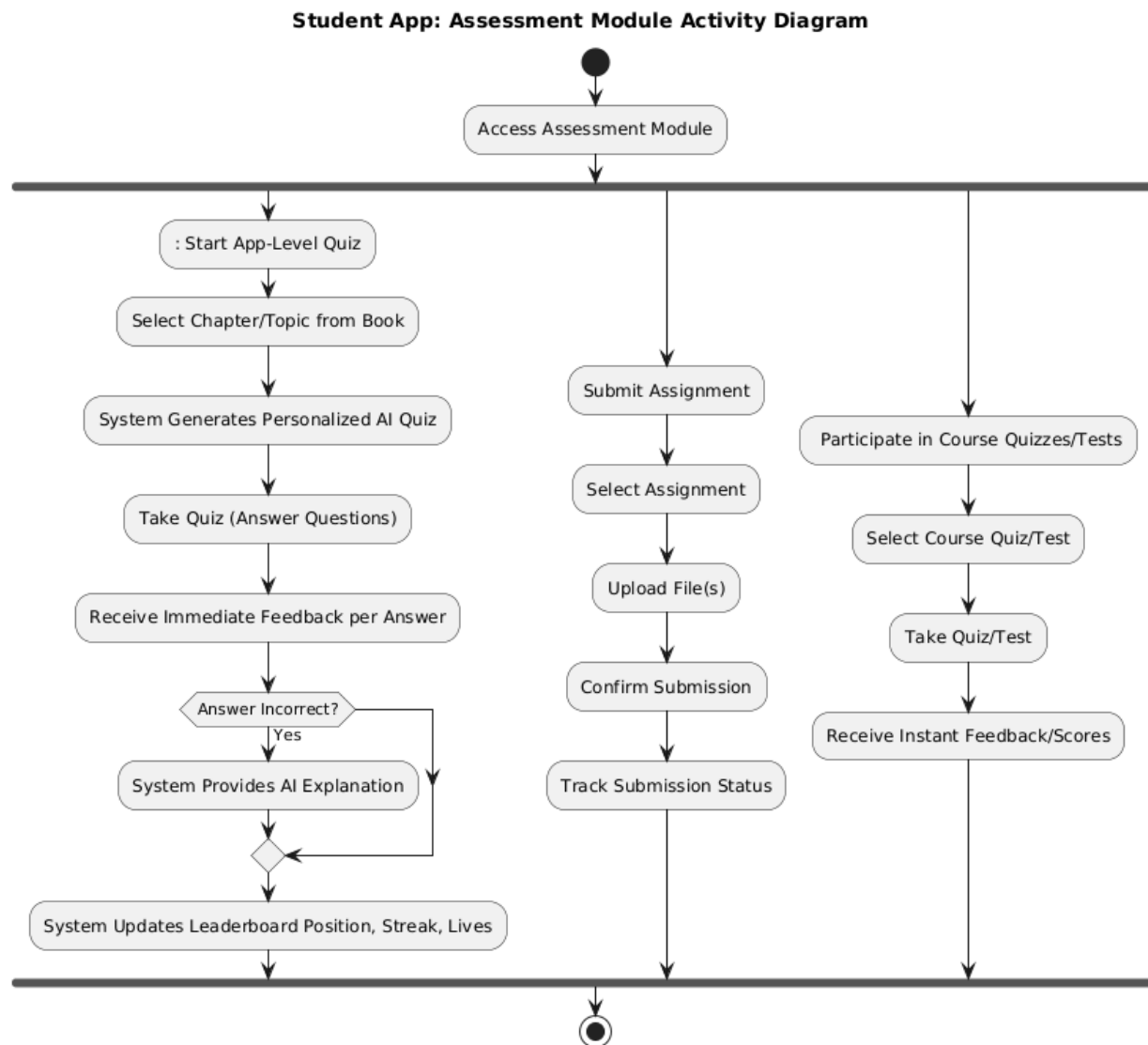
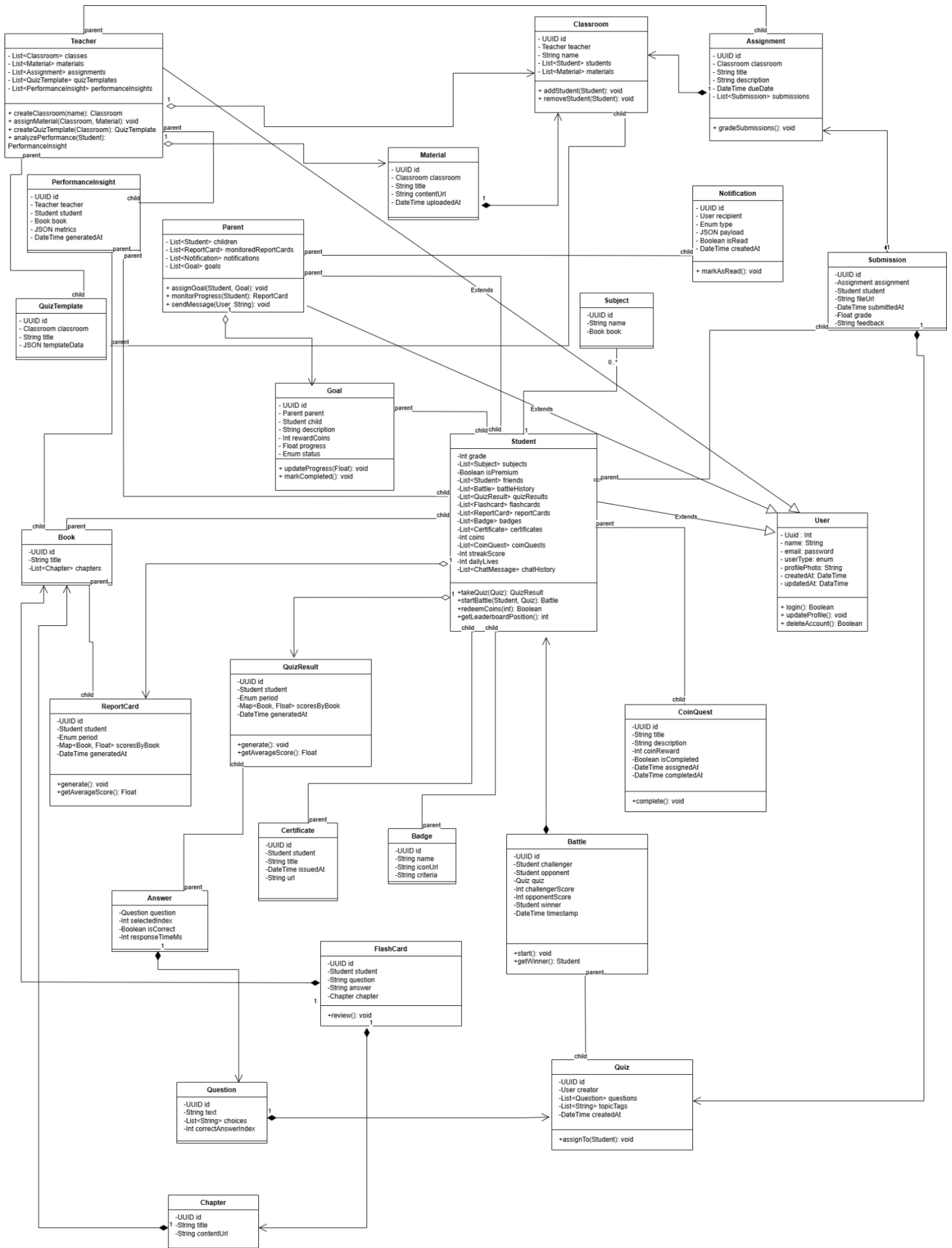


Figure 13 Activity Diagram for Assessment Module

4.2 Class Diagram

This figure shows the class diagram representing the structure of the BrainBee system, illustrating the various classes, their attributes, methods, and relationships.

Software Design Description for BrainBee



4.3 Sequence Diagrams

This subsection presents sequence diagrams depicting the sequence of interactions and messages exchanged between objects within BrainBee system during key processes.

4.3.1 Module User Management

This figure shows the sequence diagram for User Management module.

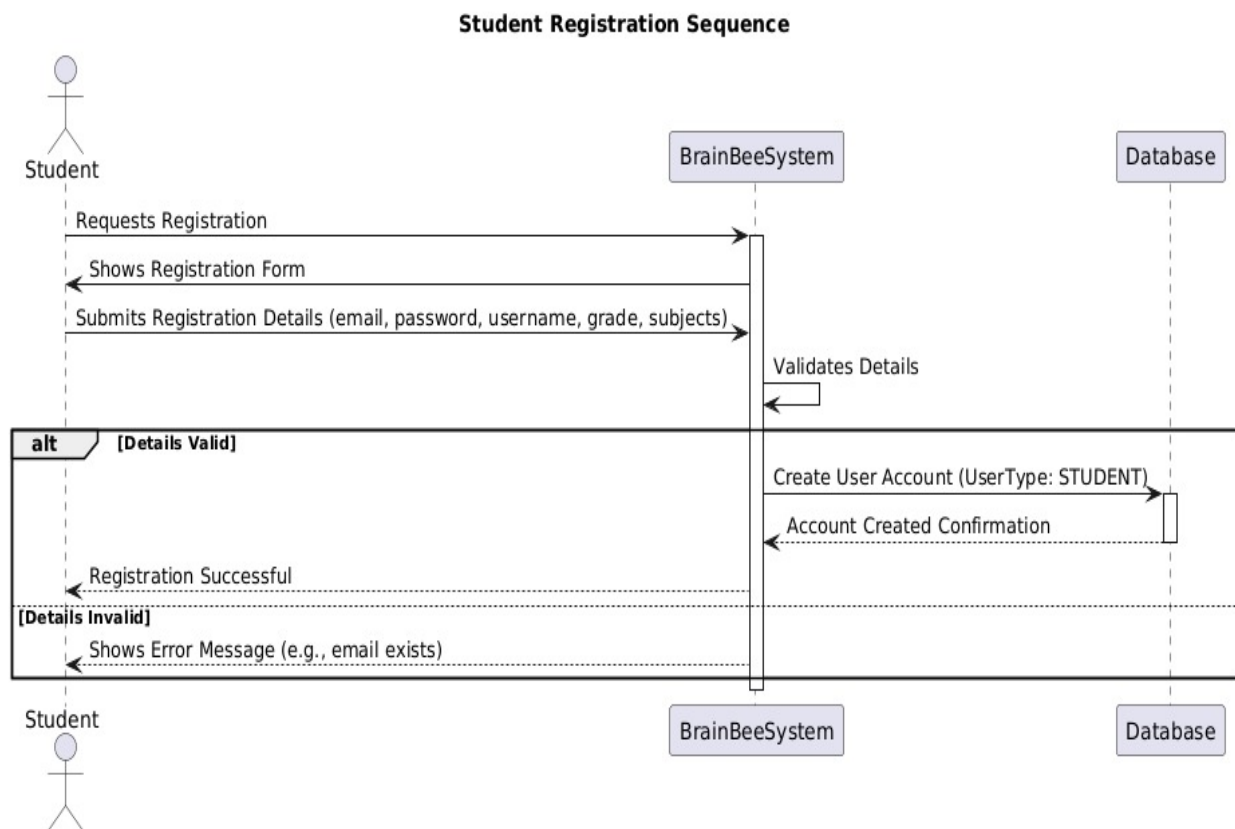


Figure 14 Sequence Diagram for User Management

4.3.2 Module Peer Competition

This figure shows the sequence diagram for Peer Competition Module

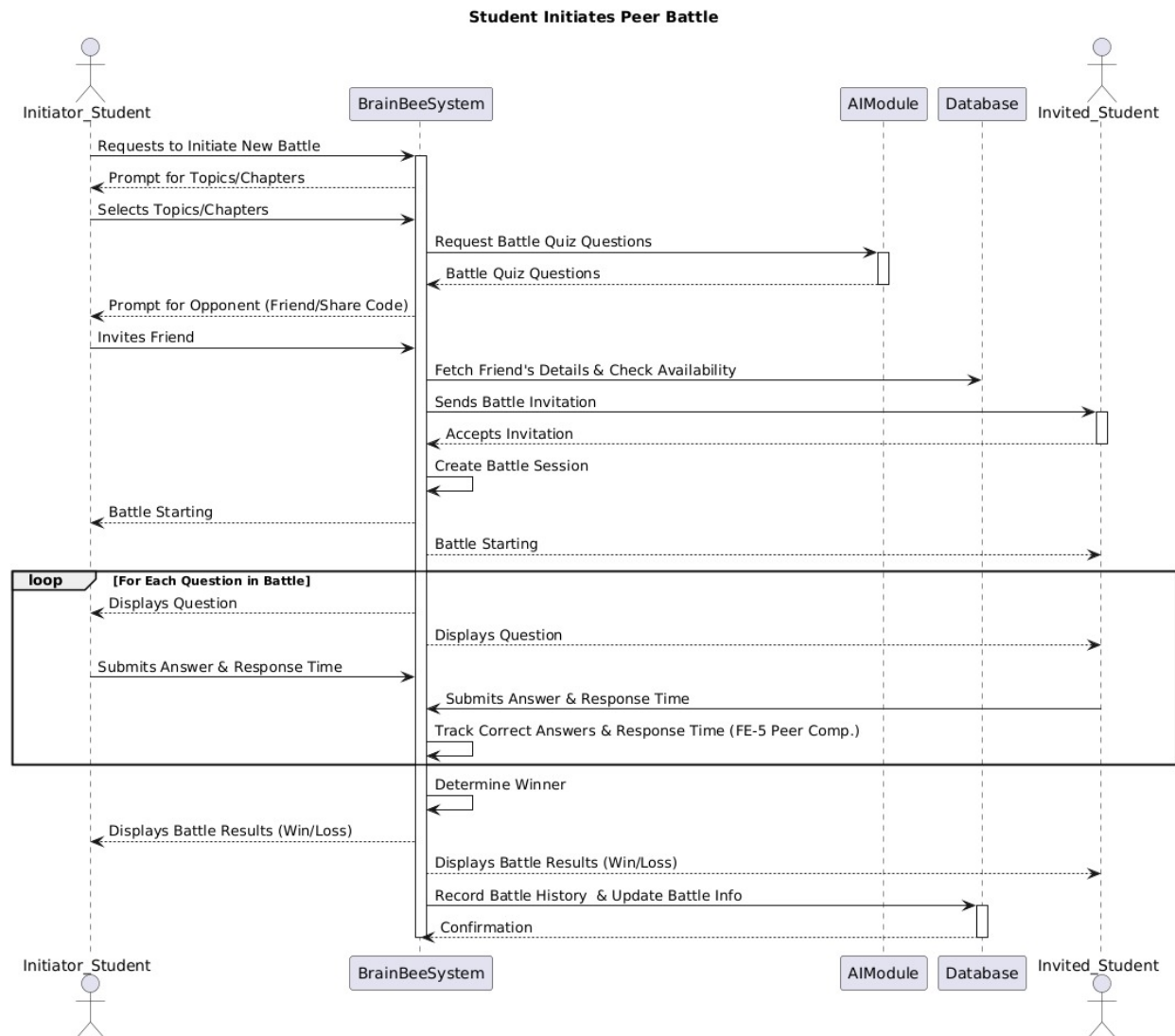


Figure 15 Sequence Diagram for Peer Competition Module

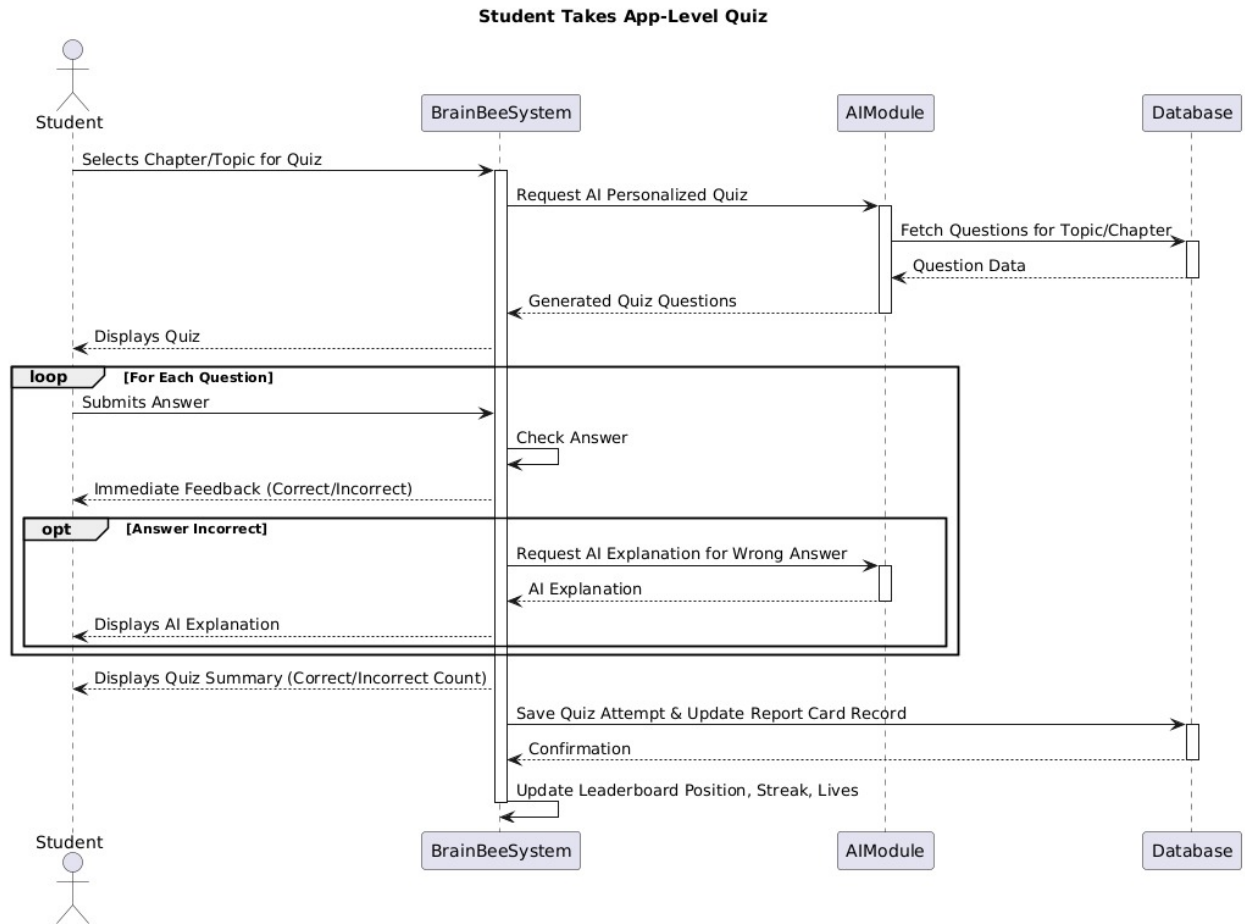


Figure 16 Sequence Diagram for Quiz

4.3.3 Module Assessment

This figure shows the sequence diagram for Assessment Module

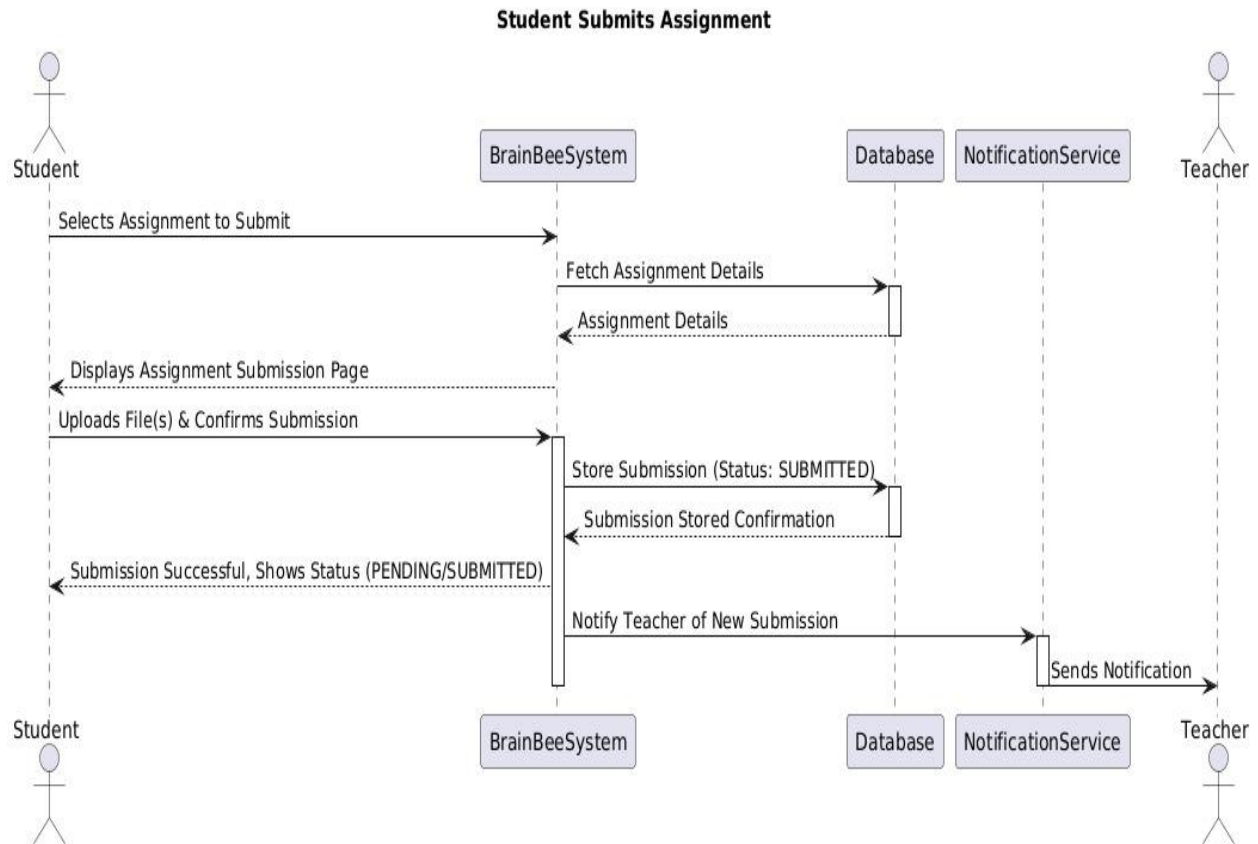


Figure 17 Sequence Diagram for Assessment Module

4.3.4 Module Performance Analytics

This figure shows the sequence diagram for erformance Analytics Module

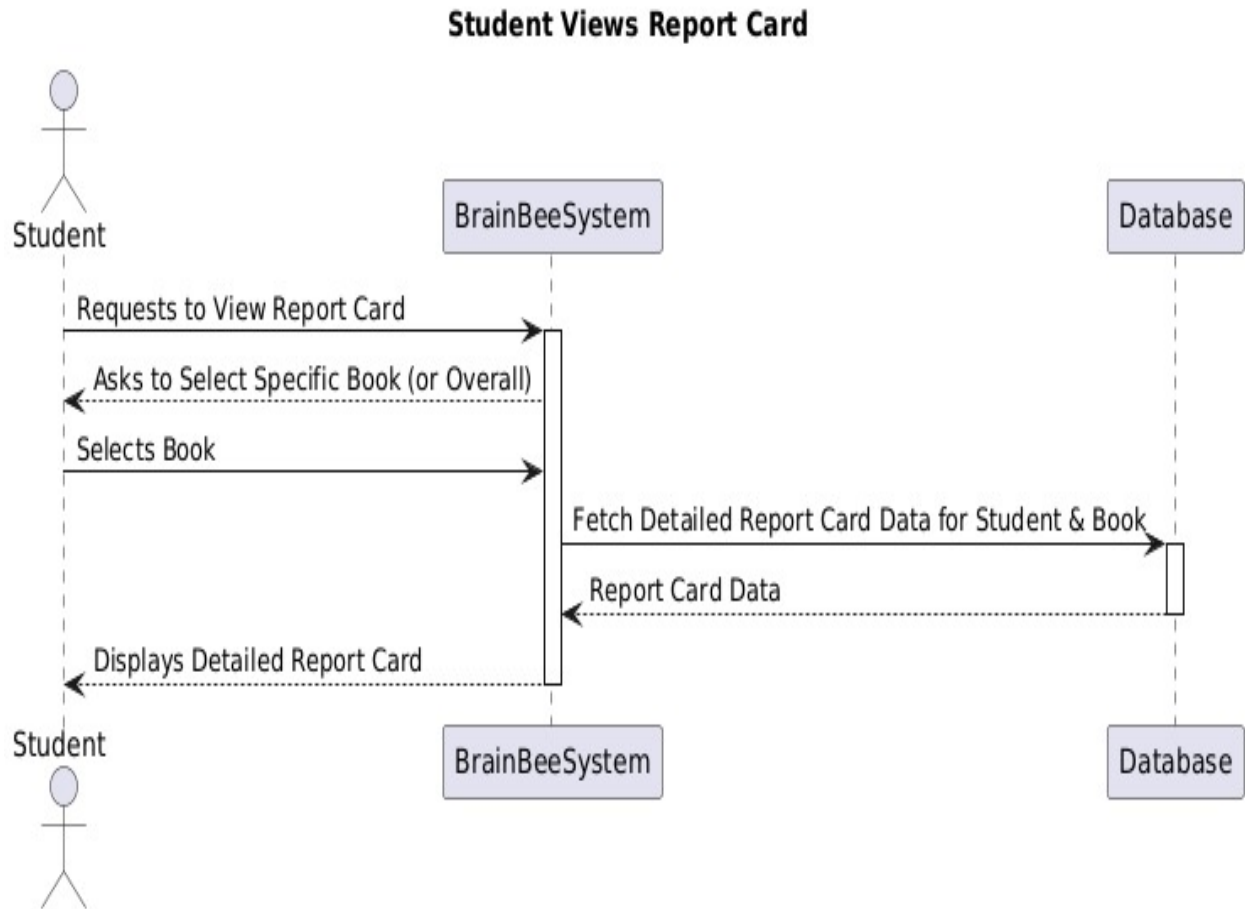


Figure 18 Sequence Diagram for Performance Analytics

4.3.5 Module AI Powered Assistant

This figure shows the sequence diagram for AI Powered Assistant Module

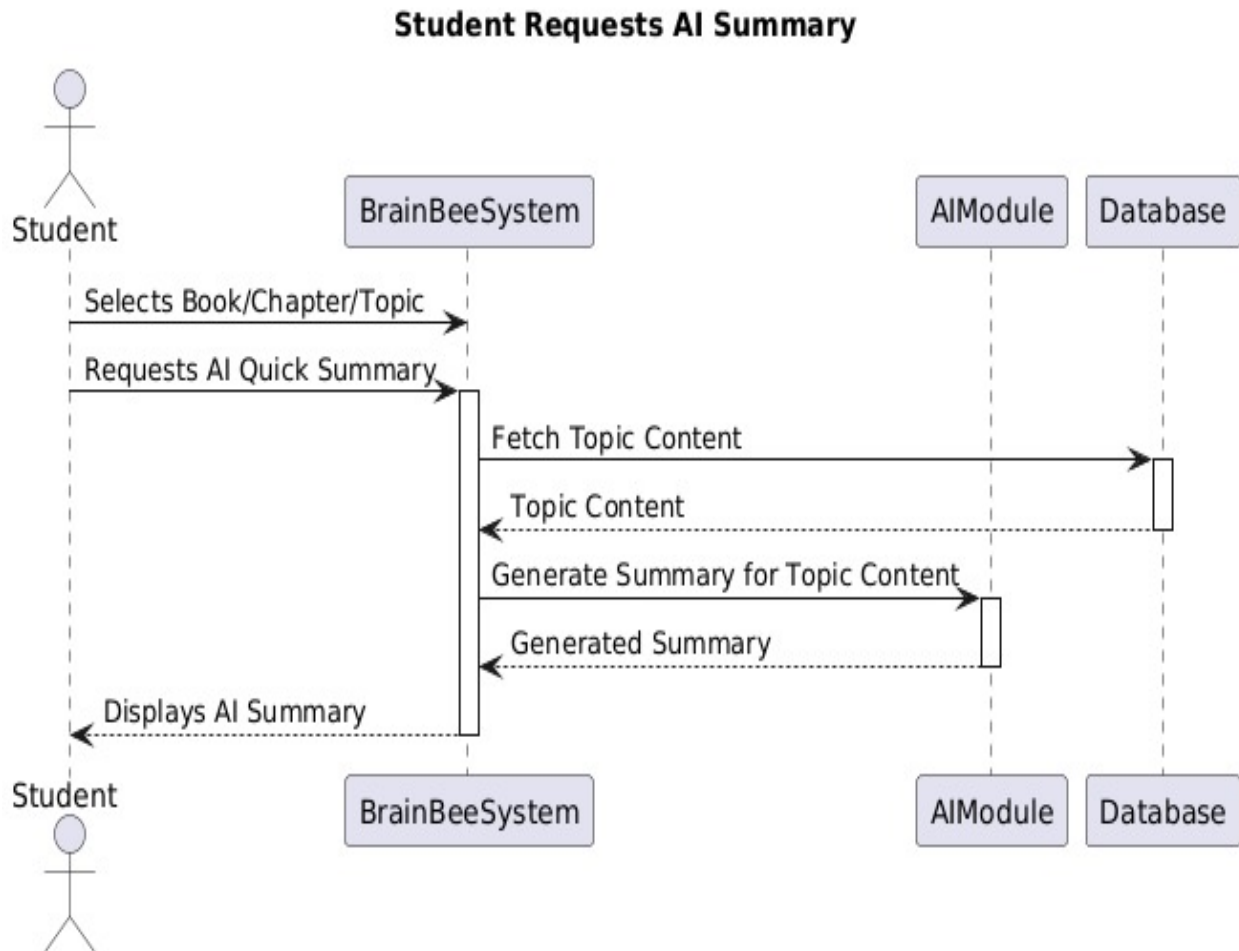


Figure 19 Sequence Diagram for AI Powered Assistant

4.3.6 Module Monitor Children

This figure shows the sequence diagram for Monitor Children

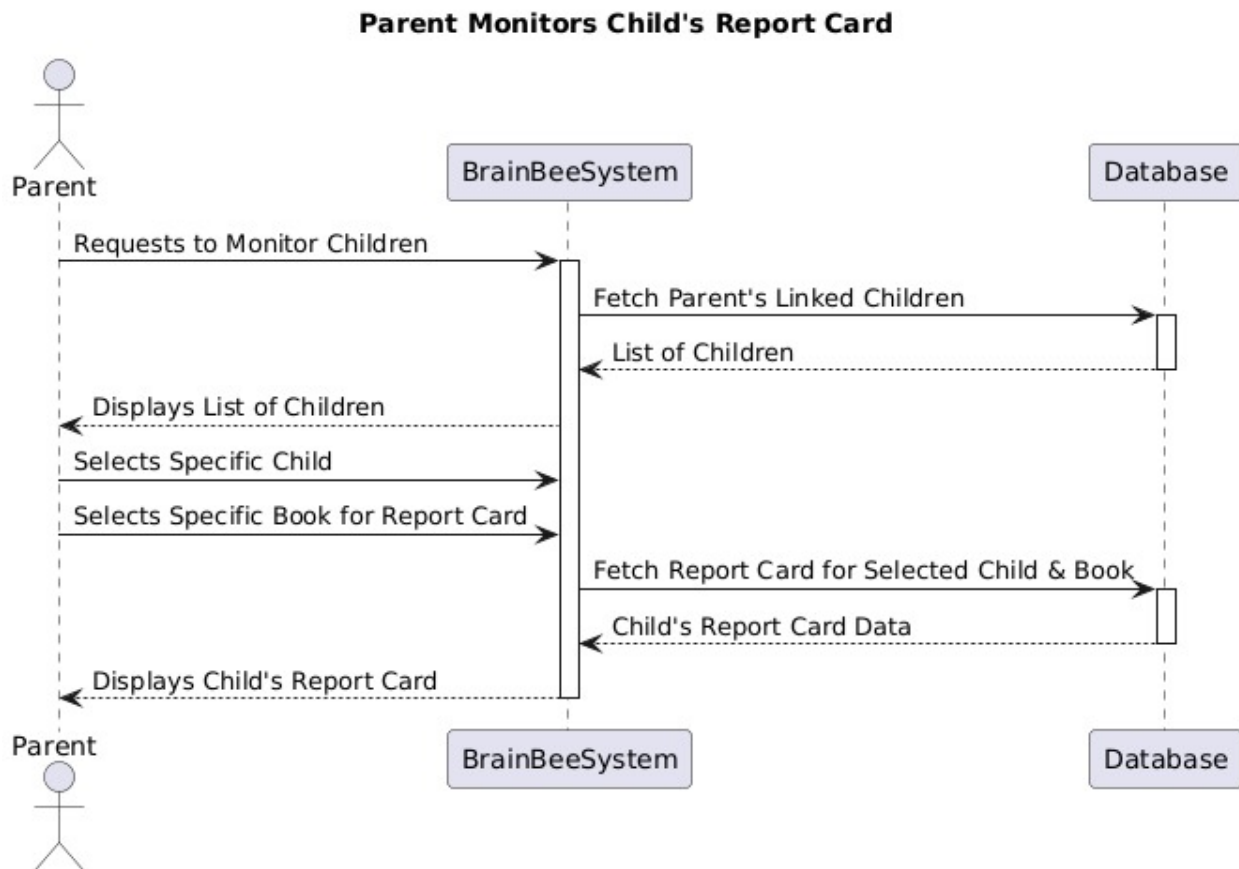


Figure 20 Sequence Diagram for Monitor Children

4.3.7 Module Goal

This figure shows the sequence diagram for Goals Module

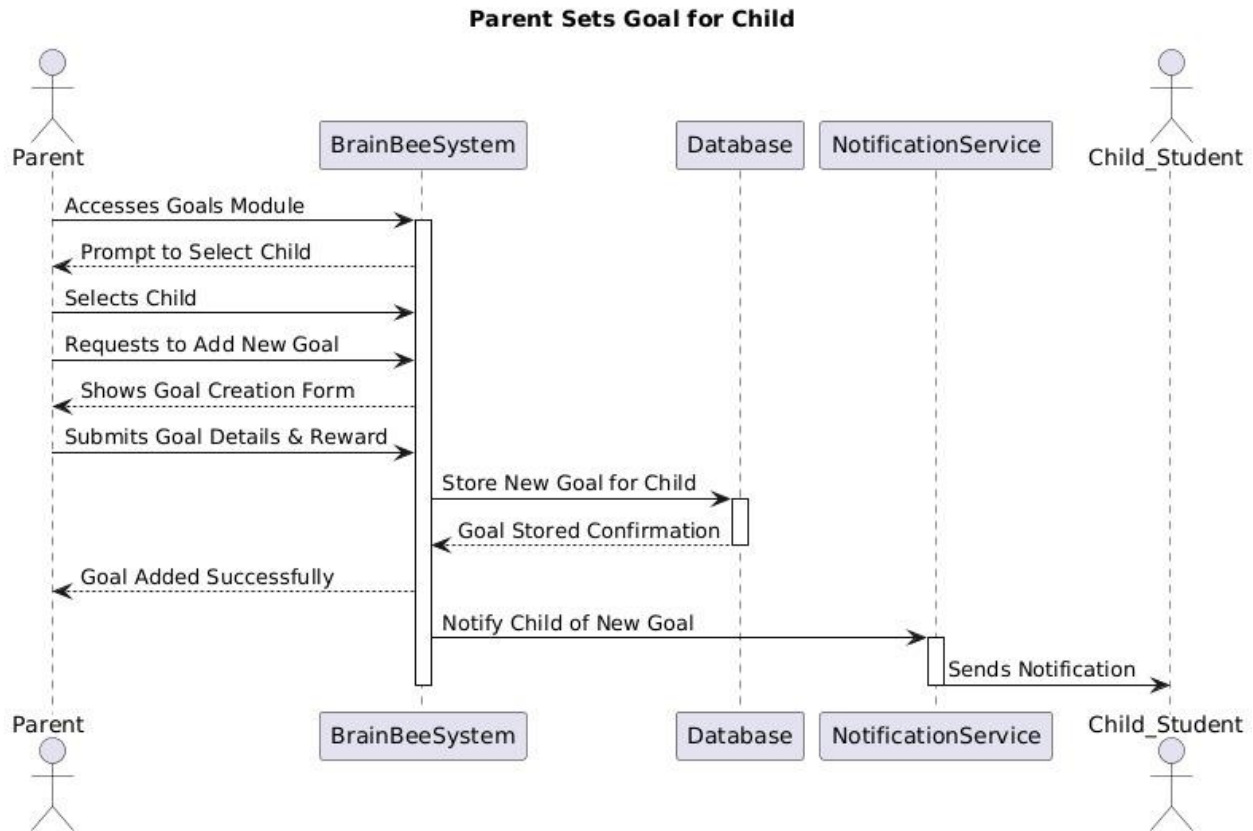


Figure 21 Sequence Diagram for Goals Module

4.3.8 Module User Management

This figure shows the sequence diagram for User Management module.

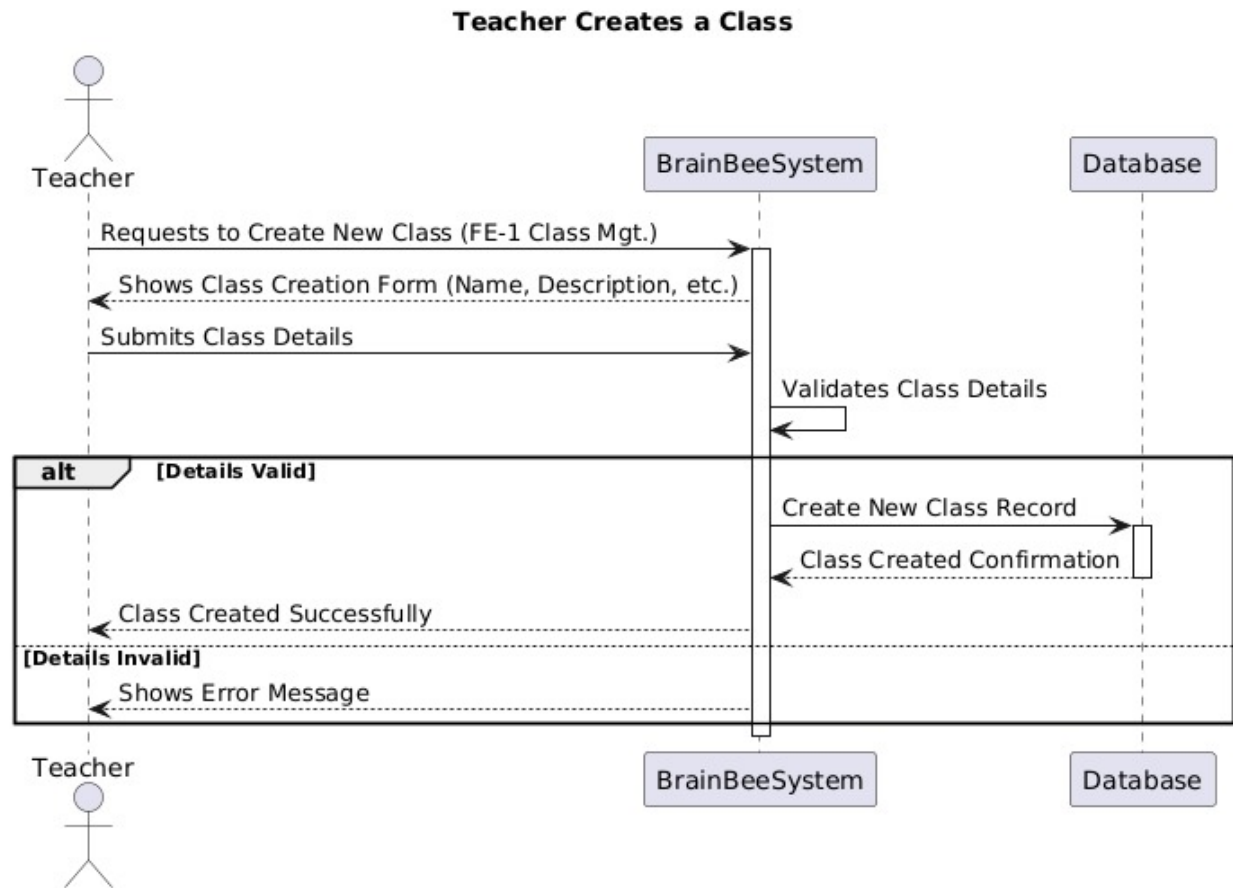


Figure 22 Sequence Diagram for User Management

4.3.9 Module Class Management

This figure shows the sequence diagram for Class Management

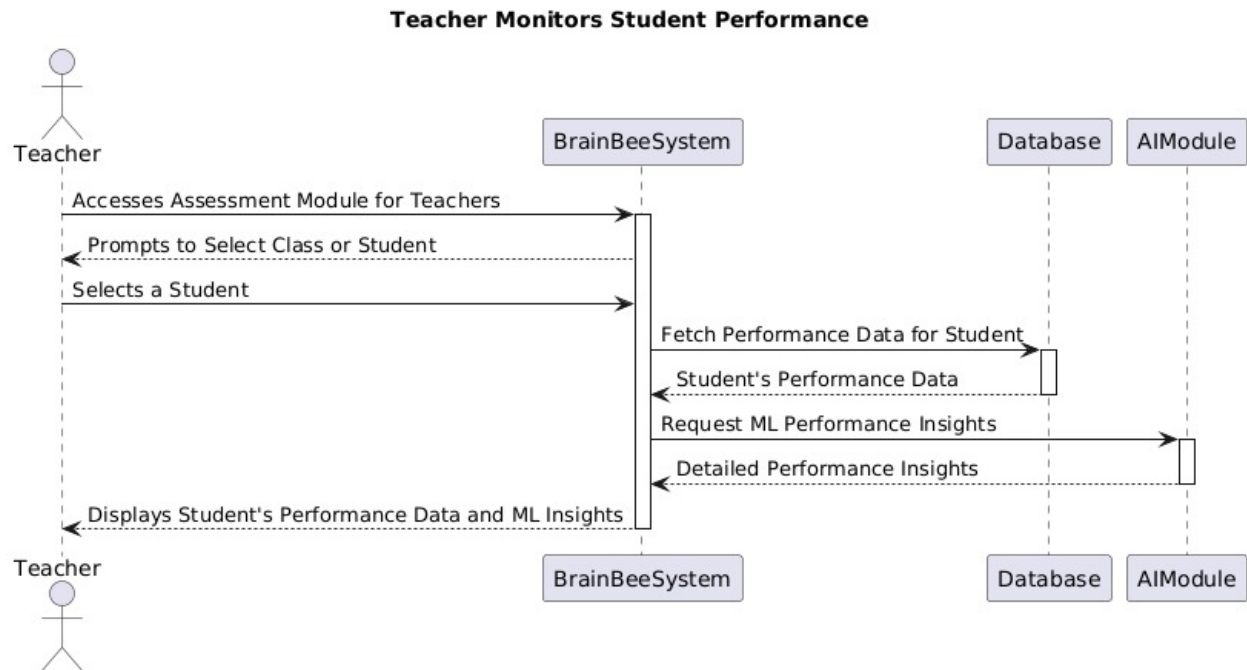


Figure 23 Sequence Diagram for Monitor Student Performance

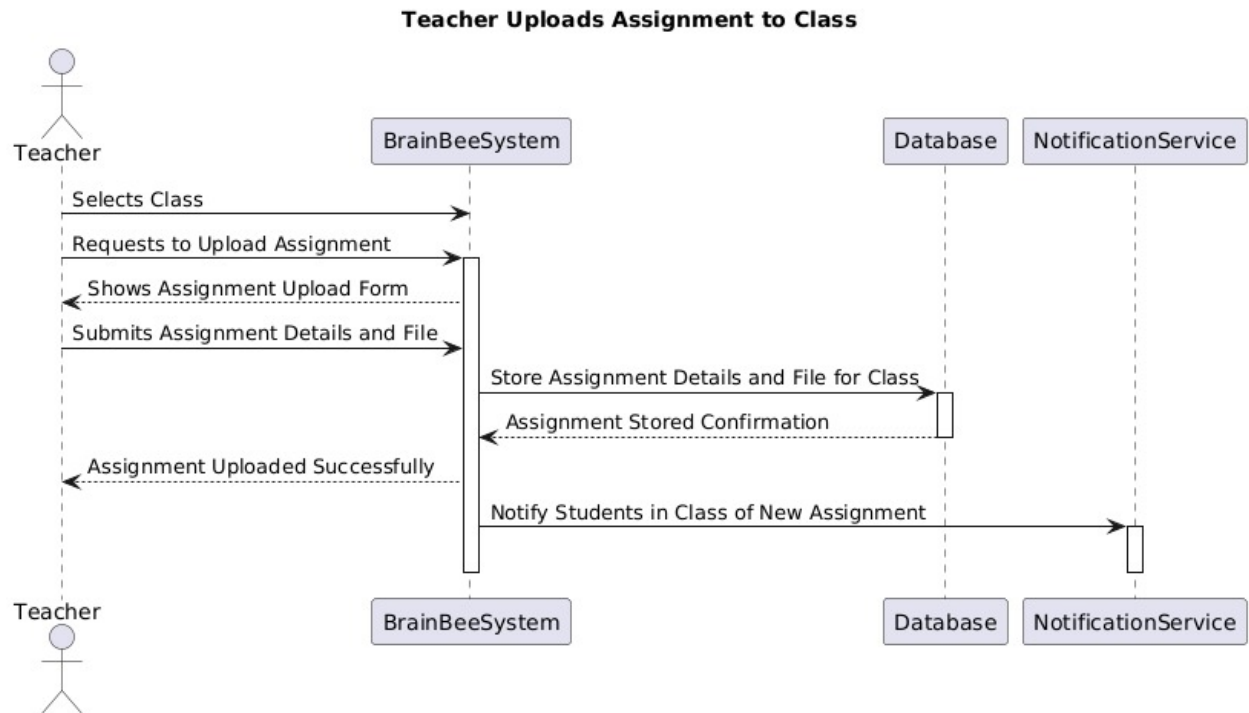


Figure 24 Sequence Diagram for Teacher Uploads assignment

4.4 State Transition Diagrams

The State Transition diagrams for our system **BrainBee** are described here.

4.4.1 State Transition Diagram for Quiz Module

This figure shows the state transition diagram for Quiz Module

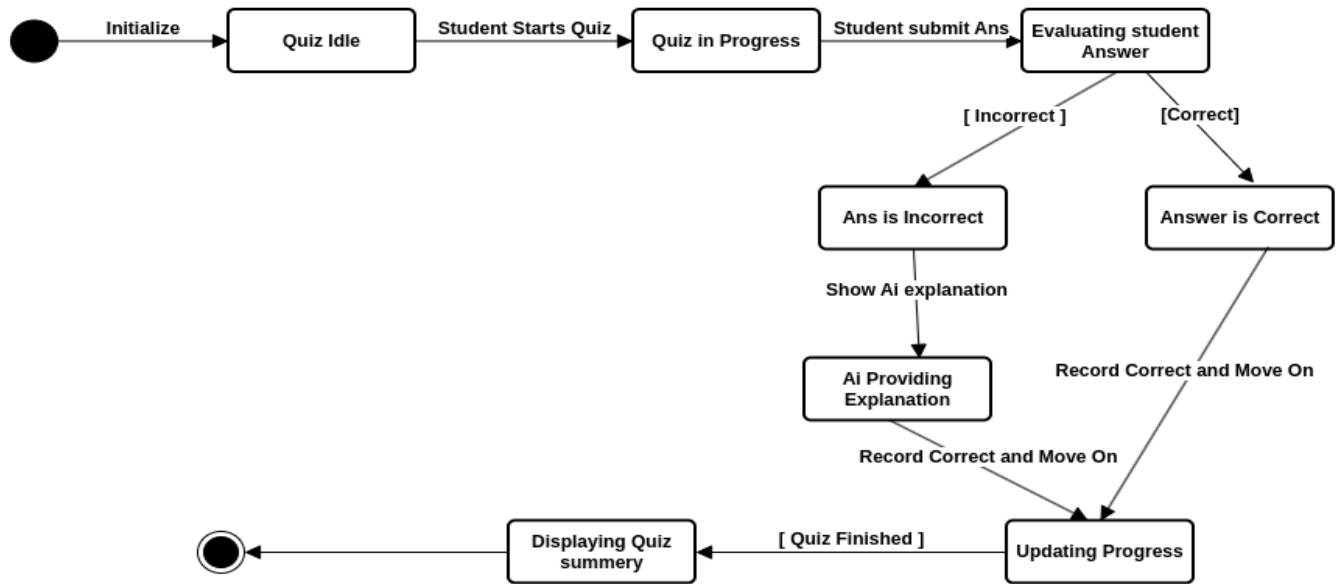


Figure 25 State Transition Diagram for Quiz

4.4.2 State Transition Diagram for Quiz Attempt

This figure shows the state transition diagram for Quiz Attempt

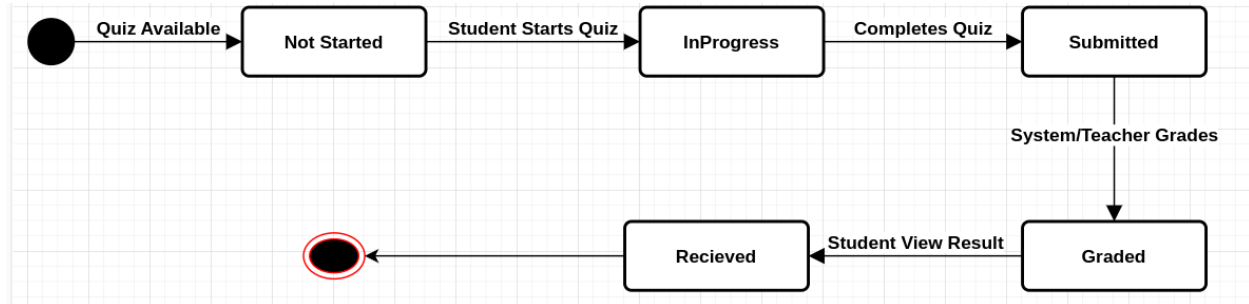


Figure 26 State Transition Diagram for Quiz Attempt

5. Data Design

This section outlines the structure and organization of the data used in **BrainBee**, detailing the schema, data models, and relationships essential for efficient data management and retrieval.

5.1 Data Dictionary

These tables show the data dictionary used in the **BrainBee** system.

User (Common Base for Student, Parent, Teacher):

Attribute	Type	Description
id	UUID	Unique user identifier
name	String	Full name of the user
email	String	Registered email address
passwordHash	String	Hashed password
userType	Enum	STUDENT, PARENT, or TEACHER
profilePhotoUrl	String?	URL to profile image
createdAt	DateTime	Account creation timestamp
updatedAt	DateTime	Last account update timestamp

Student (extends User):

Attribute	Type	Description
grade	Int	Grade level of the student
subjects	List	Subjects enrolled
isPremium	Boolean	Premium membership status
friends	List	Friends list

Software Design Description for **BrainBee**

battleHistory	List	History of quiz battles
quizResults	List	Quiz attempt records
flashcards	List	Flashcards created by the student
reportCards	List	Academic report cards
badges	List	Earned badges
certificates	List	Awarded certificates
coins	Int	Current coin balance
coinQuests	List	Active and completed coin quests
streakScore	Int	Streak count for daily activity
dailyLives	Int	Remaining lives for daily quizzes
chatHistory	List	Chat history

Parent (extends User):

Attribute	Type	Description
children	List	Associated child accounts
Monitored -ReportCards	List	Report cards being monitored
notifications	List	Received notifications
goals	List	Goals set for children

Teacher (extends User):

Attribute	Type	Description
classes	List	Assigned classes
materials	List	Teaching materials

Software Design Description for **BrainBee**

assignments	List	Given assignments
quizTemplates	List	Quiz templates created
performanceInsights	List	Performance analytics

Subject:

Attribute	Type	Description
id	UUID	Unique subject ID
name	String	Name of the subject
book	Book	Associated book

Book:

Attribute	Type	Description
id	UUID	Unique book ID
title	String	Title of the book
chapters	List	Chapters in the book

Chapter:

Attribute	Type	Description
id	UUID	Unique chapter ID
title	String	Title of the chapter
contentUrl	String	URL to content

Quiz:

Attribute	Type	Description
id	UUID	Unique quiz ID

Software Design Description for **BrainBee**

creator	User?	Creator of the quiz (or AI)
questions	List	List of questions
topicTags	List	Related topics
createdAt	DateTime	Quiz creation date

Question:

Attribute	Type	Description
id	UUID	Unique question ID
text	String	Question text
choices	List	Answer options
correctAnswerIndex	Int	Index of correct choice

QuizResult:

Attribute	Type	Description
id	UUID	Unique result ID
student	Student	Student who took the quiz
quiz	Quiz	Quiz taken
answers	List	Student answers
correctCount	Int	Number of correct answers
incorrectCount	Int	Number of incorrect answers
timestamp	DateTime	When the quiz was taken

Answer:

Attribute	Type	Description
question	Question	Associated question

Software Design Description for **BrainBee**

selectedIndex	Int	Selected answer index
isCorrect	Boolean	Whether the answer was correct
responseTimeMs	Int	Time taken to answer (ms)

Battle:

Attribute	Type	Description
id	UUID	Unique battle ID
challenger	Student	Initiating student
opponent	Student	Opponent student
quiz	Quiz	Quiz used for battle
challengerScore	Int	Score of challenger
opponentScore	Int	Score of opponent
winner	Student?	Winner of the battle
timestamp	DateTime	Battle timestamp

ReportCard:

Attribute	Type	Description
id	UUID	Unique report card ID
student	Student	Associated student
period	Enum	WEEKLY, MONTHLY, or YEARLY
scoresByBook	Map<Book, Float>	Performance by book
generatedAt	DateTime	Date of report generation

Flashcard:

Attribute	Type	Description
id	UUID	Unique flashcard ID
student	Student	Owner of the flashcard
question	String	Flashcard question
answer	String	Flashcard answer
chapter	Chapter	Linked chapter

ChatMessage:

Attribute	Type	Description
id	UUID	Unique message ID
sender	User	Message sender
recipient	User AIModel	Message recipient
content	String	Message content
timestamp	DateTime	Time of sending

CoinQuest:

Attribute	Type	Description
id	UUID	Unique quest ID
title	String	Quest title
description	String	Quest description
coinReward	Int	Coins awarded
isCompleted	Boolean	Completion status
assignedAt	DateTime	Assignment date
completedAt	DateTime?	Completion date

Badge:

Software Design Description for **BrainBee**

Attribute	Type	Description
id	UUID	Unique badge ID
name	String	Name of the badge
iconUrl	String	Badge icon URL
criteria	String	Criteria to earn the badge

Certificate:

Attribute	Type	Description
id	UUID	Unique certificate ID
student	Student	Certificate holder
title	String	Certificate title
issuedAt	DateTime	Issue date
url	String?	Optional certificate link

Notification:

Attribute	Type	Description
id	UUID	Unique notification ID
recipient	User	Notification recipient
type	Enum	QUIZ_COMPLETED, GOAL_PROGRESS, etc.
payload	JSON	Additional data
isRead	Boolean	Read status
createdAt	DateTime	Notification timestamp

Goal:

Attribute	Type	Description
id	UUID	Unique goal ID

Software Design Description for **BrainBee**

parent	Parent	Goal setter
child	Student	Goal target
description	String	Goal description
rewardCoins	Int	Reward coins
progress	Float	Progress (0.0–1.0)
status	Enum	PENDING or COMPLETED

Classroom:

Attribute	Type	Description
id	UUID	Unique classroom ID
teacher	Teacher	Class teacher
name	String	Class name
students	List	Enrolled students
materials	List	Teaching materials

Material:

Attribute	Type	Description
id	UUID	Unique material ID
classroom	Classroom	Related classroom
title	String	Material title
contentUrl	String	Content file URL
uploadedAt	DateTime	Upload time

Assignment:

Attribute	Type	Description
id	UUID	Unique assignment ID

Software Design Description for **BrainBee**

classroom	Classroom	Related class
title	String	Assignment title
description	String	Details about the assignment
dueDate	DateTime	Submission deadline
submissions	List	Student submissions

Submission:

Attribute	Type	Description
id	UUID	Unique submission ID
assignment	Assignment	Related assignment
student	Student	Student who submitted
fileUrl	String	File link
submittedAt	DateTime	Time of submission
grade	Float?	Assigned grade
feedback	String?	Teacher's feedback

QuizTemplate:

Attribute	Type	Description
id	UUID	Unique template ID
classroom	Classroom	Linked classroom
title	String	Template name
templateData	JSON	Quiz blueprint

PerformanceInsight:

Attribute	Type	Description
id	UUID	Unique insight ID

teacher	Teacher	Associated teacher
student	Student	Student being analyzed
book	Book	Targeted book
metrics	JSON	Metrics like difficulty and timing
generatedAt	DateTime	Timestamp of generation

5.2 Data Schema

5.2.1 User Schema (Base Schema)

```
const UserSchema = new mongoose.Schema({
  userID: { type: String, unique: true, required: true },
  fullName: String,
  email: { type: String, unique: true, required: true },
  phoneNumber: String,
  passwordHash: String,
  role: { type: String, enum: ['Student', 'Parent', 'Teacher'], required: true },
  accountType: { type: String, enum: ['Standard', 'Premium'], default: 'Standard' },
  profilePicture: String,
  registrationDate: { type: Date, default: Date.now },
  isActive: { type: Boolean, default: true }
});
```

5.2.2 Student Schema (Extends User)

```
const StudentSchema = new mongoose.Schema({
  userID: { type: String, ref: 'User' },
  gradeLevel: String,
```

```
    enrolledSubjects: [String],  
    quizHistory: [{ type: mongoose.Schema.Types.ObjectId, ref: 'Quiz' }],  
    leaderboardRank: Number,  
    streakScore: Number  
  });
```

5.2.3 Parent Schema (Extends User)

```
const ParentSchema = new mongoose.Schema({  
  userID: { type: String, ref: 'User' },  
  childrenIDs: [{ type: String, ref: 'Student' }],  
  notifications: [String]  
});
```

5.2.4 Teacher Schema (Extends User)

```
const TeacherSchema = new mongoose.Schema({  
  userID: { type: String, ref: 'User' },  
  classIDs: [{ type: mongoose.Schema.Types.ObjectId, ref: 'Class' }],  
  assignedQuizzes: [{ type: mongoose.Schema.Types.ObjectId, ref: 'Quiz' }],  
  uploadedMaterials: [String]  
});
```

5.2.5 Quiz Schema

```
const QuizSchema = new mongoose.Schema({  
  quizID: { type: String, unique: true },  
  createdBy: String,  
  subject: String,
```

```
topic: String,
questions: [
  {
    questionText: String,
    options: [String],
    correctAnswer: String
  }
],
generatedByAI: { type: Boolean, default: true },
attemptHistory: [{ studentID: String, score: Number, attemptedAt: Date }],
dateCreated: { type: Date, default: Date.now }
});
```

5.2.6 Battle Schema

```
const BattleSchema = new mongoose.Schema({
  battleID: { type: String, unique: true },
  challengerID: String,
  opponentID: String,
  quizData: { type: Object }, // Could embed quiz structure
  winnerID: String,
  battleDate: Date,
  durationSeconds: Number,
  winLossRecord: {
    [String]: { type: String, enum: ['Win', 'Loss', 'Draw'] }
  }
});
```

5.2.7 FlashCard Schema

```
const FlashCardSchema = new mongoose.Schema({  
  cardID: { type: String, unique: true },  
  studentID: String,  
  subject: String,  
  topic: String,  
  content: String,  
  createdAt: { type: Date, default: Date.now }  
});
```

5.2.8 Assignment Schema

```
const AssignmentSchema = new mongoose.Schema({  
  assignmentID: { type: String, unique: true },  
  classID: String,  
  uploadedBy: String,  
  description: String,  
  deadline: Date,  
  submissions: [{  
    studentID: String,  
    fileURL: String,  
    status: { type: String, enum: ['Pending', 'Submitted', 'Graded'] }  
  }]  
});
```

5.2.9 Reward Schema

```
const RewardSchema = new mongoose.Schema({  
  rewardID: { type: String, unique: true },
```



```
type: { type: String, enum: ['Badge', 'Certificate', 'CoinQuest'] },  
name: String,  
awardedTo: String,  
awardedBy: String,  
dateAwarded: { type: Date, default: Date.now },  
coinValue: Number  
});
```

5.2.10 Goal Schema

```
const GoalSchema = new mongoose.Schema({  
  goalID: { type: String, unique: true },  
  parentID: String,  
  studentID: String,  
  description: String,  
  status: { type: String, enum: ['Pending', 'Completed', 'Failed'], default: 'Pending' },  
  rewardCoins: Number,  
  dueDate: Date  
});
```

5.2.11 Report Card Schema

```
const ReportCardSchema = new mongoose.Schema({  
  reportCardID: { type: String, unique: true },  
  studentID: String,  
  bookName: String,  
  score: Number,  
  weakAreas: [String],  
  evaluationDate: { type: Date, default: Date.now }
```

```
});
```

5.2.12 Chat History Schema

```
const ChatHistorySchema = new mongoose.Schema({  
  chatID: { type: String, unique: true },  
  studentID: String,  
  topic: String,  
  messages: [{  
    sender: { type: String, enum: ['student', 'AI'] },  
    text: String,  
    timestamp: { type: Date, default: Date.now }  
  }],  
  createdAt: { type: Date, default: Date.now }  
});
```

5.2.13 Class Schema

```
const ClassSchema = new mongoose.Schema({  
  classID: { type: String, unique: true },  
  teacherID: String,  
  className: String,  
  studentList: [{ type: String, ref: 'Student' }],  
  materials: [{  
    title: String,  
    fileURL: String,  
    uploadedAt: Date  
  }]  
});
```

6. Human Interface Design

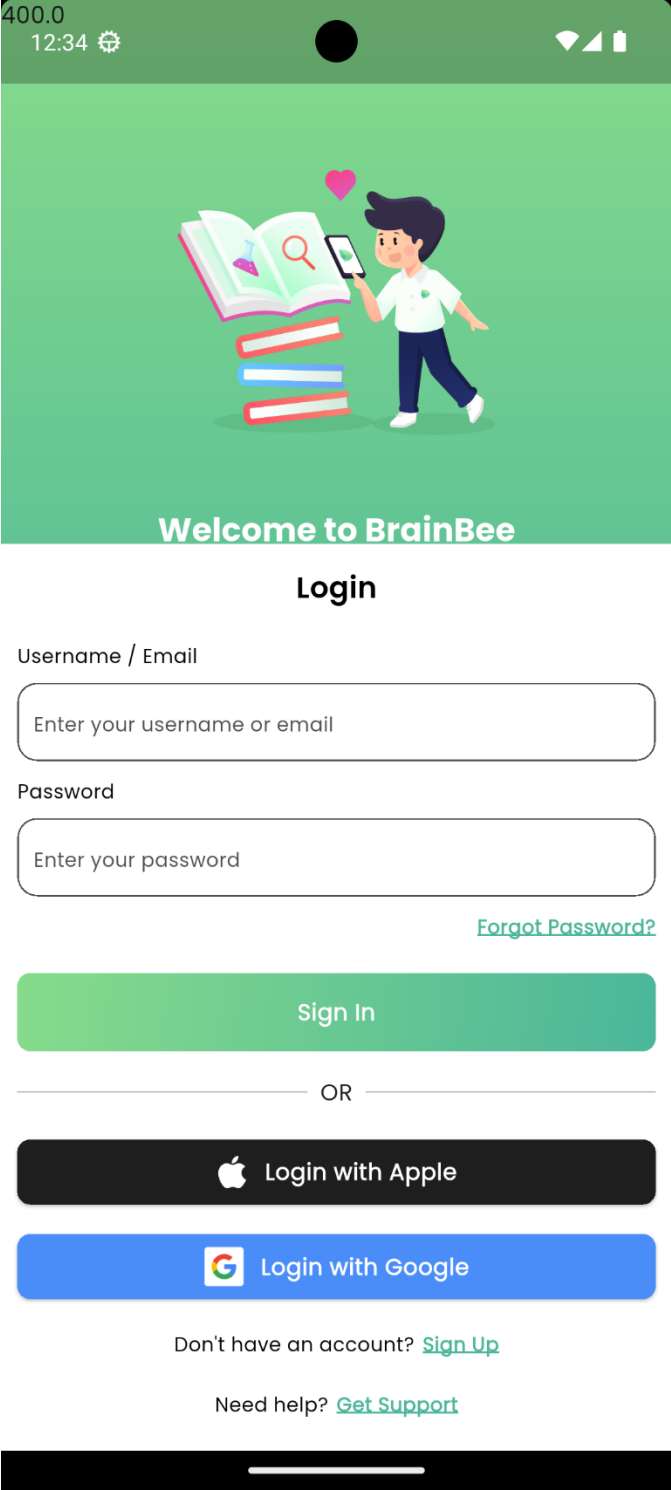
This section outlines the user interface design principles and components for **BrainBee**, ensuring an intuitive and seamless experience for both mobile app users and admin dashboard operators.

6.1 Screen Images

This subsection presents the screen images showing the user interface for the SafarRehnuma platform.

6.1.1 Student App

6.1.1.1 Login screen



The image shows a mobile app login screen for 'BrainBee'. At the top, there's a status bar with '400.0', '12:34', and icons for settings, signal, and battery. Below this is a green header area with an illustration of a student holding a smartphone, with a stack of books and a magnifying glass icon. The text 'Welcome to BrainBee' is centered below the illustration. The main section is titled 'Login' and contains two input fields: 'Username / Email' and 'Password'. Below the password field is a link 'Forgot Password?'. A large green 'Sign In' button is positioned below the links. A horizontal line with 'OR' in the center separates the 'Sign In' button from the social login options. There are two buttons for social login: 'Login with Apple' (black with white Apple logo) and 'Login with Google' (blue with white Google logo). At the bottom, there are two links: 'Don't have an account? Sign Up' and 'Need help? Get Support'. The entire screen is framed by a black border at the bottom, representing the home indicator bar.

400.0
12:34

Welcome to BrainBee

Login

Username / Email

Enter your username or email

Password

Enter your password

[Forgot Password?](#)

Sign In

OR

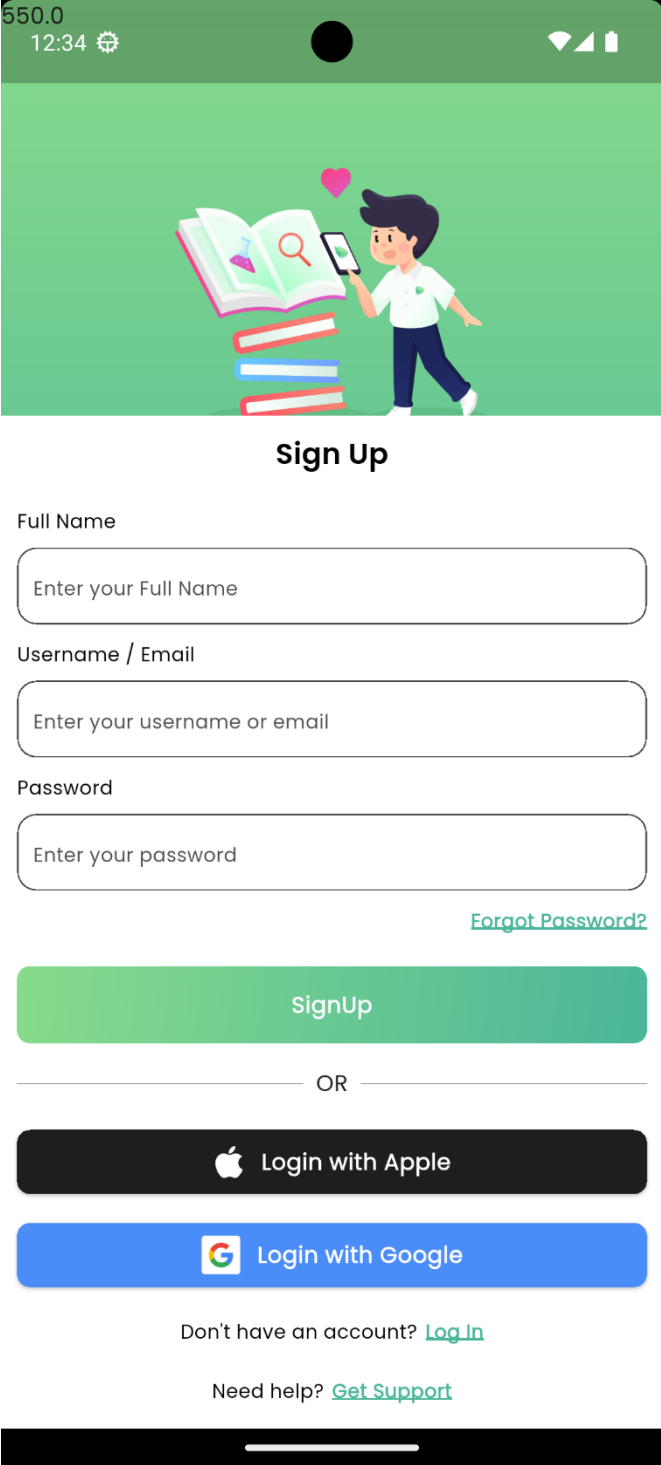
 Login with Apple

 Login with Google

Don't have an account? [Sign Up](#)

Need help? [Get Support](#)

6.1.1.2 SignUp screen



550.0
12:34

sign Up

Full Name

Enter your Full Name

Username / Email

Enter your username or email

Password

Enter your password

[Forgot Password?](#)

SignUp

OR

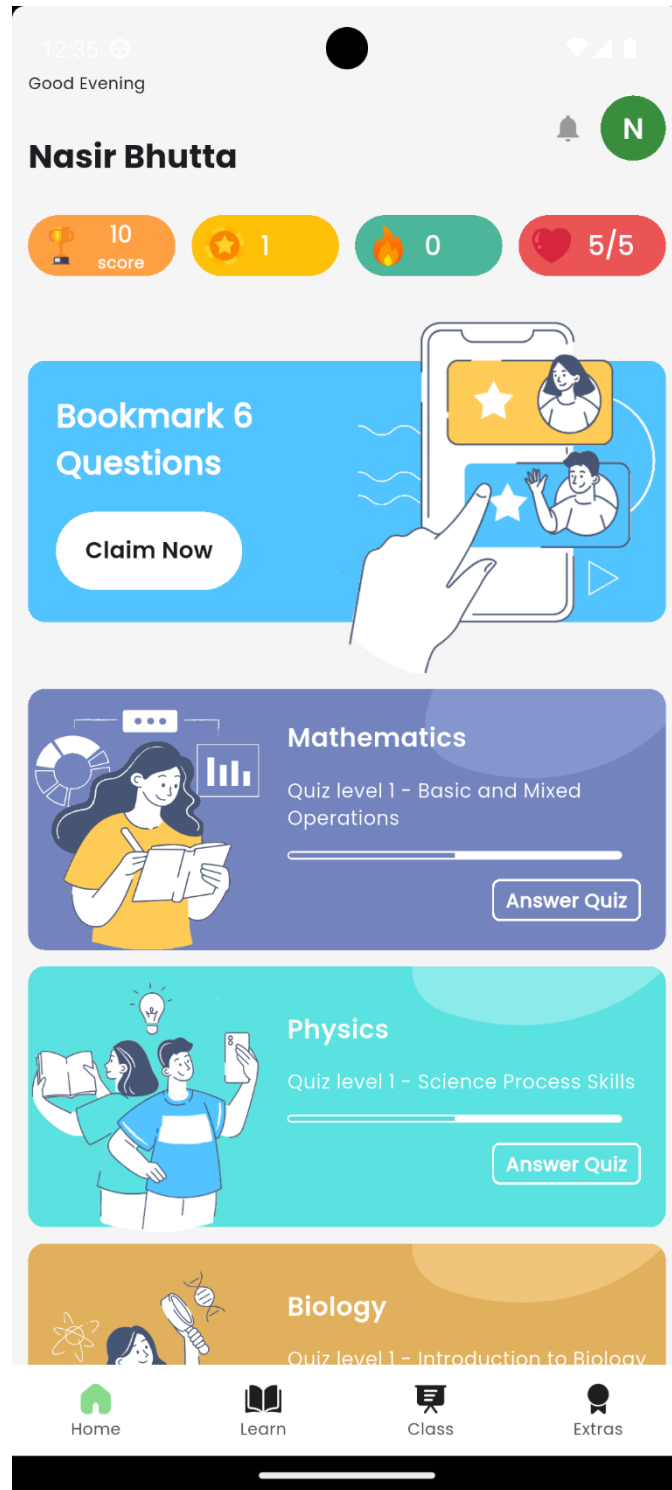
 Login with Apple

 Login with Google

Don't have an account? [Log In](#)

Need help? [Get Support](#)

6.1.1.3 Dashboard



6.1.1.4 Goals Screen

12:35

← Edit Daily Goals

Pick a goal

- ☒ **Casual**
2 Quizzes & Estimated 7 minutes daily
- ☐ **Regular**
6 Quizzes & Estimated 21 minutes daily
- ☐ **Serious**
10 Quizzes & Estimated 35 minutes daily
- ☐ **Epic**
14 Quizzes & Estimated 49 minutes daily

When can we remind you?
Pick days and time

Time Reminder Notification

Show Notifications ☒

eg: Remind me on X time?

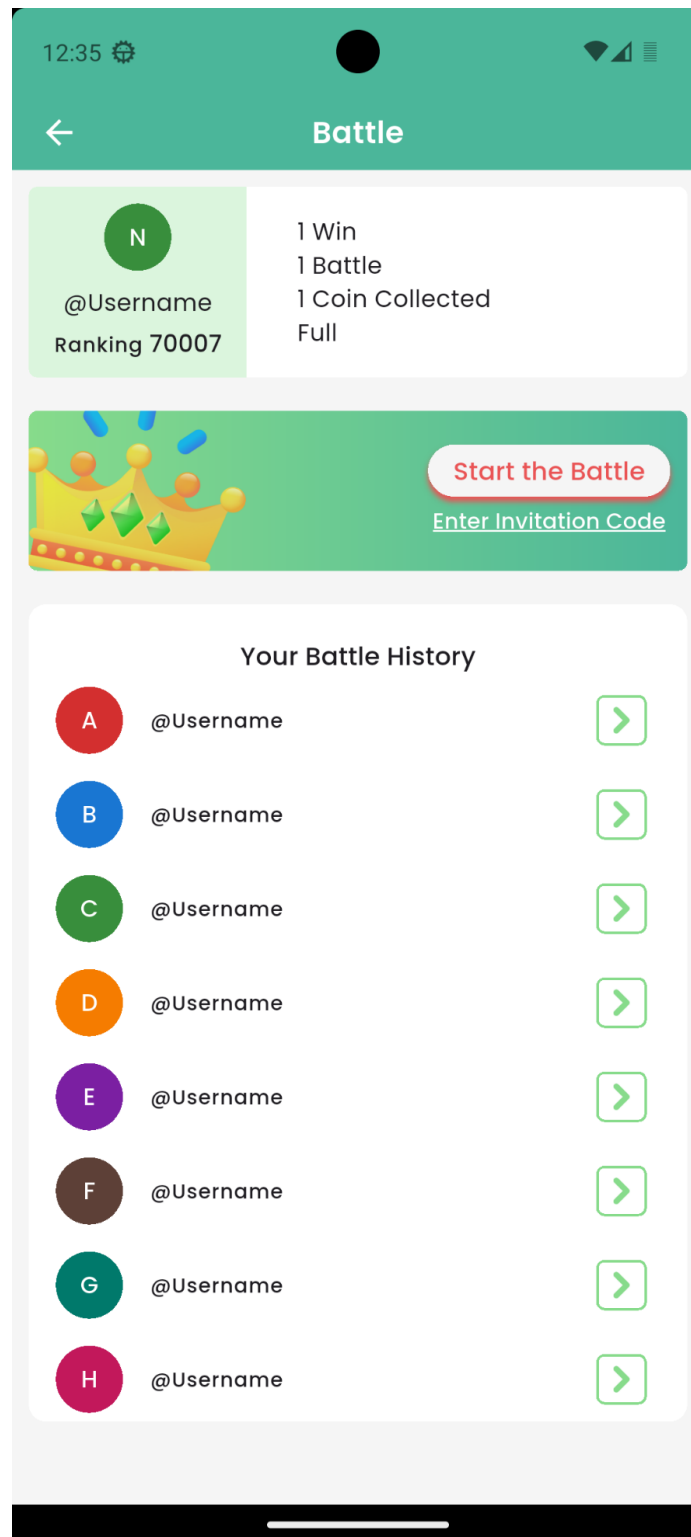
+ Add Reminder

20:00 

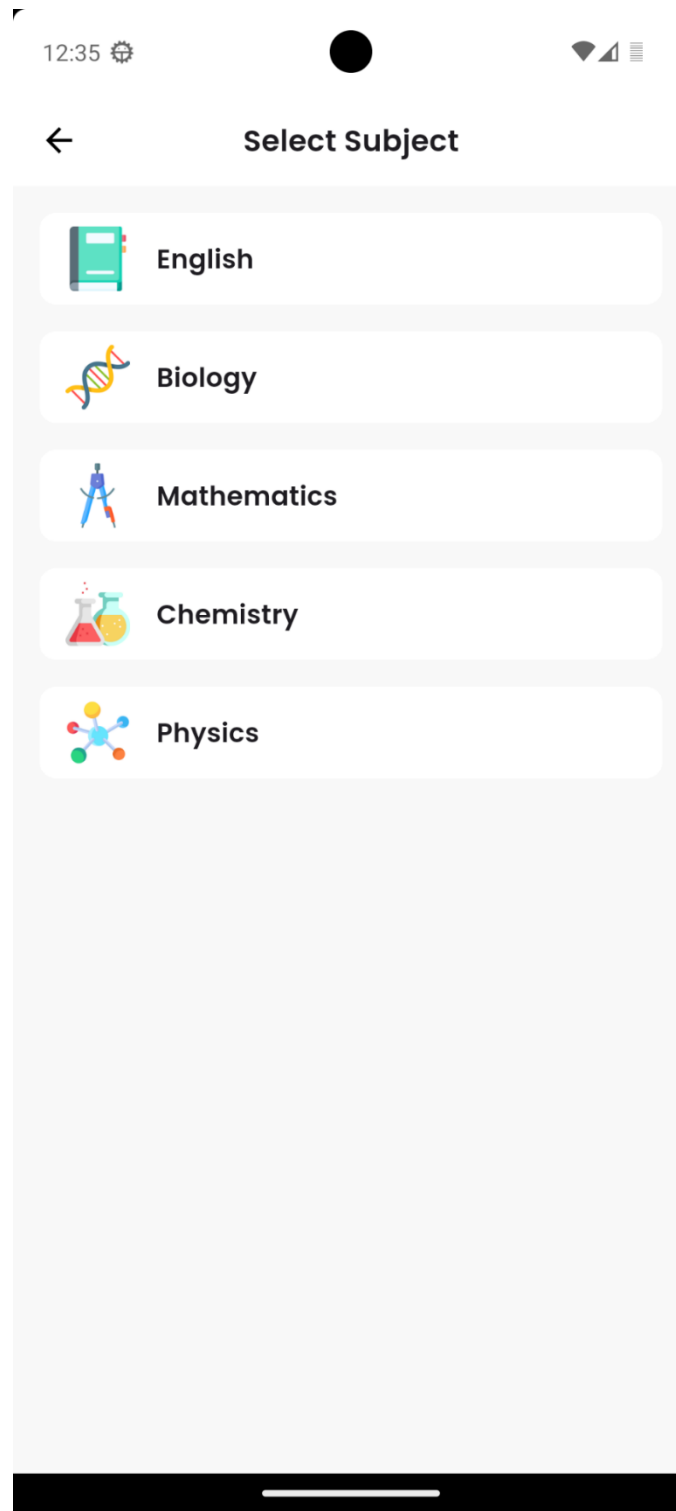
We will remind you at this time

Save

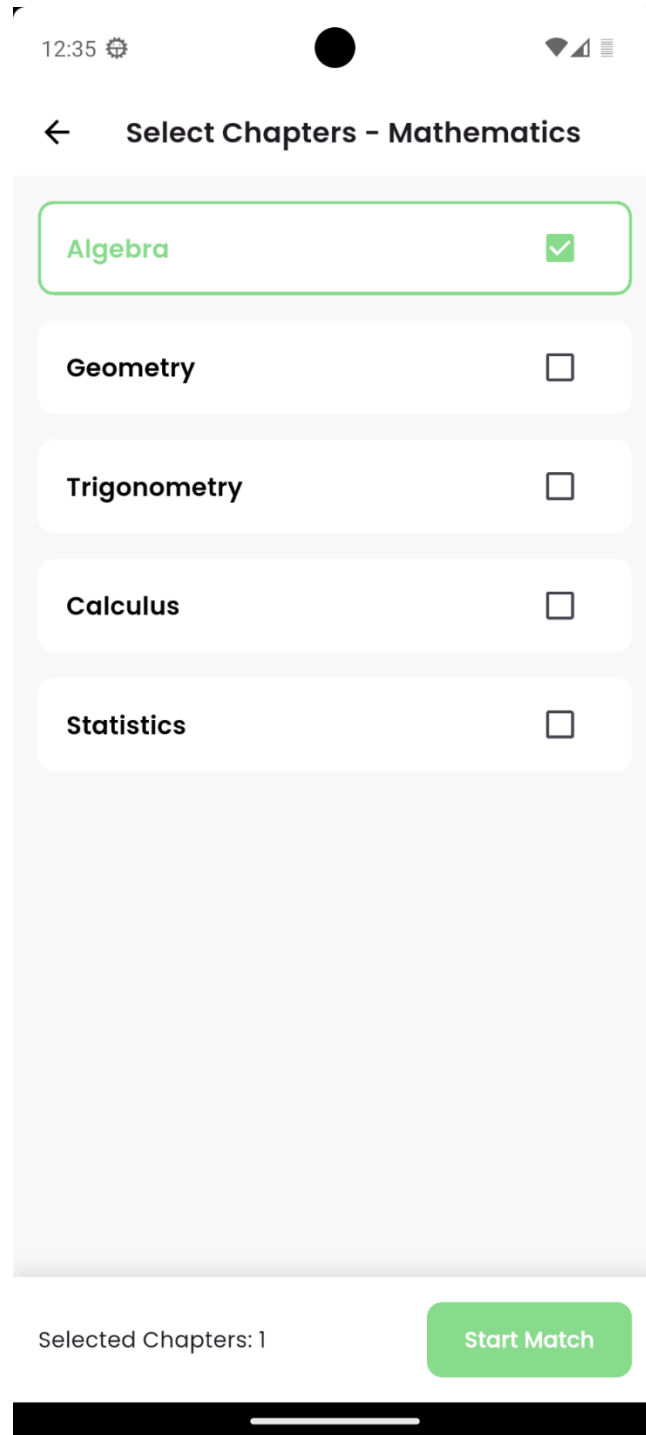
6.1.1.5 Battle Screen



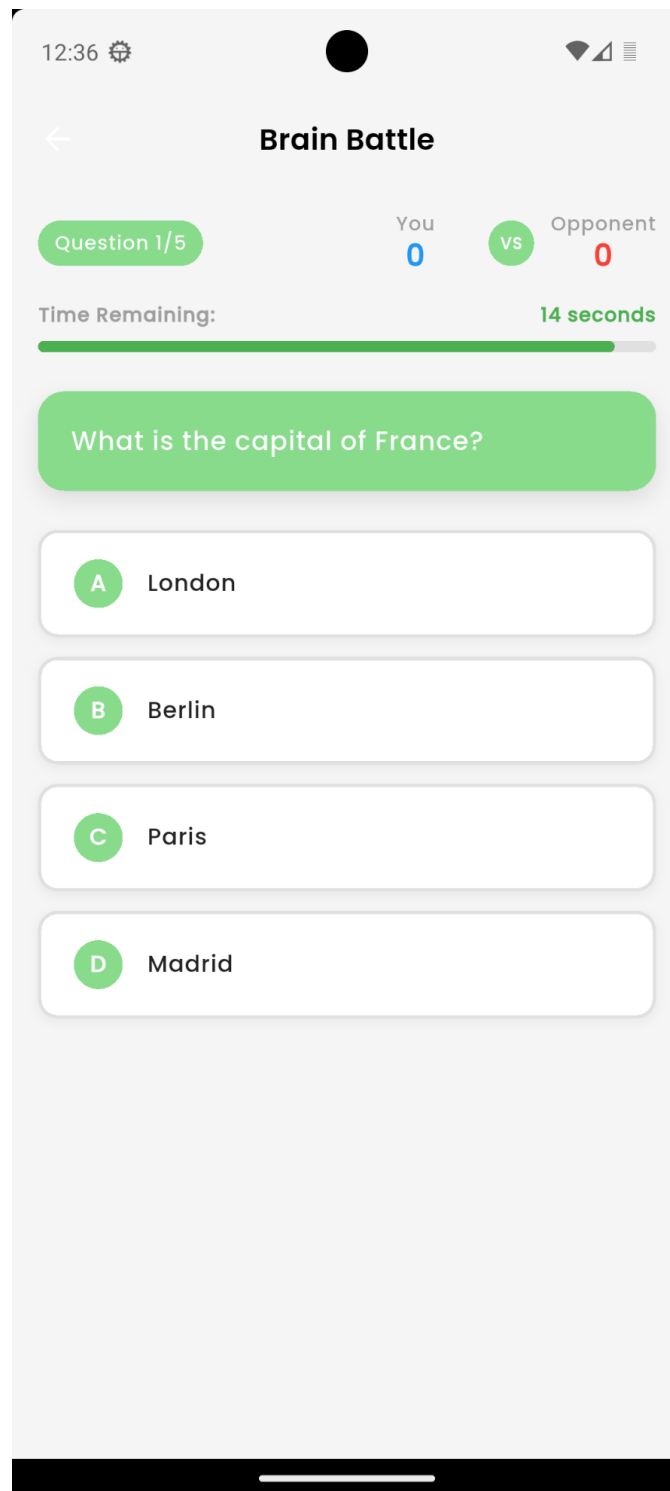
6.1.1.6 Select Subject for Battle:



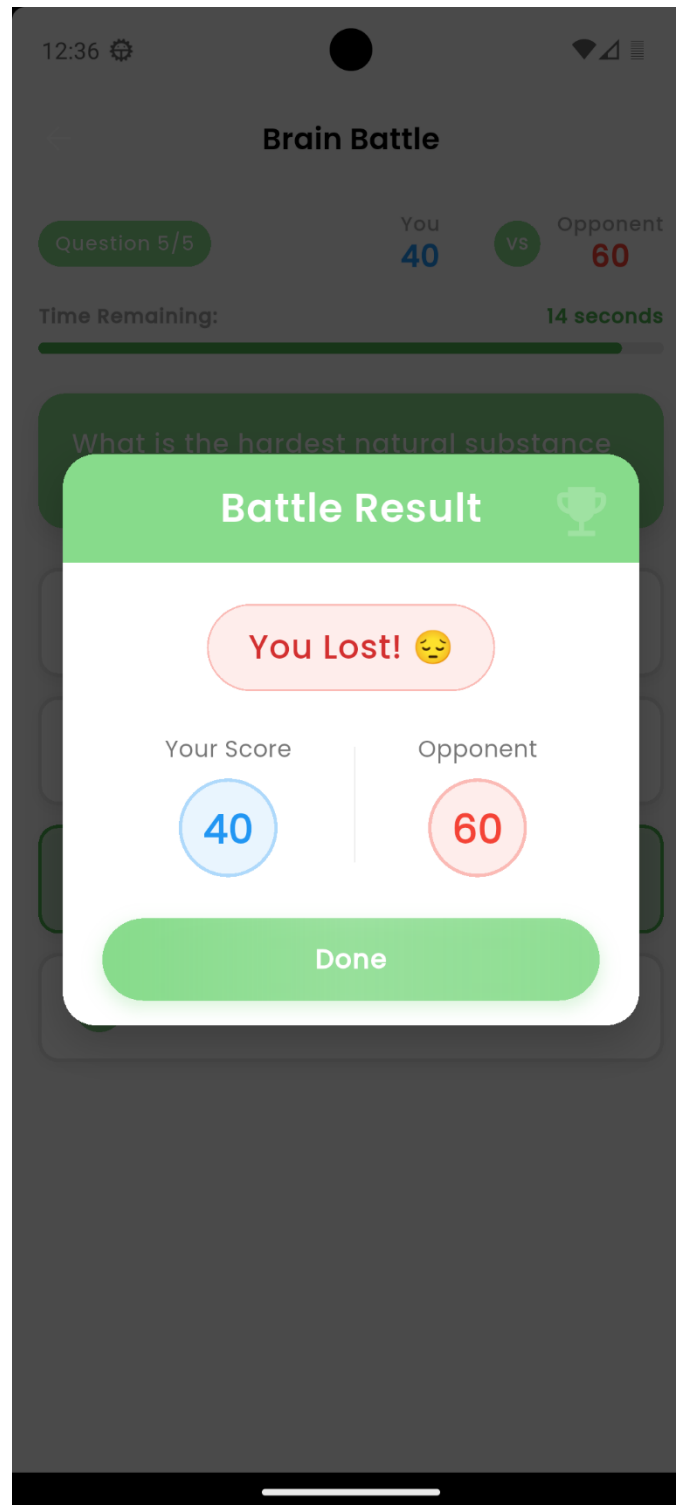
6.1.1.7 Select Chapter For Battle:



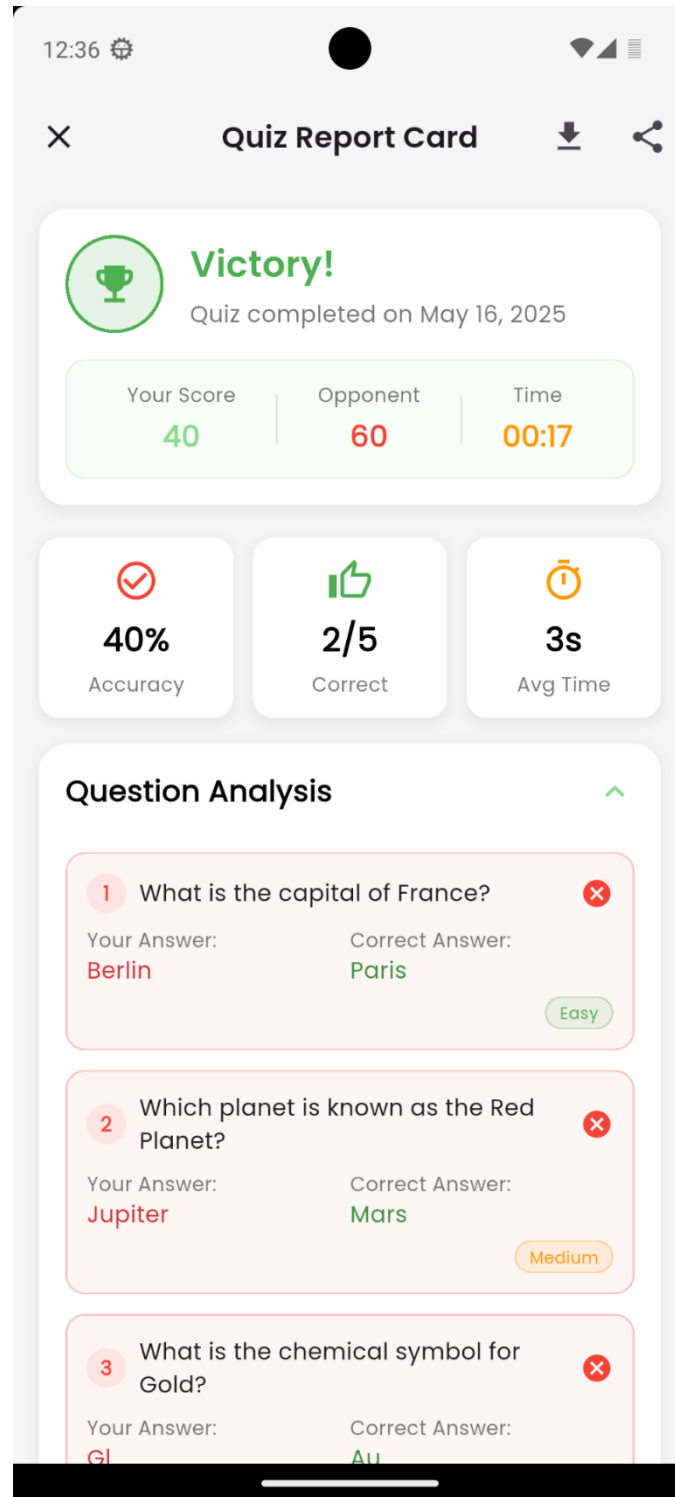
6.1.1.8 Quiz Battle Screen:



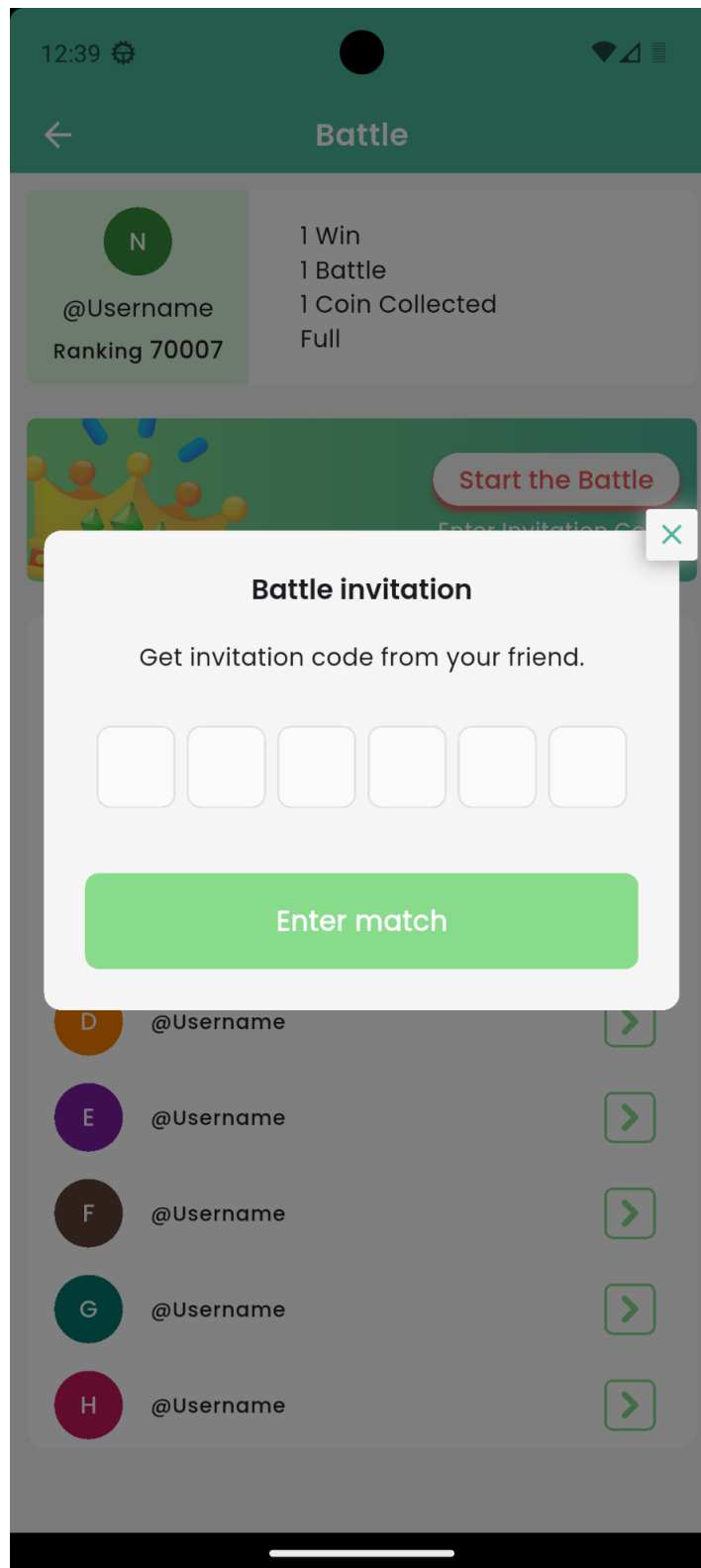
6.1.1.9 Result Screen:



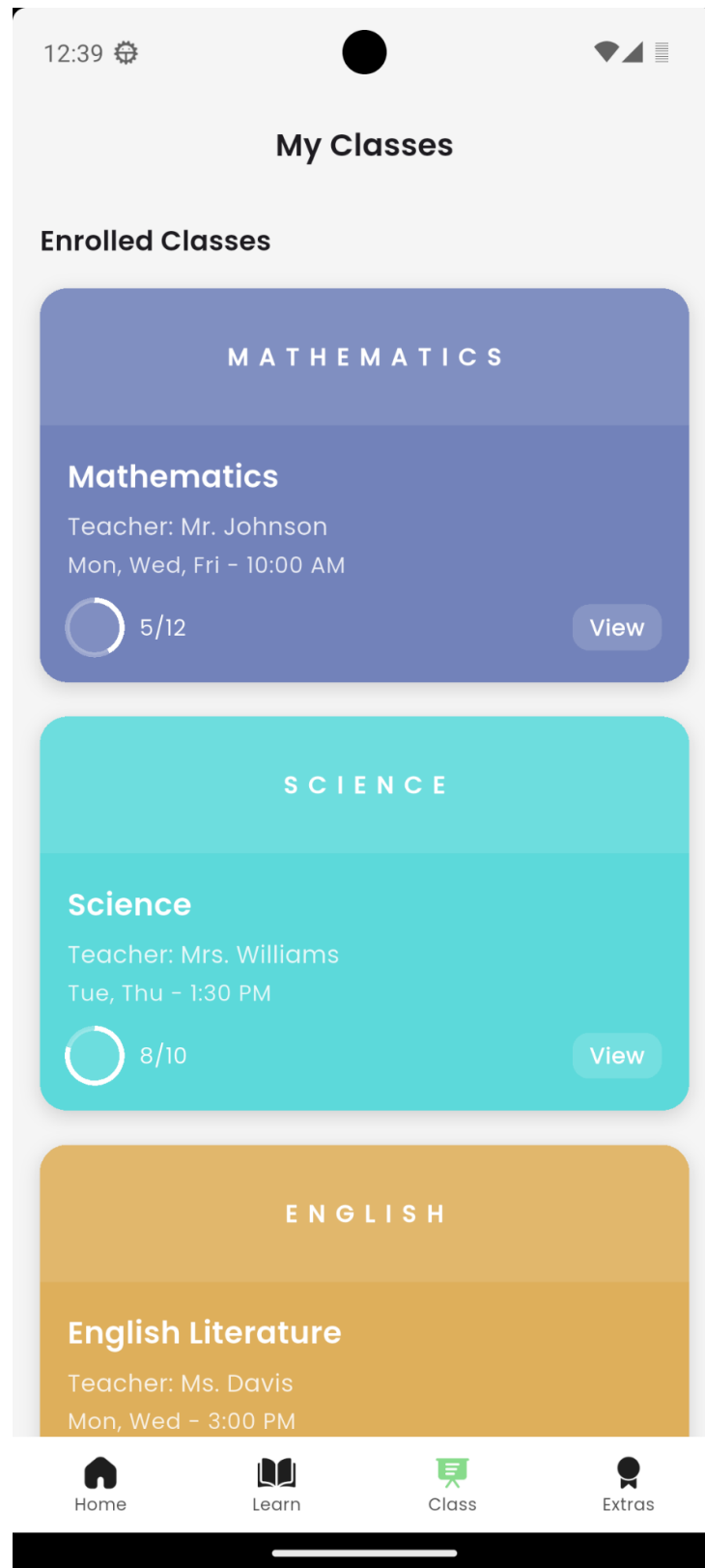
6.1.1.10 Battle Report Card:



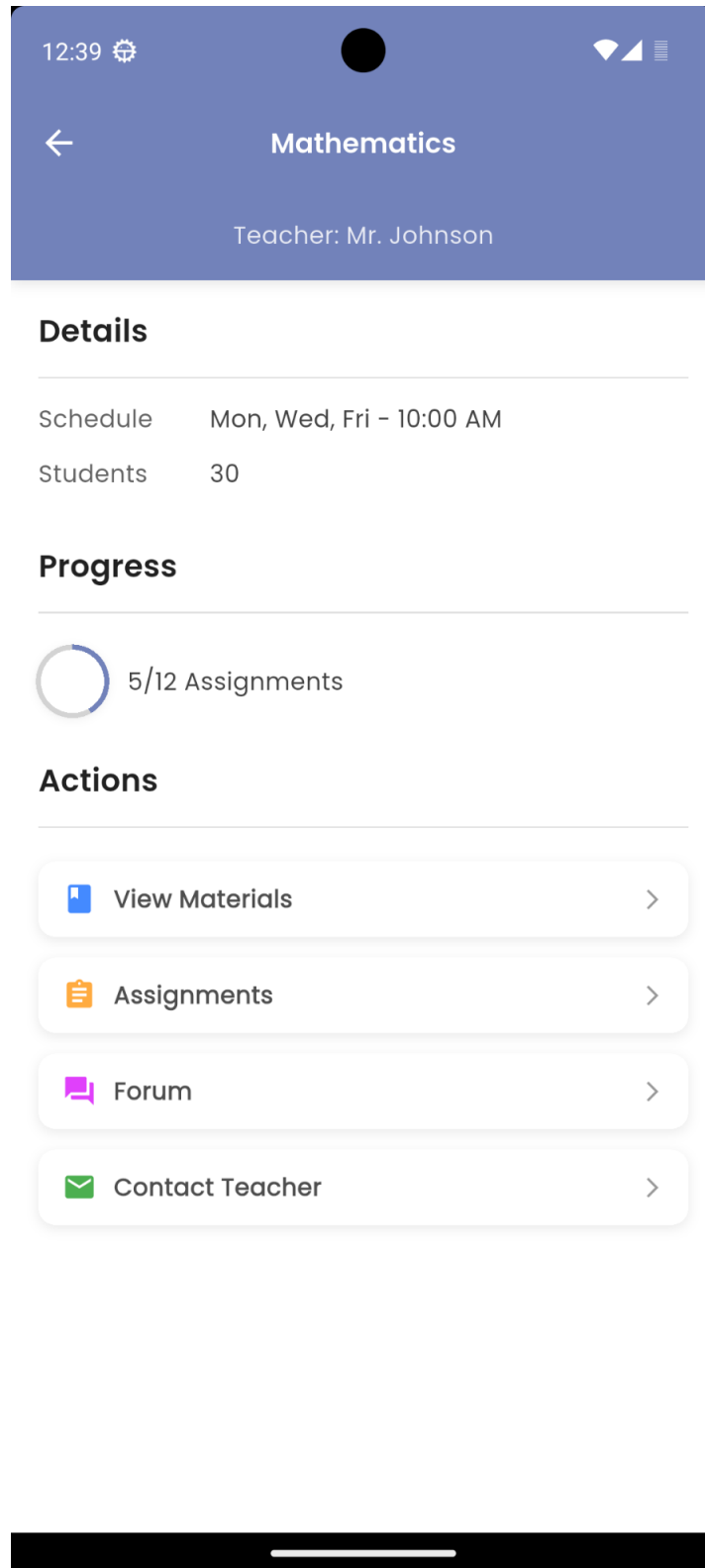
6.1.1.11 Battle invitation Screen:



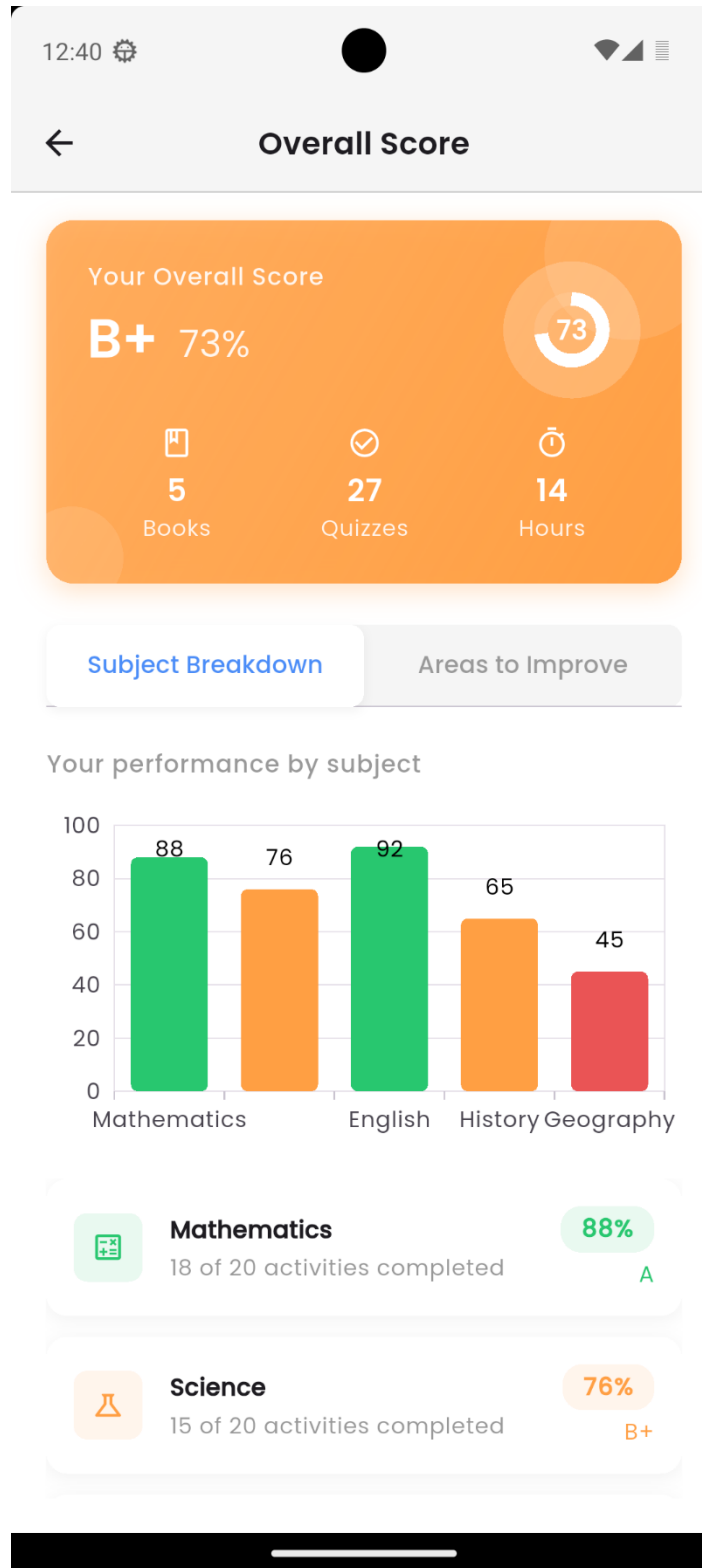
6.1.1.12 Enrolled Classes:



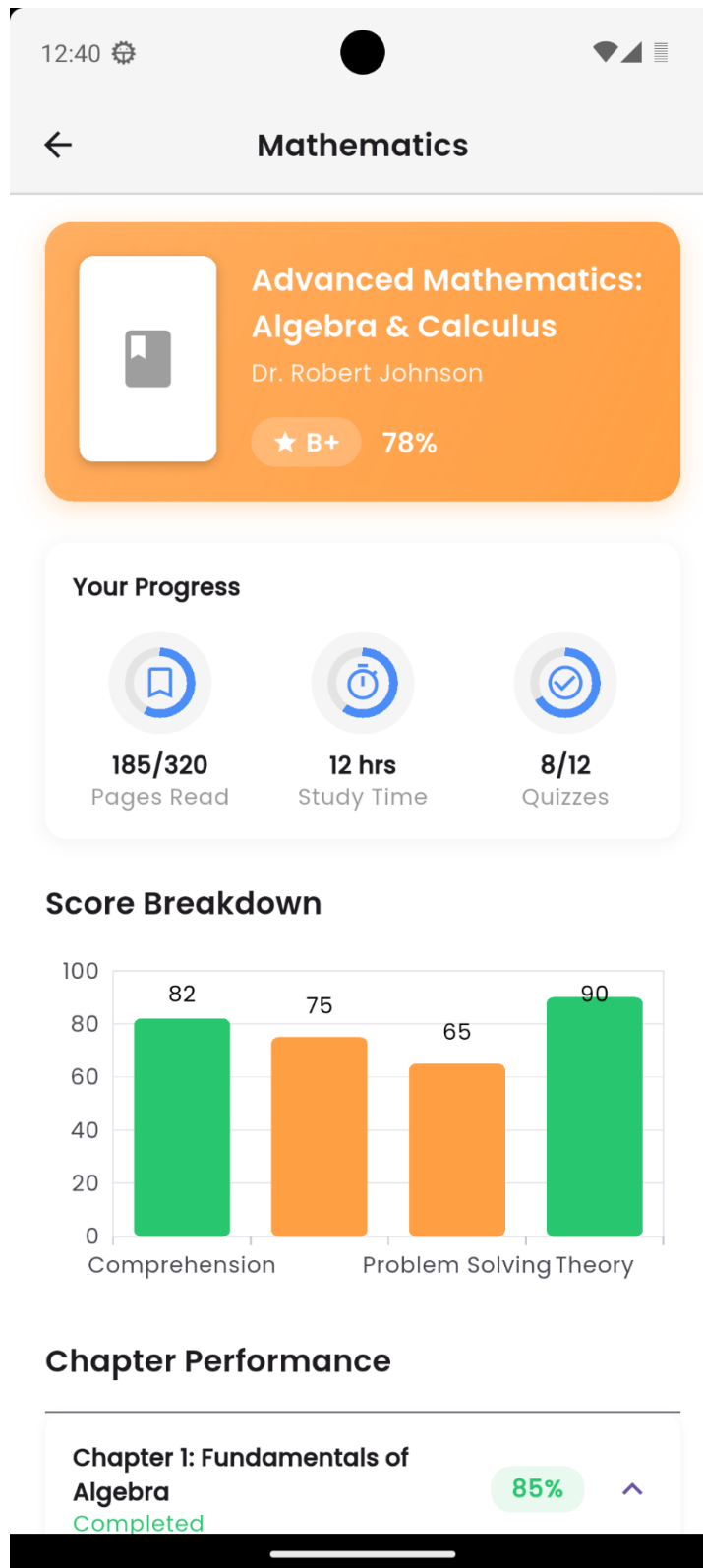
6.1.1.13 Class Detail Screen:



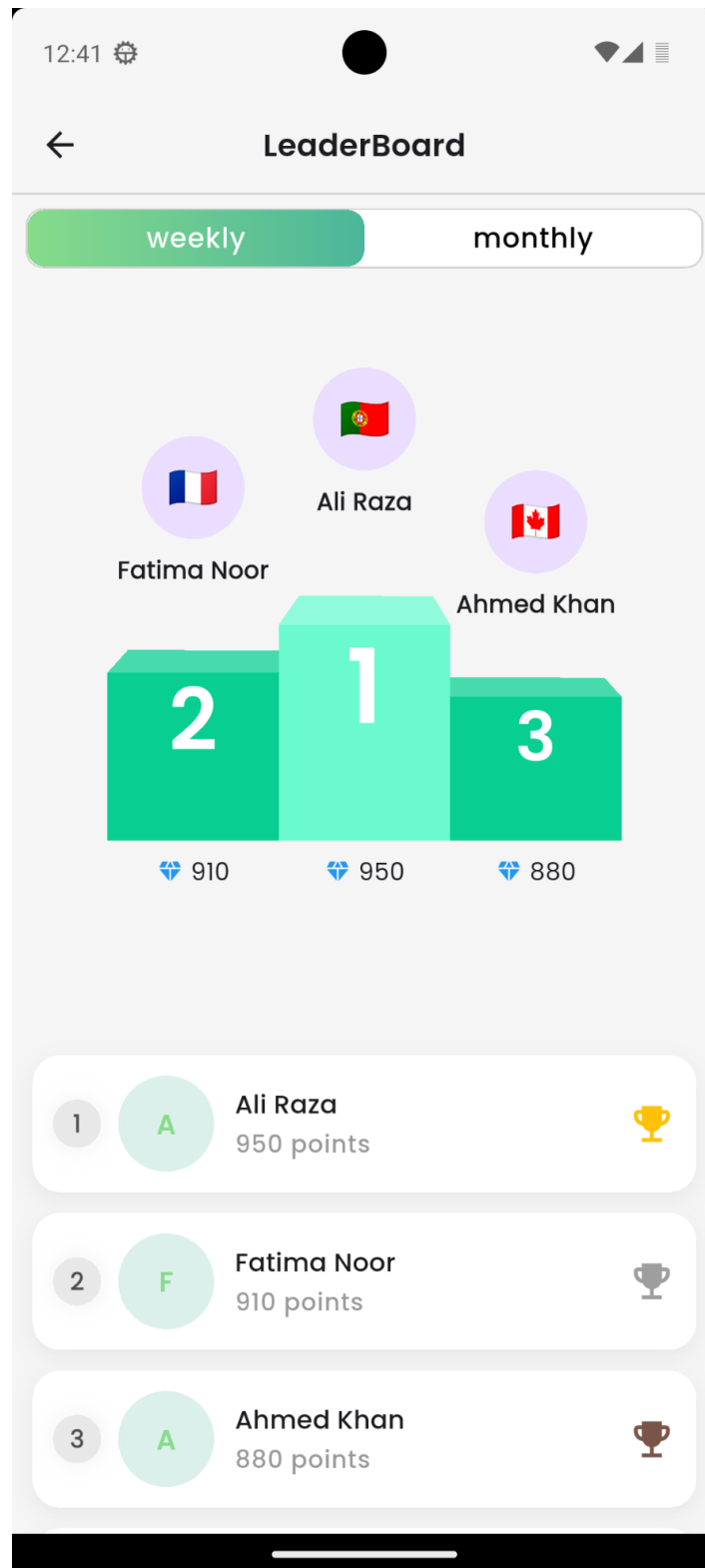
6.1.1.14 Overall Score screen of in app Quizzes:



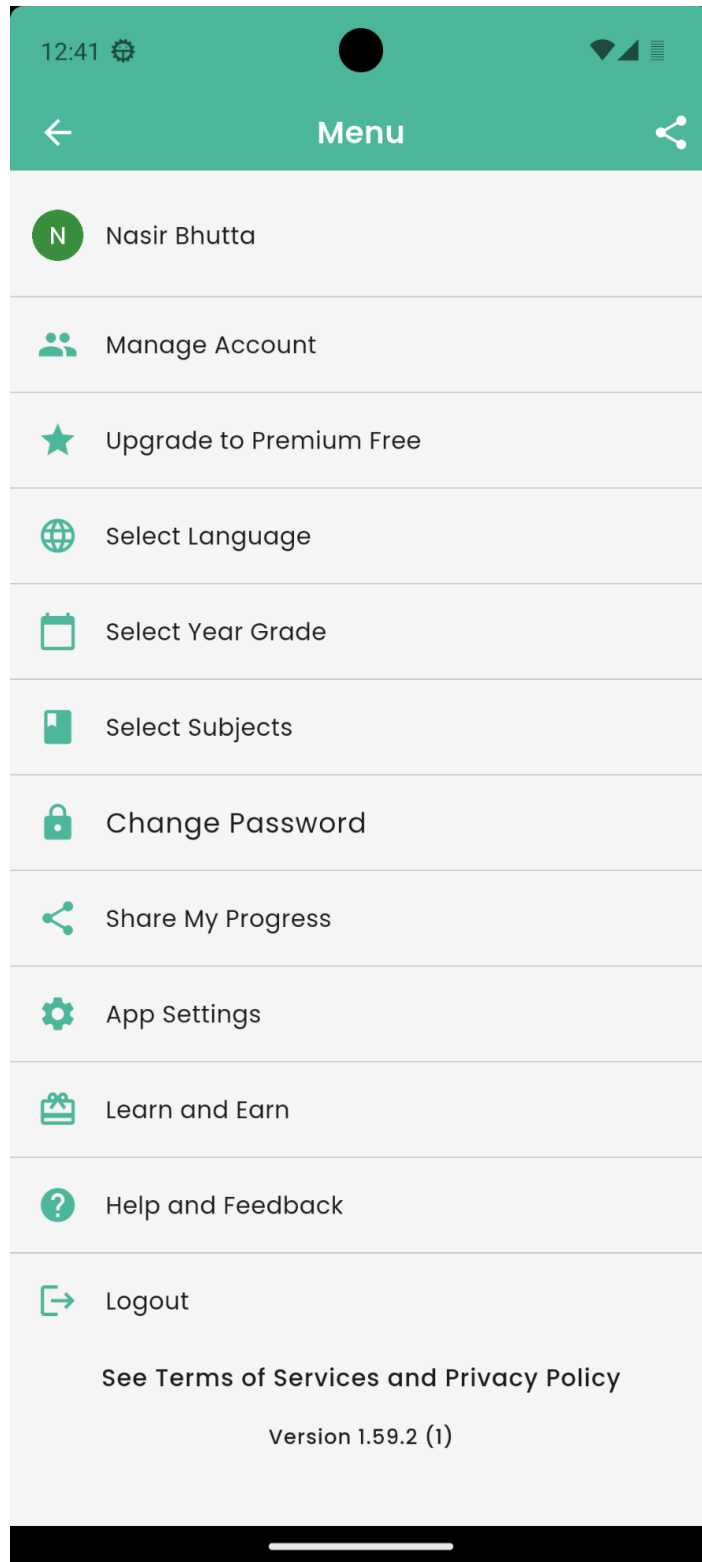
6.1.1.15 Book wise Score:



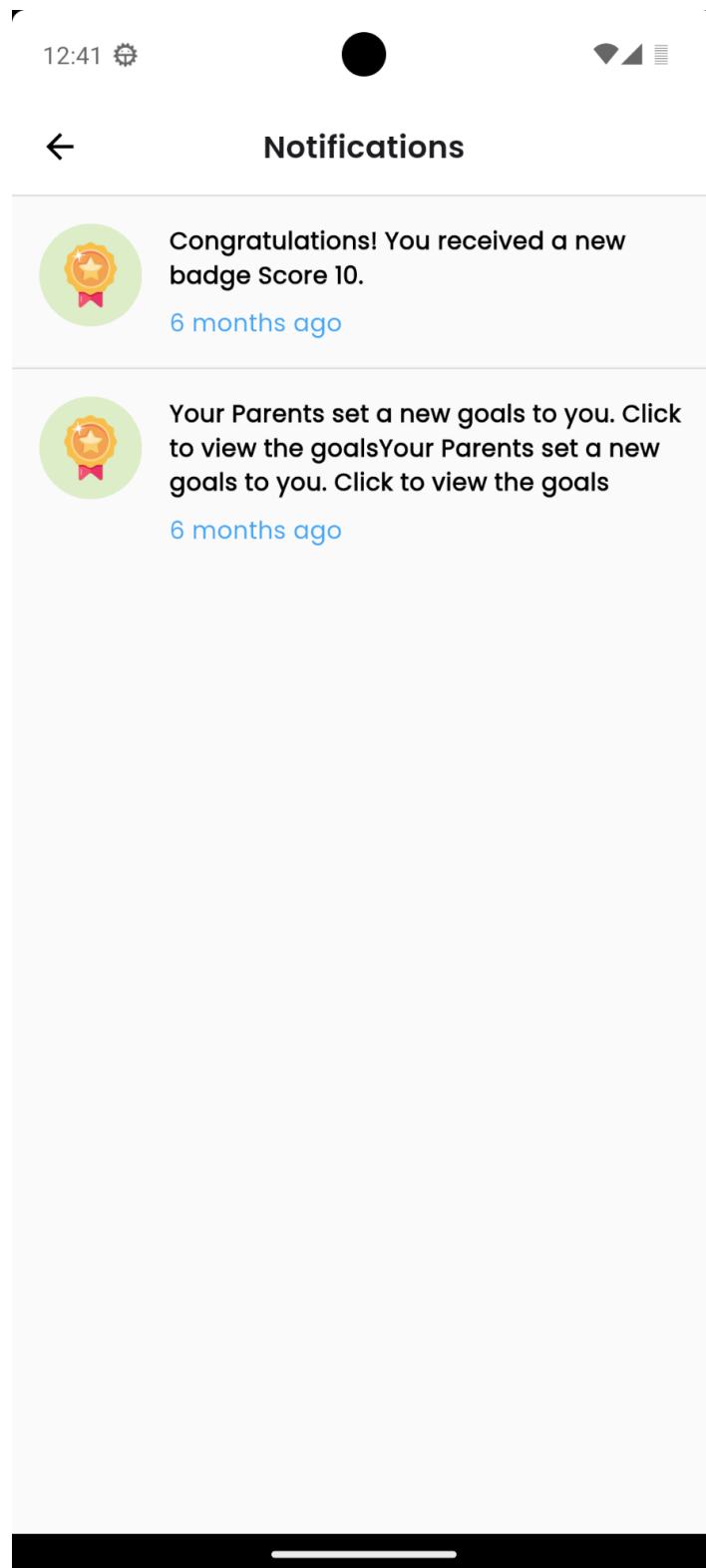
6.1.1.16 LeaderBoard:



6.1.1.17 Menu



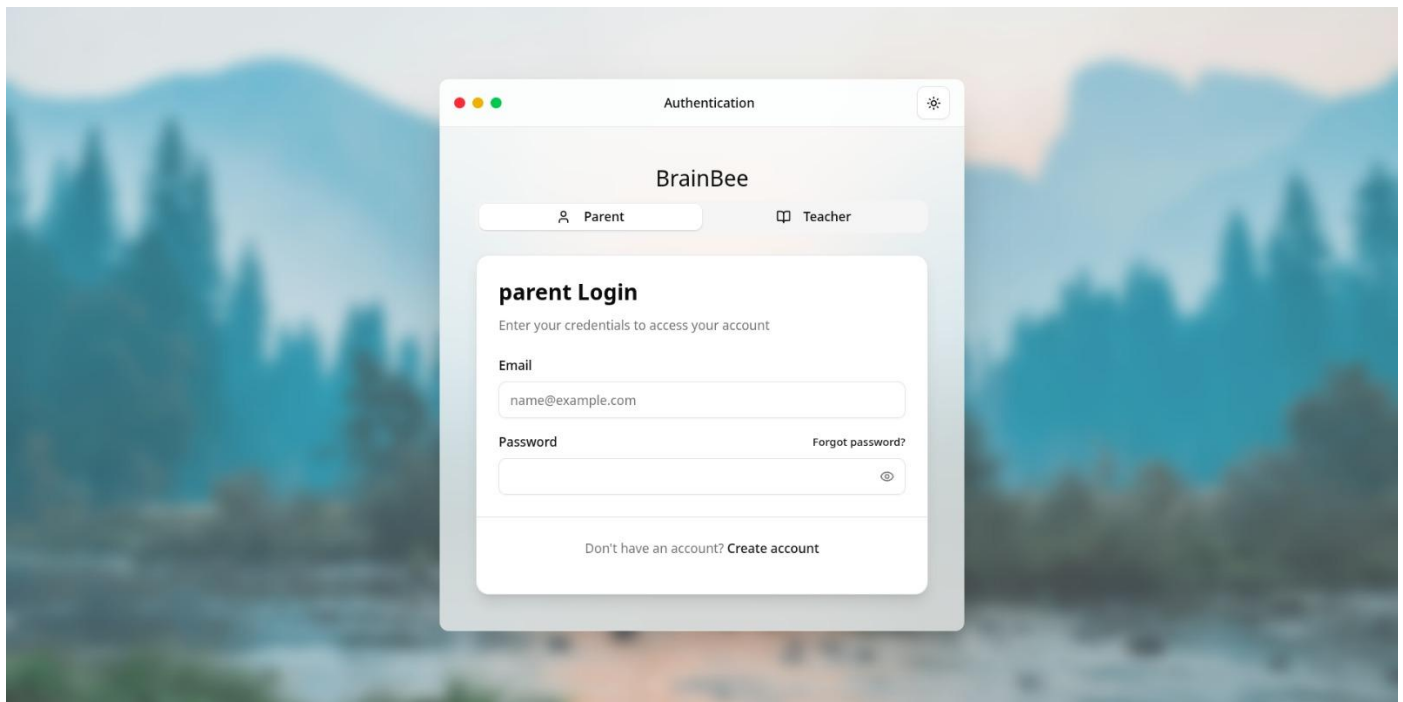
6.1.1.18 Notifications:



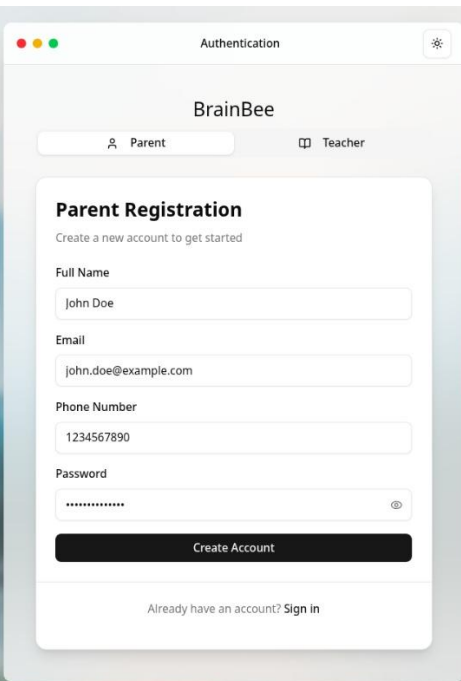
6.1.2 Parent and Teacher App:

This Section shows the screens of the parent and teacher application.

6.1.2.1 Login Page



6.1.2.2 Signup



The image shows a web application window titled "Authentication" with a "BrainBee" logo. Below the logo are two tabs: "Parent" (selected) and "Teacher". The "Parent Registration" form is displayed, featuring a subtitle "Create a new account to get started". The form contains five input fields: "Full Name" (with "John Doe"), "Email" (with "john.doe@example.com"), "Phone Number" (with "1234567890"), and "Password" (with masked characters and a strength indicator). A "Create Account" button is at the bottom of the form. Below the form, a link "Already have an account? Sign in" is visible. The background of the window is a blurred landscape with mountains and trees.

Authentication

BrainBee

Parent Teacher

Parent Registration

Create a new account to get started

Full Name
John Doe

Email
john.doe@example.com

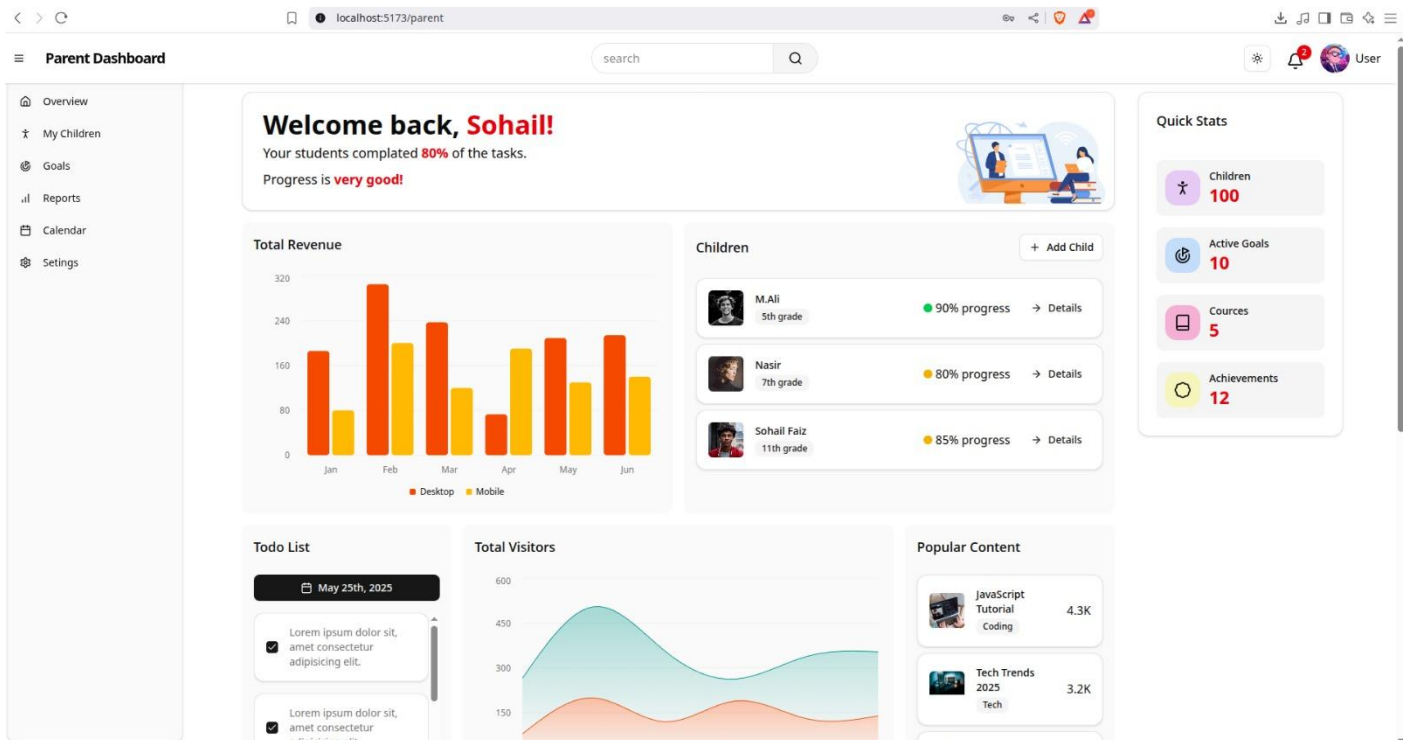
Phone Number
1234567890

Password

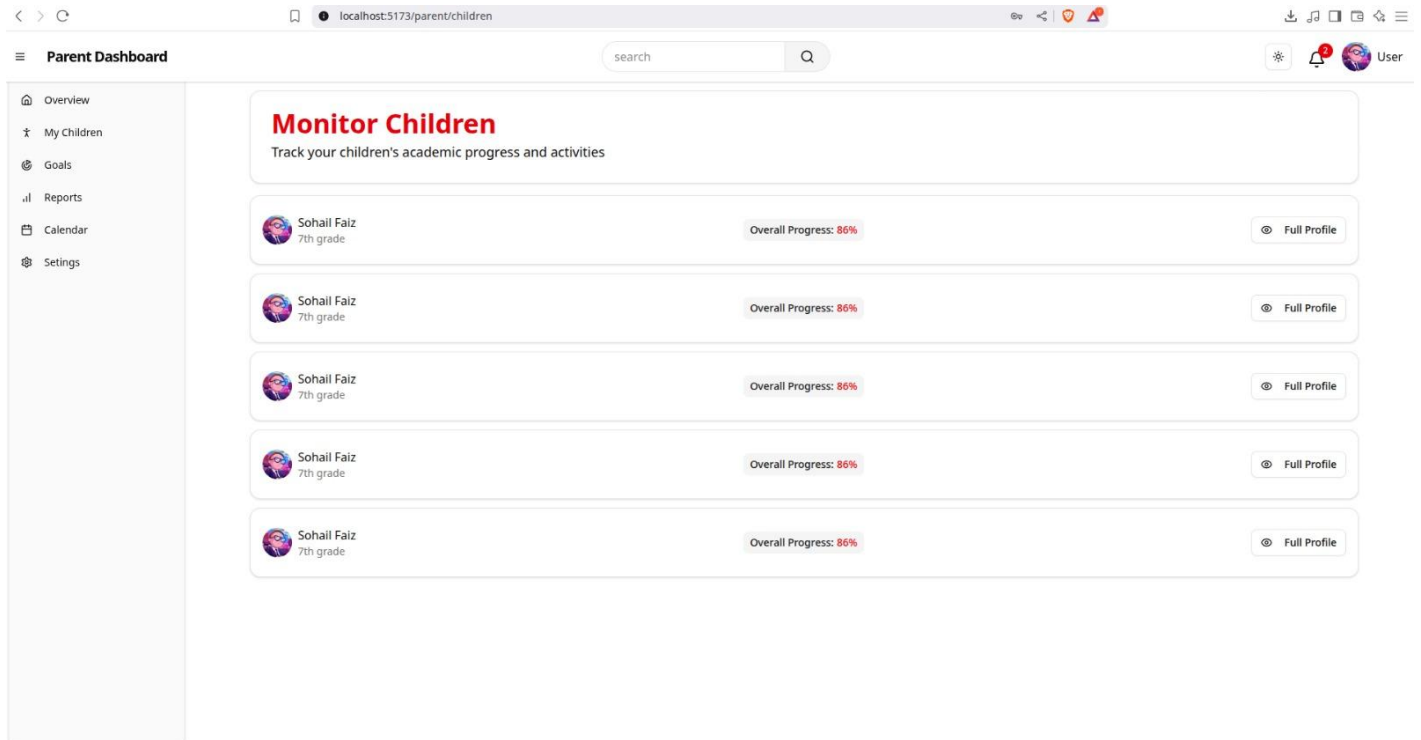
Create Account

Already have an account? [Sign in](#)

6.1.2.3 Dashboard



6.1.2.4 Monitor Children



6.2 Screen Objects and Actions

This subsection outlines the different screens and user interactions present in the **BrainBee** application.

- 6.2.1 Login Screen:** Allows existing users to access their account using their username/email and password or via social logins (Apple/Google). Provides options for password recovery and new user sign-up.
- 6.2.2 Sign Up Screen:** Enables new users to create an account by providing their full name, username/email, and password, or by using social logins. Offers a link to log in for existing users.
- 6.2.3 Home/Dashboard Screen:** Displays user's progress (score, achievements), features promotional banners like "Bookmark Questions," and lists available subjects (Mathematics, Physics, Biology) for quizzing. Provides main navigation via bottom tabs.
- 6.2.4 Battle Overview Screen:** Shows the user's battle statistics (wins, rank) and a history of past battles. Allows users to start a new battle or join one using an invitation code.
- 6.2.5 Select Subject Screen (for Battle):** Presents a list of academic subjects (English, Biology, Mathematics, etc.) for the user to choose from before starting a quiz or battle.
- 6.2.6 Select Chapters Screen:** Allows the user to refine their quiz/battle by selecting specific chapters (e.g., Algebra, Geometry) within the previously chosen subject.
- 6.2.7 Brain Battle Screen:** Displays the live quiz interface showing the current question, multiple-choice answers, user and opponent scores, question progress, and a countdown timer.
- 6.2.8 Battle Result Popup:** A modal dialog that appears after a battle, indicating whether the user won or lost, and showing their score versus the opponent's score.

- 6.2.9 Quiz Report Card Screen:** Provides a detailed summary of a completed quiz, including the overall result (Victory!), scores, time taken, accuracy, and a question-by-question analysis of answers.
- 6.2.10 Battle Invitation Popup:** Allows a user to join a private battle by entering a 6-digit invitation code obtained from a friend.
- 6.2.11 Select Subject Screen (for Flashcards):** Lists available subjects (English, Biology, etc.) and indicates the number of flashcards available for each, allowing users to choose a subject for flashcard study.
- 6.2.12 My Classes Screen:** Displays a list of classes the user is enrolled in, showing the subject, teacher, schedule, and progress on assignments for each class.
- 6.2.13 Class Details Screen :** Shows detailed information for a specific enrolled class, including schedule, student count, assignment progress, and actions like viewing materials or contacting the teacher.
- 6.2.14 Overall Score Screen :** Presents the user's overall academic score and grade, along with statistics (books, quizzes, hours) and a bar chart breaking down performance by subject.
- 6.2.15 Subject Details Screen:** Provides an in-depth look at the user's progress within a specific subject (e.g., Advanced Mathematics), including pages read, study time, quiz scores, and a skill-based score breakdown.
- 6.2.16 Leaderboard Screen:** Displays rankings of users based on points, with options to view weekly or monthly leaderboards, showcasing top performers and their scores.
- 6.2.17 Menu Screen:** Provides access to various account settings (manage account, change password, select grade/subjects), app preferences (language, settings), help, and logout options.

6.2.18 Notifications Screen: Lists important updates and alerts for the user, such as earned badges, new goals set by parents, or other app-related information.

7. Implementation

This section covers the practical implementation aspects of SafarRehnuma, encompassing algorithmic implementations, integration with external interfaces, and the development of user interfaces for seamless interaction.

7.1 Algorithms

This subsection explains the algorithms implemented within **BrainBee**, highlighting their roles in optimizing routes, calculating dynamic pricing, and managing ride requests efficiently.

7.1.1 Get Flashcards

Input: Book ID, Chapter ID

Output: List of dynamically generated flashcards

```
function GetFlashcards(bookID, chapterID)
```

```
    db ← connect_to_MongoDB()
```

```
    chaptersCollection ← db.getCollection("Chapters")
```

```
    chapter ← chaptersCollection.findOne({ "bookID": bookID, "chapterID": chapterID })
```

```
    if chapter is null or chapter.content is null then
```

```
        throw "Chapter not found or content is missing"
```

```
    end if
```

```
    flashcards ← AI.GenerateFlashcards(chapter.content)
```

```
    return flashcards
```

```
end function
```

7.1.2 Match Peer For Battle

Input: Student ID

Output: Matching result message

```
function MatchPeerForBattle(studentID)
    waitingList ← GetWaitingPeers()
    for each peer in waitingList do
        if IsCompatible(peer, studentID) then
            CreateBattleRoom(studentID, peer)
            return "Matched with " + peer.name
    return "No match found. Please wait."
end
```

7.1.3 Generate Quiz

Input: Chapter ID, Difficulty Level

Output: Quiz (list of questions)

```
function GenerateQuiz(chapterID, difficultyLevel)
    allQuestions ← GetQuestions(chapterID)
    suitable ← FilterByDifficulty(allQuestions, difficultyLevel)
    quiz ← RandomSelect(suitable, 10)
    return quiz
end
```

7.1.4 Assign Goal

Input: Child ID, Goal Text, Deadline

Output: Goal assignment confirmation

```
function AssignGoal(childID, goalText, deadline)
```

```
    goal ← CreateGoal(goalText, deadline)
    AssignToChild(childID, goal)
    return "Goal assigned"
end
```

7.1.5 Enroll In Class

Input: Student ID, Class ID

Output: Enrollment status

```
function EnrollInClass(studentID, classID)
    class ← GetClass(classID)
    if class.capacity > class.enrolled then
        Enroll(studentID, classID)
        return "Enrolled"
    return "Class Full"
end
```

7.1.6 Calculate Progress

Input: Student ID, Subject ID

Output: Progress percentage

```
function CalculateProgress(studentID, subjectID)
    totalTasks ← GetAllTasks(subjectID)
    completed ← GetStudentCompletions(studentID, subjectID)
    progress ← (completed / totalTasks) * 100
    return progress
end
```

7.1.7 Check Rewards

Input: Child ID

Output: Rewards issued (if any)

```
function CheckRewards(childID)
    goals ← GetGoals(childID)
    for goal in goals do
        if goal.status == "Completed" and goal.rewardGiven == false then
            GiveReward(childID, goal.reward)
            MarkRewardGiven(goal)
    end
```

7.1.8 Evaluate Quiz

Input: Quiz ID, Student Answers

Output: Score and feedback

```
function EvaluateQuiz(quizID, studentAnswers)
    correctAnswers ← GetCorrectAnswers(quizID)
    score ← 0
    for i in range(len(correctAnswers)) do
        if studentAnswers[i] == correctAnswers[i] then
            score += 1
    feedback ← GenerateFeedback(score)
    return score, feedback
end
```

7.1.9 Get Parent Dashboard

Input: Parent ID

Output: Dashboard data

```
function GetParentDashboard(parentID)
    children ← GetChildren(parentID)
    dashboard ← []
    for child in children do
        stats ← GetChildStats(child.id)
        dashboard.append(stats)
    return dashboard
end
```

7.1.10 Generate AI Flashcards

Input: Chapter Text

Output: Flashcards (Q&A pairs)

```
function GenerateAIFlashcards(chapterText)
    keyPoints ← ExtractKeyConcepts(chapterText)
    flashcards ← []
    for point in keyPoints do
        question ← GenerateQuestion(point)
        answer ← GenerateAnswer(point)
        flashcards.append((question, answer))
    return flashcards
end
```

7.1.11 Recommend Content

Input: User ID

Output: Recommended content list

```
function RecommendContent(userID)
    history ← GetUserHistory(userID)
    similarUsers ← FindSimilarUsers(userID)
    preferredTopics ← AnalyzeWeakAreas(history)
    recommended ← []
    for topic in preferredTopics do
        content ← FetchContent(topic)
        recommended.append(content)
    return recommended
end
```

7.1.12 Match For Battle

Input: Student ID

Output: Matched Peer or Waiting Message

```
function MatchForBattle(studentID)
    userLevel ← GetUserSkillLevel(studentID)
    availablePeers ← GetWaitingStudents()
    matchedPeer ← null
    for peer in availablePeers do
        if abs(GetUserSkillLevel(peer) - userLevel) ≤ threshold then
            matchedPeer ← peer
            break
    if matchedPeer ≠ null then
```

```
CreateBattleRoom(studentID, matchedPeer)
    return matchedPeer
return "No suitable peer found. Waiting in queue."
end
```

7.1.13 Calculate Overall Score

Input: Student ID

Output: Overall score percentage

```
function CalculateOverallScore(studentID)
    books ← GetAllBooksAttempted(studentID)
    totalScore ← 0
    totalQuestions ← 0
    for book in books do
        (score, questions) ← GetBookScore(studentID, book.id)
        totalScore += score
        totalQuestions += questions
    if totalQuestions == 0 then
        return 0
    return (totalScore / totalQuestions) * 100
end
```

7.1.14 Calculate Book-Wise Scores

Input: Student ID

Output: Dictionary of book-wise percentages

```
function CalculateBookWiseScores(studentID)
    books ← GetAllBooksAttempted(studentID)
```

```
scores ← {}  
for book in books do  
    (score, questions) ← GetBookScore(studentID, book.id)  
    if questions > 0 then  
        percentage ← (score / questions) * 100  
        scores[book.title] ← percentage  
    return scores  
end
```

7.1.15 Recommend Content (Hybrid)

Input: User ID

Output: Ranked list of recommended items

```
function RecommendContent(user_id)  
    user_profile ← GetUserPreferences(user_id)  
    similar_users ← FindSimilarUsersUsingCosineSimilarity(user_profile)  
    content_based_recs ← RecommendBasedOnUserProfile(user_profile)  
    collaborative_recs ← RecommendFromSimilarUsers(similar_users)  
    combined_recs ← MergeAndRank(content_based_recs, collaborative_recs)  
    return combined_recs  
end
```

7.1.16 Generate Explanation

Input: Topic Text

Output: Detailed Explanation

```
function GenerateExplanation(topic_text)  
    keywords ← ExtractKeyConcepts(topic_text)
```

```
    explanation ← []
    for keyword in keywords do
        step_explanation ← UseGPTOrT5ToExplain(keyword)
        explanation.append(step_explanation)
    return explanation
end
```

7.1.17 Generate Summary

Input: Content

Output: Summary

```
function GenerateSummary(content)
    cleaned_text ← PreprocessText(content)
    summary ← RunPegasusOrT5Summarizer(cleaned_text)
    return summary
end
```

7.1.18 Check and Reward Goal

Input: User ID, Goal ID

Output: Reward given (Boolean)

```
function CheckAndRewardGoal(user_id, goal_id)
    goal ← GetGoalDetails(goal_id)
    progress ← GetUserProgress(user_id, goal.book, goal.chapter)
    if progress ≥ goal.target then
        reward ← AssignReward(user_id)
        return true
    else
```

```
    return false
```

```
end
```

7.2 External APIs/SDKs

Describe the third-party APIs/SDKs used in the project implementation in the following table. Few examples of APIs are provided in the table.

Name of API/SDK or Library	Description	Purpose of Usage	API Endpoint/Function Used
Cloudinary	Image and video management solution	Uploading user-generated images (e.g., avatars, notes) to cloud storage	https://api.cloudinary.com/v1_1/demo/image/upload
Firebase Cloud Messaging (FCM)	Messaging and push notification service	Sending notifications to users (e.g., quiz results, live class reminders)	FirebaseMessaging.onMessage, FirebaseMessaging.send()
MongoDB Atlas Search	Full-text search engine integrated with MongoDB	Searching books, topics, and flashcards across user data	db.collection.aggregate([{ \$search: { ... } }])
scikit-learn (Decision Tree)	ML library for classification and regression	Classifying student performance and guiding personalized learning paths	DecisionTreeClassifier().fit(), .predict()
scikit-learn (TF-IDF)	Term Frequency–Inverse Document Frequency model	Identifying important keywords for AI flashcard and summary generation	TfidfVectorizer().fit_transform()
pyirt / IRT Model	Item Response Theory model	Adaptive quiz difficulty and estimating student ability levels	irt_model.fit(), irt_model.predict()
spaCy / Hugging Face Transformers	Natural Language Processing (NLP) libraries	Summarizing educational content, generating explanations and paraphrased notes	nlp_model(text), transformers.pipeline("summarization")
Reinforcement Learning (Q-learning / Custom)	Reward-based ML model	Used in dynamic battle matchmaking and progress optimization	env.step(), agent.learn()

8. Testing and Evaluation

8.1 Unit testing:

N o.	Test Case/Test Script	Attribute and Value	Expected Result	Actual Result	Result
1	Validate email format with valid email	`isValidEmail("test@example.com")`	Function returns `true`	`true`	Pass
2	Validate email format with invalid email (no @)	`isValidEmail("testexample.com")`	Function returns `false`	`false`	Pass
3	Validate email format with empty input	`isValidEmail("")`	Function returns `false`	`false`	Pass
7	Check if user exists with existing email	`userExists("existing@email.com")`	Function returns `true`	`true`	Pass
9	Check if user exists with non-existent email	`userExists("nonexistent@email.com")`	Function returns `false`	`false`	Pass
10	Create a new user	`createUser(validUniqueData)`	Success status/ID	Success status/ID	Pass

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	with valid unique data				
11	Create a user with duplicate email	`createUser(existingEmail)`	Error: duplicate email	Error: duplicate email	Pass
12	Create a user with invalid email format	`createUser(invalidEmail)`	Error: invalid email	Error: invalid email	Pass
13	Create a user with weak password	`createUser(weakPassword)`	Error: invalid password	Error: invalid password	Pass
17	Validate password with length above minimum	Password validation ($\geq$ min length)	Logic returns `true`	`true`	Pass
18	Validate password with length below minimum	Password validation ($<$ min length)	Logic returns `false`	`false`	Pass
19	Validate username with allowed characters	Username validation (allowed chars)	Logic returns `true`	`true`	Pass
20	Validate username with disallowed characters	Username validation (disallowed chars)	Logic returns `false`	`false`	Pass
21	Render input field with placeholder	Render `InputField` with placeholder "Enter username"	Input field with correct placeholder	Input field with "Enter username"	Pass

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22	Render password input with toggle functionality	Render `PasswordInput` with toggle	Text visibility toggles	Text visibility toggles	Pass
23	Render button in enabled state	Render `Button` with "Sign In", `disabled=false`	Button is active/clickable	Button is active/clickable	Pass
24	Render button in disabled state	Render `Button` with "Sign In", `disabled=true`	Button is greyed out/disabled	Button is greyed out/disabled	Pass
27	Render onboarding page with dummy content	Render `OnboardingPage` with dummy content	Displays dummy content	Displays dummy content	Pass
28	Render header with username and icons	Render `Header` with username "Nasir Bhutta" and icons	Displays username and icons	Displays "Nasir Bhutta" and icons	Pass
29	Render score widget with score and icon	Render `ScoreWidget` with score (user.score) and icon	Displays score and icon	Displays 10 and icon	Pass
30	Render subject card with dummy data	Render `SubjectCard` with (user.subject.name) data	Displays all dummy data	Displays "Mathematics" data	Pass
31	Render bottom navigation	Render `BottomNavBar`	Rendered Correctly	Rendered Correctly	Pass
32	Render modal with title and content	Render `Modal` with title, content, `isVisible=true`	Modal with title, content, 'X' button	Modal with title, content, 'X' button	Pass

33	Render battle history item with dummy data	Render `BattleHistoryItem` with dummy data	Displays initial, username, arrow	Displays initial, "@Username", arrow	Pass
34	Render subject list item with subject and icon	Render `SubjectListItem` with (user.subject.name) and book icon	Displays subject and icon	Displays (user.subject.name) and book icon	Pass
35	Render chapter item with checkbox states	Render `ChapterItem` with (subject.chapter) and `isChecked=true/false`	Correct checked/unchecked state	Correct checked/unchecked state	Pass
38	Render report card question item with dummy data	Render `ReportCardQuestionItem` with dummy data	Displays all data with 'X' for incorrect	Displays all data with 'X'	Pass
39	Render class card with dummy data	Render `ClassCard` with dummy data	Displays all Card of Classes	Displays all Card of Classes	Pass
40	Render score summary widget with dummy data	Render `ScoreSummaryWidget` with dummy data	Displays all dummy data with icons	Displays all dummy data with icons	Pass
41	Render podium item with rank 1 data	Render `PodiumItem` with rank 1 data	Displays rank, user, name, points	Displays rank 1, "Ali Raza", 950	Pass

8.2 Functional testing:

No .	Test Case/Test Script	Attribute and Value	Expected Result	Actual Result	Result
1	Verifying initial Onboarding screen display	Launch the application after a fresh install (or simulating first launch).	The first onboarding screen (Image 1 content) is displayed prominently. Dot indicators show 1/3 active.	The first onboarding screen (Image 1 content) is displayed prominently. Dot indicators show 1/3 active.	Pass
2	Swiping left on 1st Onboarding screen	On the first onboarding screen, swipe left.	User is navigated to the second onboarding screen (Image 2 content). Dot indicators update to 2/3 active.	User is navigated to the second onboarding screen (Image 2 content). Dot indicators update to 2/3 active.	Pass
3	Swiping left on 2nd Onboarding screen	On the second onboarding screen, swipe left.	User is navigated to the third (last) onboarding screen (Image 3 content). Dot indicators update to 3/3 active.	User is navigated to the third (last) onboarding screen (Image 3 content). Dot indicators update to 3/3 active.	Pass
4	Swiping right on 2nd Onboarding screen	On the second onboarding screen, swipe right.	User is navigated back to the first onboarding screen (Image 1 content). Dot indicators update to 1/3 active.	User is navigated back to the first onboarding screen (Image 1 content). Dot indicators update to 1/3 active.	Pass
5	Swiping right on 3rd Onboarding screen	On the third onboarding screen, swipe right.	User is navigated back to the second onboarding screen (Image 2 content). Dot indicators update to 2/3 active.	User is navigated back to the second onboarding screen (Image 2 content). Dot indicators update to 2/3 active.	Pass
6	Attempting to swipe left on 3rd Onboarding screen	On the third onboarding screen, swipe left.	No navigation occurs. The user remains on the third screen.	No navigation occurs. The user remains on the third screen.	Pass

7	Tapping a dot indicator to navigate	On any onboarding screen, tap one of the inactive dot indicators.	The user is directly navigated to the screen corresponding to the tapped dot. (Assumes dots are interactive, common UI pattern).	The user is directly navigated to the screen corresponding to the tapped dot. (Assumes dots are interactive, common UI pattern).	Pass
8	Entering valid Username/Email in login field	On Login screen, tap the "Username / Email" field and type "test@example.com".	The text "test@example.com" appears in the input field. Keyboard appears.	The text "test@example.com" appears in the input field. Keyboard appears.	Pass
9	Entering valid Password in login field	On Login screen, tap the "Password" field and type "password123".	The text "password123" appears in the input field, obscured by dots/asterisks. Keyboard appears.	The text "password123" appears in the input field, obscured by dots/asterisks. Keyboard appears.	Pass
10	Toggling password visibility on login screen	On Login screen, tap the password field toggle icon (if present).	The password text becomes visible (or hidden) as the icon changes state.	The password text becomes visible (or hidden) as the icon changes state.	Pass
11	Signing in with valid credentials	On Login screen, enter valid credentials. Click "Sign In".	User is successfully logged in and navigated to the Home screen (Image 4).	User is successfully logged in and navigated to the Home screen (Image 4).	Pass
12	Signing in with invalid password	On Login screen, enter valid Username/Email and invalid Password. Click "Sign In".	An error message (e.g., "Invalid credentials") is displayed on the screen. User remains on the Login screen.	An error message (e.g., "Invalid credentials") is displayed on the screen. User remains on the Login screen.	Pass

13	Signing in with non-existent user credentials	On Login screen, enter credentials for a non-existent user. Click "Sign In".	An error message (e.g., "User not found", "Invalid credentials") is displayed. User remains on the Login screen.	An error message (e.g., "User not found", "Invalid credentials") is displayed. User remains on the Login screen.	Pass
14	Signing in with empty Username/Email	On Login screen, leave Username/Email empty, enter password. Click "Sign In".	Validation error message appears next to the empty field. Button click might be prevented if client-side validation active. User remains on screen.	Validation error message appears next to the empty field. Button click might be prevented if client-side validation active. User remains on screen.	Pass
15	Signing in with empty Password	On Login screen, enter Username/Email, leave Password empty. Click "Sign In".	Validation error message appears next to the empty field. Button click might be prevented. User remains on screen.	Validation error message appears next to the empty field. Button click might be prevented. User remains on screen.	Pass
16	Signing in with both fields empty	On Login screen, leave both fields empty. Click "Sign In".	Validation error messages appear for both fields. Button click might be prevented. User remains on screen.	Validation error messages appear for both fields. Button click might be prevented. User remains on screen.	Pass
17	Clicking "Forgot Password?" link	On Login screen, click the "Forgot Password?" link.	User is navigated to a password recovery screen/flow.	User is navigated to a password recovery screen/flow.	Pass
18	Clicking "Sign Up" link	On Login screen, click the "Sign Up" link.	User is navigated to the Sign Up screen (Image 3).	User is navigated to the Sign Up screen (Image 3).	Pass
20	Clicking "Login with Google" button	On Login screen, click "Login with Google".	A system-level Google login dialog is presented, or a	A system-level Google login dialog is presented, or a	Pass

			placeholder action occurs.	placeholder action occurs.	
21	Clicking "Get Support" link	On Login screen, click the "Need help? Get Support" link.	User is navigated to a support screen or contact form (even if placeholder).	User is navigated to a support screen or contact form (even if placeholder).	Pass
22	Entering valid Full Name on Sign Up screen	On Sign Up screen, tap "Full Name" field and type "Test User".	Text appears correctly. Keyboard appears.	Text appears correctly. Keyboard appears.	Pass
23	Entering valid Username/Email on Sign Up screen	On Sign Up screen, tap "Username / Email" field and type "testuser@example.com".	Text appears correctly. Keyboard appears.	Text appears correctly. Keyboard appears.	Pass
24	Entering valid Password on Sign Up screen	On Sign Up screen, tap "Password" field and type "SecurePassword123".	Text appears correctly, obscured. Keyboard appears.	Text appears correctly, obscured. Keyboard appears.	Pass
25	Toggling password visibility on Sign Up screen	On Sign Up screen, tap password field toggle icon (if present).	Password text becomes visible/hidden. Icon changes state.	Password text becomes visible/hidden. Icon changes state.	Pass
26	Signing up with valid and unique data	On Sign Up screen, fill all fields with valid, unique data. Click "Signup".	User is successfully registered and navigated to Home screen or Login screen (as per design).	User is successfully registered and navigated to Home screen or Login screen (as per design).	Pass
27	Signing up with existing email	On Sign Up screen, fill fields with valid data but an existing email. Click "Signup".	Error message (e.g., "Email already exists") is displayed. User remains on screen.	Error message (e.g., "Email already exists") is displayed. User remains on screen.	Pass
28	Signing up with existing username	On Sign Up screen, fill fields with valid data but an existing username. Click "Signup".	Error message (e.g., "Username already exists") is displayed. User remains on screen.	Error message (e.g., "Username already exists") is displayed. User remains on screen.	Pass

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29	Signing up with invalid email format	On Sign Up screen, fill fields, enter invalid email format. Click "Signup".	Validation error message appears. User remains on screen. (Client-side or Backend validation feedback).	Validation error message appears. User remains on screen. (Client-side or Backend validation feedback).	Pass
30	Signing up with password failing strength/length rules	On Sign Up screen, fill fields, enter a password failing rules. Click "Signup".	Validation error message appears. User remains on screen. (Client-side or Backend validation feedback).	Validation error message appears. User remains on screen. (Client-side or Backend validation feedback).	Pass
31	Signing up with empty Full Name	On Sign Up screen, leave Full Name empty. Fill other fields. Click "Signup".	Validation error for Full Name appears. User remains on screen.	Validation error for Full Name appears. User remains on screen.	Pass
32	Signing up with empty Username/Email	On Sign Up screen, leave Username/Email empty. Fill other fields. Click "Signup".	Validation error for Username/Email appears. User remains on screen.	Validation error for Username/Email appears. User remains on screen.	Pass
33	Signing up with empty Password	On Sign Up screen, leave Password empty. Fill other fields. Click "Signup".	Validation error for Password appears. User remains on screen.	Validation error for Password appears. User remains on screen.	Pass
34	Signing up with all fields empty	On Sign Up screen, leave all fields empty. Click "Signup".	Validation error for all required fields appears. User remains on screen.	Validation error for all required fields appears. User remains on screen.	Pass
35	Clicking "Log In" link on Sign Up screen	On Sign Up screen, click the "Log In" link.	User is navigated to the Login screen (Image 1 or 2).	User is navigated to the Login screen (Image 1 or 2).	Pass
37	Clicking "Login with Google" button on Sign Up screen	On Sign Up screen, click "Login with Google".	A system-level Google login dialog is presented, or a placeholder action occurs.	A system-level Google login dialog is presented, or a placeholder action occurs.	Pass

38	Clicking "Get Support" link on Sign Up screen	On Sign Up screen, click the "Need help? Get Support" link.	User is navigated to a support screen or contact form (even if placeholder).	User is navigated to a support screen or contact form (even if placeholder).	Pass
39	Verifying Welcome greeting and Username display on Home screen	On Home screen, observe the greeting and username in the header.	The greeting (e.g., "Good Evening") is displayed. The logged-in user's name ("Nasir Bhutta") is displayed below it.	The greeting (e.g., "Good Evening") is displayed. The logged-in user's name ("Nasir Bhutta") is displayed below it.	Pass
40	Verifying Header Icons presence on Home screen	On Home screen, observe the top right of the header.	A notification bell icon and a user initial/profile icon ('N') are displayed.	A notification bell icon and a user initial/profile icon ('N') are displayed.	Pass
41	Verifying Score widget display on Home screen	On Home screen, observe the Score widget.	The widget displays an icon (trophy), a number (10), and the text "score".	The widget displays an icon (trophy), a number (10), and the text "score".	Pass
42	Verifying Coins widget display on Home screen	On Home screen, observe the Coins widget.	The widget displays an icon (star), a number (1), and the icon text.	The widget displays an icon (star), a number (1), and the icon text.	Pass
43	Verifying Streak widget display on Home screen	On Home screen, observe the Streak widget.	The widget displays an icon (flame), a number (0), and the icon text.	The widget displays an icon (flame), a number (0), and the icon text.	Pass
44	Verifying Lives widget display on Home screen	On Home screen, observe the Lives widget.	The widget displays an icon (heart), a number (5/5), and the icon text.	The widget displays an icon (heart), a number (5/5), and the icon text.	Pass
45	Clicking Score widget on Home screen	On Home screen, click the Score widget (trophy).	The Score & Goals modal (Image 5) is displayed on top	The Score & Goals modal (Image 5) is displayed on top	Pass

			of the Home screen.	of the Home screen.	
46	Clicking Coins widget on Home screen	On Home screen, click the Coins widget (star).	The Coins modal (Image 7) is displayed.	The Coins modal (Image 7) is displayed.	Pass
47	Clicking Streak widget on Home screen	On Home screen, click the Streak widget (flame).	The Streak modal (Image 8) is displayed.	The Streak modal (Image 8) is displayed.	Pass
48	Clicking Lives widget on Home screen	On Home screen, click the Lives widget (heart).	The Lives modal (Image 9) is displayed.	The Lives modal (Image 9) is displayed.	Pass
49	Verifying Bookmark card display on Home screen	On Home screen, observe the Bookmark card.	A card titled "Bookmark 6 Questions" with a "Claim Now" button is displayed. An illustration is present.	A card titled "Bookmark 6 Questions" with a "Claim Now" button is displayed. An illustration is present.	Pass
50	Clicking "Claim Now" on Bookmark card on Home screen	On Home screen, click the "Claim Now" button on the Bookmark card.	A placeholder action occurs (e.g., a toast message "Claimed!" or navigation to a bookmarks screen). The button might become disabled or change text.	A placeholder action occurs (e.g., a toast message "Claimed!" or navigation to a bookmarks screen). The button might become disabled or change text.	Pass
51	Verifying Subject cards display on Home screen	On Home screen, scroll down and observe the subject cards.	Subject cards for Mathematics, Physics, and Biology (and potentially others if scrollable) are displayed, showing subject name, level, description, progress bar, and "Answer Quiz" button.	Subject cards for Mathematics, Physics, and Biology (and potentially others if scrollable) are displayed, showing subject name, level, description, progress bar, and "Answer Quiz" button.	Pass

			Illustrations are present.	Illustrations are present.	
52	Clicking "Answer Quiz" on Mathematics card on Home screen	On Home screen, click the "Answer Quiz" button on the Mathematics card.	User is navigated to a quiz screen related to Mathematics (e.g., Brain Battle screen or a specific quiz entry point). Context (subject: Mathematics) should be passed.	User is navigated to a quiz screen related to Mathematics (e.g., Brain Battle screen or a specific quiz entry point). Context (subject: Mathematics) should be passed.	Pass
53	Clicking "Answer Quiz" on Physics card on Home screen	On Home screen, click the "Answer Quiz" button on the Physics card.	User is navigated to a quiz screen related to Physics. Context (subject: Physics) should be passed.	User is navigated to a quiz screen related to Physics. Context (subject: Physics) should be passed.	Pass
54	Clicking "Answer Quiz" on Biology card on Home screen	On Home screen, click the "Answer Quiz" button on the Biology card.	User is navigated to a quiz screen related to Biology. Context (subject: Biology) should be passed.	User is navigated to a quiz screen related to Biology. Context (subject: Biology) should be passed.	Pass
55	Clicking notification icon in header on Home screen	On Home screen, click the notification bell icon.	User is navigated to the Notifications screen (Image 31).	User is navigated to the Notifications screen (Image 31).	Pass
56	Clicking user initial/profile icon in header on Home screen	On Home screen, click the 'N' icon.	User is navigated to the Menu screen (Image 30).	User is navigated to the Menu screen (Image 30).	Pass
57	Clicking "Home" when on Home screen	On Home screen, tap "Home" in the bottom nav.	No navigation occurs. The Home screen remains active. (Unless designed to refresh).	No navigation occurs. The Home screen remains active. (Unless designed to refresh).	Pass

58	Clicking "Learn" from Home screen	On Home screen, tap "Learn" in the bottom nav.	The "Learn" icon becomes active. User is navigated to a Learn-related screen (likely subject selection, similar to Image 12 or 22).	The "Learn" icon becomes active. User is navigated to a Learn-related screen (likely subject selection, similar to Image 12 or 22).	Pass
59	Clicking "Class" from Home screen	On Home screen, tap "Class" in the bottom nav.	The "Class" icon becomes active. User is navigated to the My Classes screen (Image 23).	The "Class" icon becomes active. User is navigated to the My Classes screen (Image 23).	Pass
60	Clicking "Extras" from Home screen	On Home screen, tap "Extras" in the bottom nav.	The "Extras" icon becomes active. The Extras menu modal (Image 10) is displayed on top of the Home screen.	The "Extras" icon becomes active. The Extras menu modal (Image 10) is displayed on top of the Home screen.	Pass
61	Clicking "Home" from Class screen	Navigate to Class screen (FT_BottomNav_059). Tap "Home" in the bottom nav.	The "Home" icon becomes active. User is navigated back to the Home screen (Image 4).	The "Home" icon becomes active. User is navigated back to the Home screen (Image 4).	Pass
62	Clicking "Learn" from Class screen	Navigate to Class screen. Tap "Learn" in the bottom nav.	The "Learn" icon becomes active. User is navigated to the Learn screen.	The "Learn" icon becomes active. User is navigated to the Learn screen.	Pass
63	Clicking "Extras" from Class screen	Navigate to Class screen. Tap "Extras" in the bottom nav.	The "Extras" icon becomes active. The Extras modal appears over the Class screen.	The "Extras" icon becomes active. The Extras modal appears over the Class screen.	Pass
64	Verifying Extras modal content	Trigger Extras modal	The modal displays icons and text for "Battle", "Practice",	The modal displays icons and text for "Battle", "Practice",	Pass

			"FlashCards", and "Books". An 'X' close button is in the corner.	"FlashCards", and "Books". An 'X' close button is in the corner.	
65	Clicking "Battle" in Extras modal	Open Extras modal. Click "Battle".	Modal closes. User is navigated to the Battle Lobby screen (Image 11).	Modal closes. User is navigated to the Battle Lobby screen (Image 11).	Pass
66	Clicking "Practice" in Extras modal	Open Extras modal. Click "Practice".	Modal closes. User is navigated to a Practice subject selection screen (similar to Image 12).	Modal closes. User is navigated to a Practice subject selection screen (similar to Image 12).	Pass
67	Clicking "FlashCards" in Extras modal	Open Extras modal. Click "FlashCards".	Modal closes. User is navigated to the Flashcard subject selection screen (Image 22).	Modal closes. User is navigated to the Flashcard subject selection screen (Image 22).	Pass
68	Clicking "Books" in Extras modal	Open Extras modal. Click "Books".	Modal closes. User is navigated to a Books/Study Materials screen (even if dummy).	Modal closes. User is navigated to a Books/Study Materials screen (even if dummy).	Pass
69	Clicking 'X' to close Extras modal	Open Extras modal. Click the 'X' button.	Modal is dismissed. User returns to the screen from which the modal was opened.	Modal is dismissed. User returns to the screen from which the modal was opened.	Pass
70	Tapping outside Extras modal to close (if applicable)	Open Extras modal. Tap on the dimmed background outside the modal area.	If implemented, modal is dismissed. User returns to the screen below.	If implemented, modal is dismissed. User returns to the screen below.	Pass
71	Verifying Battle Lobby screen layout and content	Navigate to Battle Lobby (FT_Extras_65).	Screen displays header with back button ("<") and title "Battle". User info area (initial, username, stats). Banner. "Start	Screen displays header with back button ("<") and title "Battle". User info area (initial, username, stats). Banner. "Start	Pass

			the Battle", "Enter Invitation Code" buttons. "Your Battle History" section title and dummy list.	the Battle", "Enter Invitation Code" buttons. "Your Battle History" section title and dummy list.	
72	Clicking "Start the Battle" button on Battle Lobby	On Battle Lobby, click "Start the Battle".	User is navigated to the Select Subject screen (Image 12).	User is navigated to the Select Subject screen (Image 12).	Pass
73	Clicking "Enter Invitation Code" button on Battle Lobby	On Battle Lobby, click "Enter Invitation Code".	The "Battle invitation" modal (Image 21) is displayed on top.	The "Battle invitation" modal (Image 21) is displayed on top.	Pass
74	Clicking a Battle History list item on Battle Lobby	On Battle Lobby, click one of the dummy items in "Your Battle History".	User is navigated to a Battle Report Card or history details screen (Image 19/20 equivalent) related to that history item.	User is navigated to a Battle Report Card or history details screen (Image 19/20 equivalent) related to that history item.	Pass
75	Clicking back button on Battle Lobby	On Battle Lobby, click the "<" back button.	User is navigated back to the previous screen (likely Home).	User is navigated back to the previous screen (likely Home).	Pass
76	Verifying Battle Invitation modal content	Open Battle Invitation modal (FT_BattleLobby_73).	Modal displays title "Battle invitation", instruction text, input fields for code, and "Enter match" button. 'X' close button is present.	Modal displays title "Battle invitation", instruction text, input fields for code, and "Enter match" button. 'X' close button is present.	Pass
77	Entering code in Invitation modal fields	Open Invitation modal. Tap fields and type text.	Text appears in the fields. Keyboard appears.	Text appears in the fields. Keyboard appears.	Pass
78	Clicking "Enter match"	Open Invitation modal. Enter text (valid/invalid	A placeholder action occurs	A placeholder action occurs	Pass

	in Invitation modal (dummy action)	format) in fields. Click "Enter match".	(e.g., message "Attempting to join match with code..."). Navigation might be attempted to a dummy battle screen or error message displayed based on dummy logic.	(e.g., message "Attempting to join match with code..."). Navigation might be attempted to a dummy battle screen or error message displayed based on dummy logic.	
79	Clicking 'X' to close Battle Invitation modal	Open Invitation modal. Click the 'X' button.	Modal is dismissed. User returns to the Battle Lobby screen.	Modal is dismissed. User returns to the Battle Lobby screen.	Pass
80	Verifying Select Subject screen content (from Battle)	Navigate to Select Subject from Battle (FT_BattleLobby_72).	Screen displays header ("Select Subject") and list of subjects (English, Biology, etc.) with icons.	Screen displays header ("Select Subject") and list of subjects (English, Biology, etc.) with icons.	Pass
81	Clicking a subject on Select Subject (from Battle)	On Select Subject screen, click "Mathematics".	The "Battle Mode" dialog (Image 13) is displayed, mentioning the selected subject "Mathematics".	The "Battle Mode" dialog (Image 13) is displayed, mentioning the selected subject "Mathematics".	Pass
82	Clicking back button on Select Subject	On Select Subject, click "<".	User is navigated back to the Battle Lobby screen.	User is navigated back to the Battle Lobby screen.	Pass
83	Verifying Battle Mode dialog content	Open Battle Mode dialog (FT_SelectSubjectBattle_81).	Dialog displays title "Battle Mode", question "How would you like to compete...", and buttons "By Chapter", "Whole Book".	Dialog displays title "Battle Mode", question "How would you like to compete...", and buttons "By Chapter", "Whole Book".	Pass

			'X' close button is present.	'X' close button is present.	
84	Clicking "By Chapter" in Battle Mode dialog	Open Battle Mode dialog. Click "By Chapter".	Modal closes. User is navigated to the "Select Chapters - Mathematics" screen (Image 15).	Modal closes. User is navigated to the "Select Chapters - Mathematics" screen (Image 15).	Pass
85	Clicking "Whole Book" in Battle Mode dialog	Open Battle Mode dialog. Click "Whole Book".	Modal closes. The "Invite Friends" dialog (Image 14) is displayed.	Modal closes. The "Invite Friends" dialog (Image 14) is displayed.	Pass
86	Clicking 'X' to close Battle Mode dialog	Open Battle Mode dialog. Click the 'X'.	Modal is dismissed. User returns to the Select Subject screen.	Modal is dismissed. User returns to the Select Subject screen.	Pass
87	Verifying Invite Friends dialog content	Open Invite Friends dialog (FT_BattleModeDialog_85).	Dialog displays title "Invite Friends", question "Are you ready?", and buttons "Share invitation code", "Random Match". 'X' close button is present.	Dialog displays title "Invite Friends", question "Are you ready?", and buttons "Share invitation code", "Random Match". 'X' close button is present.	Pass
88	Clicking "Share invitation code" in dialog (dummy action)	Open Invite Friends dialog. Click "Share invitation code".	A system-level share sheet or placeholder appears (e.g., message "Sharing code...").	A system-level share sheet or placeholder appears (e.g., message "Sharing code...").	Pass
89	Clicking "Random Match" in dialog (dummy action)	Open Invite Friends dialog. Click "Random Match".	Modal closes. User is navigated to the Brain Battle screen (Image 16), simulating a random match start.	Modal closes. User is navigated to the Brain Battle screen (Image 16), simulating a random match start.	Pass

90	Clicking 'X' to close Invite Friends dialog	Open Invite Friends dialog. Click the 'X'.	Modal is dismissed. User returns to the Select Subject screen (where Battle Mode dialog originated).	Modal is dismissed. User returns to the Select Subject screen (where Battle Mode dialog originated).	Pass
91	Verifying Select Chapters screen layout and content	Navigate to Select Chapters (FT_BattleModeDialog_84).	Screen displays header ("Select Chapters - Mathematics") with back button. A list of chapters with checkboxes is shown (Algebra, Geometry, etc.). Below, "Selected Chapters: X" count. "Start Match" button below, initially disabled.	Screen displays header ("Select Chapters - Mathematics") with back button. A list of chapters with checkboxes is shown (Algebra, Geometry, etc.). Below, "Selected Chapters: X" count. "Start Match" button below, initially disabled.	Pass
92	Clicking a chapter checkbox (select) on Select Chapters	On Select Chapters, click the "Algebra" checkbox.	The checkbox becomes checked. The "Selected Chapters" count increases by 1. The "Start Match" button becomes enabled.	The checkbox becomes checked. The "Selected Chapters" count increases by 1. The "Start Match" button becomes enabled.	Pass
93	Clicking a chapter checkbox (deselect) on Select Chapters	On Select Chapters, click the "Algebra" checkbox which is currently checked.	The checkbox becomes unchecked. The "Selected Chapters" count decreases by 1. The "Start Match" button might become disabled if count is 0.	The checkbox becomes unchecked. The "Selected Chapters" count decreases by 1. The "Start Match" button might become disabled if count is 0.	Pass

94	Selecting multiple chapters on Select Chapters	On Select Chapters, select "Algebra", then "Geometry".	Both checkboxes are checked. "Selected Chapters" count is 2. "Start Match" button remains enabled.	Both checkboxes are checked. "Selected Chapters" count is 2. "Start Match" button remains enabled.	Pass
95	Clicking "Start Match" with chapters selected on Select Chapters	On Select Chapters, ensure at least one chapter is selected. Click "Start Match".	User is navigated to the Brain Battle screen (Image 16), simulating a match starting with questions from selected chapters.	User is navigated to the Brain Battle screen (Image 16), simulating a match starting with questions from selected chapters.	Pass
96	Attempting to click "Start Match" with no chapters selected on Select Chapters	On Select Chapters, ensure no chapters are selected. Attempt to click the "Start Match" button.	The button does not respond to clicks (it's disabled).	The button does not respond to clicks (it's disabled).	Pass
97	Clicking back button on Select Chapters	On Select Chapters, click "<".	User is navigated back to the Select Subject screen.	User is navigated back to the Select Subject screen.	Pass
98	Verifying Brain Battle screen initial state	Navigate to Brain Battle	Screen displays header ("Brain Battle", back button), question number ("Question 1/5"), scores ("You 0 vs Opponent 0"), Time Remaining bar (full) and text (e.g., "15 seconds"), question text, and 4 answer options (A, B, C, D).	Screen displays header ("Brain Battle", back button), question number ("Question 1/5"), scores ("You 0 vs Opponent 0"), Time Remaining bar (full) and text (e.g., "15 seconds"), question text, and 4 answer options (A, B, C, D).	Pass

99	Selecting an Answer Option on Brain Battle	On Brain Battle, click one of the answer options (e.g., Option C).	The clicked option is highlighted. The screen progresses to the next question after a brief delay. Your score and opponent score may update (dummy). Question number increments. Timer resets.	The clicked option is highlighted. The screen progresses to the next question after a brief delay. Your score and opponent score may update (dummy). Question number increments. Timer resets.	Pass
100	Observing Time Remaining decrement on Brain Battle	On Brain Battle, watch the Time Remaining bar and text.	The bar visually decreases and the text counts down towards 0.	The bar visually decreases and the text counts down towards 0.	Pass
101	Letting timer run out on a question on Brain Battle	On Brain Battle, wait for the timer to reach 0 without selecting an answer.	The UI indicates time is up (e.g., bar empty, text 0s). The screen progresses to the next question after a brief delay. Your score does not increase for this question.	The UI indicates time is up (e.g., bar empty, text 0s). The screen progresses to the next question after a brief delay. Your score does not increase for this question.	Pass
102	Completing the Battle (answering all 5 questions) on Brain Battle	Proceed through the Brain Battle, answering all 5 questions or letting the timer expire.	After the 5th question, the Brain Battle screen transitions to the Battle Result modal (Image 18).	After the 5th question, the Brain Battle screen transitions to the Battle Result modal (Image 18).	Pass
103	Clicking back button during Battle on Brain Battle	On Brain Battle screen, click the "<" back button.	A confirmation dialog (e.g., "Are you sure you want to leave? Your progress will be lost.") is displayed. (Common pattern for	A confirmation dialog (e.g., "Are you sure you want to leave? Your progress will be lost.") is displayed. (Common pattern for	Pass

			ongoing activities).	ongoing activities).	
104	Confirming leaving Battle in dialog on Brain Battle	If leaving dialog appears (FT_BrainBattle_103), click "Yes" or "Leave".	The user is navigated back to the previous screen (e.g., Battle Lobby). The battle state is abandoned.	The user is navigated back to the previous screen (e.g., Battle Lobby). The battle state is abandoned.	Pass
105	Canceling leaving Battle in dialog on Brain Battle	If leaving dialog appears (FT_BrainBattle_103), click "No" or "Cancel".	The dialog is dismissed. User remains on the Brain Battle screen.	The dialog is dismissed. User remains on the Brain Battle screen.	Pass
106	Verifying Battle Result modal content	Complete a battle (FT_BrainBattle_102).	Modal displays title "Battle Result", a result trophy icon, a win/loss/draw status ("You Lost!"). It shows "Your Score" and "Opponent Score" with numerical values. A "Done" button is present.	Modal displays title "Battle Result", a result trophy icon, a win/loss/draw status ("You Lost!"). It shows "Your Score" and "Opponent Score" with numerical values. A "Done" button is present.	Pass
107	Clicking "Done" button in Battle Result modal	On Battle Result modal, click "Done".	Modal is dismissed. User is navigated to the Quiz Report Card screen (Image 19).	Modal is dismissed. User is navigated to the Quiz Report Card screen (Image 19).	Pass
108	Verifying Report Card header and summary section	Navigate to Quiz Report Card (FT_BattleResult_107 or FT_BattleLobby_74).	Screen displays header ("Quiz Report Card", back button, download, share icons). A summary section shows result status (Victory/Loss), completion date/time, Your	Screen displays header ("Quiz Report Card", back button, download, share icons). A summary section shows result status (Victory/Loss), completion date/time, Your	Pass

			Score, Opponent Score, and Time Taken (dummy data).	Score, Opponent Score, and Time Taken (dummy data).	
109	Verifying Performance Summary section on Report Card	On Report Card, observe Performance Summary section.	Section displays "Accuracy" (%), "Correct" (count), and "Avg Time" (seconds) (dummy data).	Section displays "Accuracy" (%), "Correct" (count), and "Avg Time" (seconds) (dummy data).	Pass
110	Verifying Question Analysis section layout on Report Card	On Report Card, observe Question Analysis section.	Section is titled "Question Analysis" and shows a list of questions. Each item has a number, question text, "Your Answer", "Correct Answer", and difficulty.	Section is titled "Question Analysis" and shows a list of questions. Each item has a number, question text, "Your Answer", "Correct Answer", and difficulty.	Pass
111	Verifying Question Analysis item (Incorrect answer) on Report Card	On Report Card, find an item for an incorrectly answered question (e.g., Question 1 in Image 19).	The question text, "Your Answer" (e.g., "Berlin"), and "Correct Answer" (e.g., "Paris") are displayed. An 'X' icon or red indicator is shown next to the question. Difficulty is shown.	The question text, "Your Answer" (e.g., "Berlin"), and "Correct Answer" (e.g., "Paris") are displayed. An 'X' icon or red indicator is shown next to the question. Difficulty is shown.	Pass
112	Verifying Question Analysis item (Correct answer) on Report Card	On Report Card, find an item for a correctly answered question (if dummy data includes one).	The question text, "Your Answer", and "Correct Answer" are displayed (they should be the same). A checkmark icon	The question text, "Your Answer", and "Correct Answer" are displayed (they should be the same). A checkmark icon	Pass

			or green indicator is shown. Difficulty is shown.	or green indicator is shown. Difficulty is shown.	
113	Verifying Performance Metrics section on Report Card	On Report Card, scroll to Performance Metrics section (Image 20).	Section titled "Performance Metrics". Bar graphs for Accuracy, Speed, and Score are displayed with dummy percentages/values and change indicators.	Section titled "Performance Metrics". Bar graphs for Accuracy, Speed, and Score are displayed with dummy percentages/values and change indicators.	Pass
114	Verifying Comparison with Past Quizzes section on Report Card	On Report Card, scroll to Comparison with Past Quizzes section (Image 20).	Section titled "Comparison with Past Quizzes". A chart (bar graph) compares current vs past quizzes (dummy data). A table below shows quiz details (Date, Score, Accuracy, Time/Q) for current and past quizzes (dummy data).	Section titled "Comparison with Past Quizzes". A chart (bar graph) compares current vs past quizzes (dummy data). A table below shows quiz details (Date, Score, Accuracy, Time/Q) for current and past quizzes (dummy data).	Pass
115	Clicking back button on Report Card	On Report Card, click "<".	User is navigated back to the screen from which the report card was accessed (likely Battle Lobby or Home).	User is navigated back to the screen from which the report card was accessed (likely Battle Lobby or Home).	Pass
116	Clicking Download icon on Report Card	On Report Card, click the download icon in the header.	A placeholder action related to downloading the report occurs (e.g., a toast, a modal, or	A placeholder action related to downloading the report occurs (e.g., a toast, a modal, or	Pass

			initiating download - dummy logic).	initiating download - dummy logic).	
117	Clicking Share icon on Report Card	On Report Card, click the share icon in the header.	A system-level share sheet or placeholder appears to share the report (dummy logic).	A system-level share sheet or placeholder appears to share the report (dummy logic).	Pass
118	Navigating to Learn subject selection	Tap "Learn" in bottom nav (FT_BottomNav_58).	User is navigated to a Select Subject screen (similar to Image 12), titled "Select Subject", listing subjects with icons.	User is navigated to a Select Subject screen (similar to Image 12), titled "Select Subject", listing subjects with icons.	Pass
119	Clicking a subject on Learn selection screen	On Learn subject selection, click "Physics".	User is navigated to a Physics-specific learning screen (e.g., chapter list, learning materials, quiz list) (even if dummy). Context (subject: Physics) passed.	User is navigated to a Physics-specific learning screen (e.g., chapter list, learning materials, quiz list) (even if dummy). Context (subject: Physics) passed.	Pass
120	Clicking back button on Learn subject selection	On Learn subject selection, click "<".	User is navigated back to the Home screen.	User is navigated back to the Home screen.	Pass
121	Navigating to Flashcard subject selection	Tap "FlashCards" in Extras modal (FT_Extras_67).	User is navigated to the Select Subject screen for Flashcards (Image 22), titled "Select Subject". List of subjects with icons and flashcard counts (dummy).	User is navigated to the Select Subject screen for Flashcards (Image 22), titled "Select Subject". List of subjects with icons and flashcard counts (dummy).	Pass
122	Clicking a subject on Flashcard	On Flashcard selection (Image 22), click "English".	User is navigated to the Flashcards screen for	User is navigated to the Flashcards screen for	Pass

	selection screen		English (even if dummy flashcards). Context (subject: English) passed.	English (even if dummy flashcards). Context (subject: English) passed.	
123	Clicking back button on Flashcard subject selection	On Flashcard selection (Image 22), click "<".	User is navigated back to the screen from which the Flashcards screen was opened (likely Home or Battle Lobby if accessed via Home->Extras->Flashcards).	User is navigated back to the screen from which the Flashcards screen was opened (likely Home or Battle Lobby if accessed via Home->Extras->Flashcards).	Pass
124	Navigating to My Classes screen	Tap "Class" in bottom nav (FT_BottomNav_59).	User is navigated to the My Classes screen (Image 23), titled "My Classes". "Enrolled Classes" title and a list of class cards are displayed.	User is navigated to the My Classes screen (Image 23), titled "My Classes". "Enrolled Classes" title and a list of class cards are displayed.	Pass
125	Verifying My Classes list content on My Classes screen	On My Classes screen, observe the list of class cards.	Cards for Mathematics, Science, and English Literature are shown, each with dummy teacher, schedule, and progress (e.g., 5/12).	Cards for Mathematics, Science, and English Literature are shown, each with dummy teacher, schedule, and progress (e.g., 5/12).	Pass
126	Clicking a class card on My Classes screen	On My Classes screen, click the "Mathematics" class card.	User is navigated to the Class Details screen for Mathematics (Image 24). Context (class: Mathematics) passed.	User is navigated to the Class Details screen for Mathematics (Image 24). Context (class: Mathematics) passed.	Pass

127	Clicking back button on My Classes screen	On My Classes screen, click "<".	User is navigated back to the Home screen.	User is navigated back to the Home screen.	Pass
128	Verifying Class Details screen layout and content	Navigate to Class Details (FT_Classes_126).	Screen displays header ("Mathematics", back button), Teacher name ("Mr. Johnson"). "Details" section (Schedule, Students - dummy). "Progress" section (Assignments completion - dummy). "Actions" section with buttons "View Materials", "Assignments", "Forum", "Contact Teacher".	Screen displays header ("Mathematics", back button), Teacher name ("Mr. Johnson"). "Details" section (Schedule, Students - dummy). "Progress" section (Assignments completion - dummy). "Actions" section with buttons "View Materials", "Assignments", "Forum", "Contact Teacher".	Pass
129	Clicking "View Materials" button on Class Details	On Class Details, click "View Materials".	User is navigated to a Class Materials screen (even if dummy content).	User is navigated to a Class Materials screen (even if dummy content).	Pass
130	Clicking "Assignments" button on Class Details	On Class Details, click "Assignments".	User is navigated to a Class Assignments screen (even if dummy content).	User is navigated to a Class Assignments screen (even if dummy content).	Pass
131	Clicking "Forum" button on Class Details	On Class Details, click "Forum".	User is navigated to a Class Forum screen (even if dummy content).	User is navigated to a Class Forum screen (even if dummy content).	Pass
132	Clicking "Contact Teacher" button on Class Details	On Class Details, click "Contact Teacher".	A contact interface (e.g., email composer pre-filled, message form	A contact interface (e.g., email composer pre-filled, message form	Pass

			modal/screen - dummy) appears, targeting the teacher.	modal/screen - dummy) appears, targeting the teacher.	
133	Clicking back button on Class Details	On Class Details, click "<".	User is navigated back to the My Classes screen.	User is navigated back to the My Classes screen.	Pass
134	Navigating to Overall Score screen	Navigate via Menu (FT_Menu_162).	User is navigated to Overall Score screen (Image 25), titled "Overall Score". Header back button present. Overall Score section, Subject Breakdown/Area s to Improve tabs (Subject Breakdown active), Subject Breakdown content (chart, list).	User is navigated to Overall Score screen (Image 25), titled "Overall Score". Header back button present. Overall Score section, Subject Breakdown/Area s to Improve tabs (Subject Breakdown active), Subject Breakdown content (chart, list).	Pass
135	Verifying Overall Score section content on Overall Score	On Overall Score, observe top section.	Displays "Your Overall Score" (Grade, Percentage, Score Ring) and summary metrics (Books, Quizzes, Hours) with icons (dummy data).	Displays "Your Overall Score" (Grade, Percentage, Score Ring) and summary metrics (Books, Quizzes, Hours) with icons (dummy data).	Pass
136	Verifying Subject Breakdown tab content on Overall Score	On Overall Score, ensure "Subject Breakdown" tab is active.	Displays performance bar chart by subject (Mathematics, English, History Geography - dummy data). List of subjects below with completion stats and grade (dummy data).	Displays performance bar chart by subject (Mathematics, English, History Geography - dummy data). List of subjects below with completion stats and grade (dummy data).	Pass

137	Switching to "Areas to Improve" tab on Overall Score	On Overall Score, click the "Areas to Improve" tab.	The "Areas to Improve" tab becomes active. The content below the tabs updates to display areas needing improvement (Image 26).	The "Areas to Improve" tab becomes active. The content below the tabs updates to display areas needing improvement (Image 26).	Pass
138	Verifying "Areas to Improve" tab content on Overall Score	On Overall Score, ensure "Areas to Improve" tab is active.	Section titled "Areas that need improvement". Lists specific areas (Geography - Map Reading, Mathematics - Algebra) with percentages (dummy), improvement suggestions, and "Practice Now" buttons.	Section titled "Areas that need improvement". Lists specific areas (Geography - Map Reading, Mathematics - Algebra) with percentages (dummy), improvement suggestions, and "Practice Now" buttons.	Pass
139	Clicking "Practice Now" button on Areas to Improve on Overall Score	On Overall Score ("Areas to Improve" tab), click "Practice Now" for "Geography - Map Reading".	User is navigated to a practice screen/quiz specifically for Geography Map Reading (even if dummy). Context passed.	User is navigated to a practice screen/quiz specifically for Geography Map Reading (even if dummy). Context passed.	Pass
140	Clicking a subject item in "Subject Breakdown" on Overall Score	On Overall Score ("Subject Breakdown" tab), click the "Mathematics" list item.	User is navigated to the Subject Performance screen for Mathematics (Image 27). Context passed.	User is navigated to the Subject Performance screen for Mathematics (Image 27). Context passed.	Pass
141	Clicking back button on Overall Score	On Overall Score, click "<".	User is navigated back to the screen from which the Overall Score	User is navigated back to the screen from which the Overall Score	Pass

			was accessed (likely Menu).	was accessed (likely Menu).	
14 2	Verifying Subject Performance screen layout and content	Navigate to Subject Performance (FT_OverallScore_140 or FT_Home_52 etc. if subjects on Home navigate here).	Screen displays header (Subject Title, back button). Course card (title, instructor, grade, percentage - dummy). Progress metrics (Pages Read, Study Time, Quizzes - dummy). Score Breakdown chart (dummy). Chapter Performance list (collapsed, dummy data). Personalized Recommendations (dummy).	Screen displays header (Subject Title, back button). Course card (title, instructor, grade, percentage - dummy). Progress metrics (Pages Read, Study Time, Quizzes - dummy). Score Breakdown chart (dummy). Chapter Performance list (collapsed, dummy data). Personalized Recommendations (dummy).	Pass
14 3	Verifying Course card details on Subject Performance	On Subject Performance screen, observe the card below the header.	Displays course title ("Advanced Mathematics: Algebra & Calculus"), instructor ("Dr. Robert Johnson"), grade ("B+"), and percentage ("78%") (dummy data).	Displays course title ("Advanced Mathematics: Algebra & Calculus"), instructor ("Dr. Robert Johnson"), grade ("B+"), and percentage ("78%") (dummy data).	Pass
14 4	Verifying Progress metrics display on Subject Performance	On Subject Performance screen, observe the progress metrics section.	Displays circles with counts/values and labels ("Pages Read", "Study Time", "Quizzes") (dummy data).	Displays circles with counts/values and labels ("Pages Read", "Study Time", "Quizzes") (dummy data).	Pass

14 5	Verifying Score Breakdown chart on Subject Performance	On Subject Performance screen, observe the Score Breakdown chart.	Displays a bar chart with categories ("Comprehension", "Problem Solving Theory") and scores (dummy data).	Displays a bar chart with categories ("Comprehension", "Problem Solving Theory") and scores (dummy data).	Pass
14 6	Verifying Chapter Performance list (collapsed state) on Subject Performance	On Subject Performance screen, observe the "Chapter Performance" list.	List items for chapters are displayed (e.g., Chapter 1, Chapter 2, etc.), showing name, status (Completed, In Progress), percentage, and down arrow.	List items for chapters are displayed (e.g., Chapter 1, Chapter 2, etc.), showing name, status (Completed, In Progress), percentage, and down arrow.	Pass
14 7	Expanding a Chapter Performance item on Subject Performance	On Subject Performance screen, click a chapter item with a down arrow (e.g., "Chapter 1: Fundamentals of Algebra").	The section expands below the chapter item. Sub-topics with percentages and checkmarks appear (Basic Equations 90%, Polynomials 85%, Word Problems 80% - dummy data). The arrow icon changes to point up. A "Practice This Chapter" button appears.	The section expands below the chapter item. Sub-topics with percentages and checkmarks appear (Basic Equations 90%, Polynomials 85%, Word Problems 80% - dummy data). The arrow icon changes to point up. A "Practice This Chapter" button appears.	Pass
14 8	Collapsing an expanded Chapter Performance item on Subject Performance	On Subject Performance screen, click an expanded chapter item (with an up arrow).	The section collapses, hiding the sub-topic details and "Practice This Chapter" button. The arrow icon changes to point down.	The section collapses, hiding the sub-topic details and "Practice This Chapter" button. The arrow icon changes to point down.	Pass

149	Clicking "Practice This Chapter" button on Subject Performance	On Subject Performance screen, expand a chapter. Click the "Practice This Chapter" button.	User is navigated to a practice screen/quiz specifically for that chapter (e.g., Algebra Fundamentals) (even if dummy). Context passed.	User is navigated to a practice screen/quiz specifically for that chapter (e.g., Algebra Fundamentals) (even if dummy). Context passed.	Pass
150	Verifying Personalized Recommendations section on Subject Performance	On Subject Performance screen, scroll to the bottom.	Section titled "Personalized Recommendations" is displayed. A list of numbered recommendations with text is shown (dummy data).	Section titled "Personalized Recommendations" is displayed. A list of numbered recommendations with text is shown (dummy data).	Pass
151	Clicking back button on Subject Performance	On Subject Performance screen, click "<".	User is navigated back to the previous screen (e.g., Overall Score or Home).	User is navigated back to the previous screen (e.g., Overall Score or Home).	Pass
152	Navigating to Leaderboard screen	Navigate via Menu (FT_Menu_163).	User is navigated to Leaderboard screen (Image 29), titled "LeaderBoard". Header back button present. "weekly" and "monthly" tabs visible ("weekly" active). Top 3 podium display and list below.	User is navigated to Leaderboard screen (Image 29), titled "LeaderBoard". Header back button present. "weekly" and "monthly" tabs visible ("weekly" active). Top 3 podium display and list below.	Pass
153	Verifying "weekly" tab content on Leaderboard	On Leaderboard, ensure "weekly" tab is active.	Top 3 podium shows users ranked 1, 2, 3 with names, flags/images, and scores (dummy data). List below shows users ranked 1+ with	Top 3 podium shows users ranked 1, 2, 3 with names, flags/images, and scores (dummy data). List below shows users ranked 1+ with	Pass

			rank, initial, name, points, and trophy icons (dummy data).	rank, initial, name, points, and trophy icons (dummy data).	
154	Switching to "monthly" tab on Leaderboard	On Leaderboard, click the "monthly" tab.	The "monthly" tab becomes active. The podium and list content update to show monthly rankings (dummy data).	The "monthly" tab becomes active. The podium and list content update to show monthly rankings (dummy data).	Pass
155	Verifying "monthly" tab content on Leaderboard	On Leaderboard, ensure "monthly" tab is active.	Top 3 podium shows users ranked 1, 2, 3 with names, flags/images, and scores (dummy data). List below shows users ranked 1+ with rank, initial, name, points, and trophy icons (dummy data). (Differs from weekly data).	Top 3 podium shows users ranked 1, 2, 3 with names, flags/images, and scores (dummy data). List below shows users ranked 1+ with rank, initial, name, points, and trophy icons (dummy data). (Differs from weekly data).	Pass
156	Clicking back button on Leaderboard	On Leaderboard, click "<".	User is navigated back to the screen from which the Leaderboard was accessed (likely Menu).	User is navigated back to the screen from which the Leaderboard was accessed (likely Menu).	Pass
157	Navigating to Menu screen	Click user initial on Home (FT_Home_56).	User is navigated to Menu screen (Image 30), titled "Menu". Header back button and share icon present. User name ("Nasir Bhutta") and initial ('N'). List of menu items. Terms/Privacy	User is navigated to Menu screen (Image 30), titled "Menu". Header back button and share icon present. User name ("Nasir Bhutta") and initial ('N'). List of menu items. Terms/Privacy	Pass

			link and version info.	link and version info.	
158	Verifying Menu items list on Menu screen	On Menu screen, observe the list of items.	Items for "Manage Account", "Upgrade to Premium Free", "Select Language", "Select Year Grade", "Select Subjects", "Change Password", "Share My Progress", "App Settings", "Learn and Earn", "Help and Feedback", and "Logout" are displayed with icons and forward arrows.	Items for "Manage Account", "Upgrade to Premium Free", "Select Language", "Select Year Grade", "Select Subjects", "Change Password", "Share My Progress", "App Settings", "Learn and Earn", "Help and Feedback", and "Logout" are displayed with icons and forward arrows.	Pass
159	Clicking "Manage Account" menu item on Menu screen	On Menu screen, click "Manage Account".	User is navigated to an Account Management screen (even if dummy).	User is navigated to an Account Management screen (even if dummy).	Pass
160	Clicking "Upgrade to Premium Free" menu item on Menu screen	On Menu screen, click "Upgrade to Premium Free".	User is navigated to a premium/upgrade screen (even if dummy).	User is navigated to a premium/upgrade screen (even if dummy).	Pass
161	Clicking "Select Language" menu item on Menu screen	On Menu screen, click "Select Language".	User is navigated to a Language selection screen/modal (even if dummy).	User is navigated to a Language selection screen/modal (even if dummy).	Pass
162	Clicking "Select Year Grade" menu item on Menu screen	On Menu screen, click "Select Year Grade".	User is navigated to a Year Grade selection screen/modal (even if dummy).	User is navigated to a Year Grade selection screen/modal (even if dummy).	Pass
163	Clicking "Select	On Menu screen, click "Select Subjects".	User is navigated to a Subject	User is navigated to a Subject	Pass

	Subjects" menu item on Menu screen		selection screen (similar to Image 12/22, possibly for customizing interests) (even if dummy).	selection screen (similar to Image 12/22, possibly for customizing interests) (even if dummy).	
164	Clicking "Change Password" menu item on Menu screen	On Menu screen, click "Change Password".	User is navigated to a Change Password screen (even if dummy input fields).	User is navigated to a Change Password screen (even if dummy input fields).	Pass
165	Clicking "Share My Progress" menu item on Menu screen	On Menu screen, click "Share My Progress".	A system-level share sheet or placeholder appears to share user progress (dummy logic).	A system-level share sheet or placeholder appears to share user progress (dummy logic).	Pass
166	Clicking "App Settings" menu item on Menu screen	On Menu screen, click "App Settings".	User is navigated to an App Settings screen (even if dummy toggles/options).	User is navigated to an App Settings screen (even if dummy toggles/options).	Pass
167	Clicking "Learn and Earn" menu item on Menu screen	On Menu screen, click "Learn and Earn".	User is navigated to a Learn and Earn explanation/feature screen (even if dummy).	User is navigated to a Learn and Earn explanation/feature screen (even if dummy).	Pass
168	Clicking "Help and Feedback" menu item on Menu screen	On Menu screen, click "Help and Feedback".	User is navigated to a Help/Feedback screen or support contact options (even if dummy).	User is navigated to a Help/Feedback screen or support contact options (even if dummy).	Pass
169	Clicking "Logout" menu item on Menu screen	On Menu screen, click "Logout".	User is logged out (frontend state reset, potentially backend call). User is navigated back to the Welcome/Login screen. Session data (if any) is cleared.	User is logged out (frontend state reset, potentially backend call). User is navigated back to the Welcome/Login screen. Session data (if any) is cleared.	Pass

170	Clicking "See Terms of Services and Privacy Policy" link on Menu screen	On Menu screen, click the "See Terms of Services and Privacy Policy" link.	A screen displaying the Terms and Privacy Policy is shown, or an external link is opened in a browser (even if dummy content/URL).	A screen displaying the Terms and Privacy Policy is shown, or an external link is opened in a browser (even if dummy content/URL).	Pass
171	Verifying App Version display on Menu screen	On Menu screen, observe the bottom.	The app version number ("Version 1.59.2 (1)") is displayed.	The app version number ("Version 1.59.2 (1)") is displayed.	Pass
172	Clicking Share icon in Menu header on Menu screen	On Menu screen, click the share icon.	A system-level share sheet or placeholder appears to share the app (dummy logic).	A system-level share sheet or placeholder appears to share the app (dummy logic).	Pass
173	Clicking back button on Menu screen	On Menu screen, click "<".	User is navigated back to the Home screen.	User is navigated back to the Home screen.	Pass
174	Navigating to Notifications screen	Click notification icon on Home (FT_Home_55).	User is navigated to Notifications screen (Image 31), titled "Notifications". Header back button present. A list of notifications is displayed.	User is navigated to Notifications screen (Image 31), titled "Notifications". Header back button present. A list of notifications is displayed.	Pass
175	Verifying Notification list content on Notifications screen	On Notifications screen, observe the list.	List items are displayed, each showing an icon, title ("Congratulations!", "Your Parents set..."), description text, and timestamp ("6 months ago") (dummy data).	List items are displayed, each showing an icon, title ("Congratulations!", "Your Parents set..."), description text, and timestamp ("6 months ago") (dummy data).	Pass

Software Design Description for **BrainBee**

176	Clicking a notification item on Notifications screen	On Notifications screen, click a notification item.	An action related to the notification occurs (e.g., navigating to a relevant screen like Report Card or Goals screen, displaying more details - even if dummy). The UI reacts to the click.	An action related to the notification occurs (e.g., navigating to a relevant screen like Report Card or Goals screen, displaying more details - even if dummy). The UI reacts to the click.	Pass
177	Clicking back button on Notifications screen	On Notifications screen, click "<".	User is navigated back to the Home screen.	User is navigated back to the Home screen.	Pass
178	Testing scrolling functionality on scrollable lists/screens	On any screen with lists or content longer than the viewport (e.g., Battle History, Subject Lists, Class list, Menu, Report Card, Notifications), swipe up/down.	The content scrolls smoothly, allowing all list items or content to be visible. Scrolling reaches the top and bottom boundaries.	The content scrolls smoothly, allowing all list items or content to be visible. Scrolling reaches the top and bottom boundaries.	Pass
179	Testing keyboard dismissal after input on input screens	On a screen with an input field (Login, Signup, Invitation modal), tap a field, type, then tap outside the field or use system key to dismiss keyboard.	The on-screen keyboard is dismissed, returning the full screen view.	The on-screen keyboard is dismissed, returning the full screen view.	Pass
180	Testing back button consistency (Header)	Navigate through a sequence using header back buttons (e.g., Home -> Extras -> Battle -> Select Subject -> Select Chapters -> Battle Result -> Report Card). Press back repeatedly.	Clicking the header back button consistently navigates to the immediately preceding screen in the stack. The sequence is reversed.	Clicking the header back button consistently navigates to the immediately preceding screen in the stack. The sequence is reversed.	Pass
181	Testing back button	Use the system-level back gesture or button	From screens that are top-level	From screens that are top-level	Pass

	consistency (System/Gesture)	from various screens (e.g., Home, My Classes, Menu).	in a flow (like Home or My Classes if accessed directly), the back action should typically exit the app or go to the background. From screens deeper in a flow, it should navigate back within the app.	in a flow (like Home or My Classes if accessed directly), the back action should typically exit the app or go to the background. From screens deeper in a flow, it should navigate back within the app.	
182	Testing modal overlay behavior (cannot interact with background)	Open any modal (e.g., Score & Goals). Attempt to tap on elements on the screen visible *behind* the modal.	Taps on the background elements do not register or trigger actions on the screen behind the modal. Only interactions with the modal itself or its close mechanism are possible.	Taps on the background elements do not register or trigger actions on the screen behind the modal. Only interactions with the modal itself or its close mechanism are possible.	Pass

8.3 Business Rules Testing:

No.	Test Case/Test Script	Attribute and Value	Expected Result	Actual Result	Result
1	Testing username uniqueness during registration	Username: "testuser" for user A and user B	Registration attempt for user B fails with an error message indicating the username is already taken.	Registration attempt for user B fails with an error message indicating the username is already taken.	Pass
2	Testing email uniqueness	Email: "test@example.com" for user A and user B	Registration attempt for user B fails with an error message indicating	Registration attempt for user B fails with an error message indicating	Pass

	during registration		the email is already in use.	the email is already in use.	
3	Testing login restrictions for unregistered users	Credentials: Non-existent user; Valid user with incorrect password	Login attempts fail with an error message indicating invalid credentials or user not found. The user remains on the login screen.	Login attempts fail with an error message indicating invalid credentials or user not found. The user remains on the login screen.	Pass
4	Testing password reset email delivery for registered users	Email: Registered address; Unregistered address	For a registered email, a confirmation message is displayed, and a password reset email is sent (verified via mock email service or logs). For an unregistered email, an error message is displayed indicating the email is not found, and no email is sent.	For a registered email, a confirmation message is displayed, and a password reset email is sent (verified via mock email service or logs). For an unregistered email, an error message is displayed indicating the email is not found, and no email is sent.	Pass
5	Testing direct login with a linked Google account	User: Previously linked Google account	The user is logged in directly without being prompted to grant permissions via the Google authorization dialog.	The user is logged in directly without being prompted to grant permissions via the Google authorization dialog.	Pass
6	Testing recovery restrictions for a deleted account	Account: Deleted user credentials	Attempts to recover the account fail with a message indicating the account does not exist or cannot be recovered, unless within a defined temporary deactivation window where recovery is possible (test this edge case if applicable).	Attempts to recover the account fail with a message indicating the account does not exist or cannot be recovered, unless within a defined temporary deactivation window where recovery is possible (test this edge case if applicable).	Pass

7	Testing role switching to authorized roles only	User: Student; Teacher authorized for Student role	Attempts to switch to unauthorized roles (e.g., Teacher/Parent) fail or are not presented as options. Switching to the authorized Student role succeeds, with only authorized roles displayed as options.	Attempts to switch to unauthorized roles (e.g., Teacher/Parent) fail or are not presented as options. Switching to the authorized Student role succeeds, with only authorized roles displayed as options.	Pass
8	Testing re-authentication requirement during role switching	Credentials: Invalid; Valid	The UI prompts for authentication (e.g., password entry) during role switching. With invalid credentials, authentication fails, and the role switch is aborted or shows an error. With valid credentials, authentication succeeds, and the role switch proceeds.	The UI prompts for authentication (e.g., password entry) during role switching. With invalid credentials, authentication fails, and the role switch is aborted or shows an error. With valid credentials, authentication succeeds, and the role switch proceeds.	Pass
9	Testing new role creation option for users with a single role	User: Single role (Student), eligible for other roles	The UI displays the current role and an option to "Create New Role" or similar, provided the user meets eligibility criteria for creating another role type.	The UI displays the current role and an option to "Create New Role" or similar, provided the user meets eligibility criteria for creating another role type.	Pass
10	Testing admin notification for role creation requiring approval	Role: New role requiring admin approval	The user sees a message indicating the request is pending approval. A notification is sent to the administrator (verified via mock notification service or admin dashboard/logs).	The user sees a message indicating the request is pending approval. A notification is sent to the administrator (verified via mock notification service or admin dashboard/logs).	Pass

			The new role is not immediately active for the user.	The new role is not immediately active for the user.	
11	Testing quality of AI-generated summaries for students	Content: Selected chapter	The generated text is reasonably short, logically organized, and uses simple language appropriate for a student. (Qualitative test requiring manual review).	The generated text is reasonably short, logically organized, and uses simple language appropriate for a student. (Qualitative test requiring manual review).	Pass
12	Testing usage limits for AI-generated summaries	Summary Requests: Exceed configured daily limit	Requests succeed up to the configured daily limit. Additional requests beyond the limit fail with an error message indicating the usage limit has been reached.	Requests succeed up to the configured daily limit. Additional requests beyond the limit fail with an error message indicating the usage limit has been reached.	Pass
13	Testing accuracy and readability of AI-generated explanations	Concept: Specific curriculum topic	The explanation is factually correct according to the subject matter and is written in a way that is easy for a student to understand. (Qualitative test requiring manual review and subject matter expertise).	The explanation is factually correct according to the subject matter and is written in a way that is easy for a student to understand. (Qualitative test requiring manual review and subject matter expertise).	Pass
14	Testing single question presentation in quizzes	Quiz: Any quiz from Home screen	The quiz UI displays only the current question and its answer options. Subsequent questions are not visible until the current one is answered or skipped (if skipping is allowed).	The quiz UI displays only the current question and its answer options. Subsequent questions are not visible until the current one is answered or skipped (if skipping is allowed).	Pass

15	Testing prohibition of skipping questions in quizzes	Quiz: Any quiz, no answer selected	No "Skip" button is present. Any "Next" or progression mechanism is disabled or non-responsive until an answer option is selected. The quiz does not automatically advance unless a timer runs out (per simulated FT_BrainBattle_101).	No "Skip" button is present. Any "Next" or progression mechanism is disabled or non-responsive until an answer option is selected. The quiz does not automatically advance unless a timer runs out (per simulated FT_BrainBattle_101).	Pass
16	Testing relevance of AI-generated flashcards to selected content	Content: Specific chapter/topic	The content (terms, definitions, concepts, questions) on the flashcards corresponds to the key information within the selected chapter or topic. (Qualitative test requiring manual review and content knowledge).	The content (terms, definitions, concepts, questions) on the flashcards corresponds to the key information within the selected chapter or topic. (Qualitative test requiring manual review and content knowledge).	Pass
17	Testing personalization of AI-generated flashcards based on user performance	User A: Weak in definitions; User B: Weak in formulas	Flashcards for User A include terms/definitions they struggle with. Flashcards for User B include formulas they struggle with. Content differs based on user performance profiles. (Requires mock user data and AI logic).	Flashcards for User A include terms/definitions they struggle with. Flashcards for User B include formulas they struggle with. Content differs based on user performance profiles. (Requires mock user data and AI logic).	Pass
18	Testing battle invitations for online and available friends only	Friend Status: Offline; In another battle; Online and available	Invitations to offline friends or friends in another battle fail, with an error message or unavailable friends	Invitations to offline friends or friends in another battle fail, with an error message or unavailable friends	Pass

			greyed out/hidden in the UI. Invitations to online and available friends succeed, and the invitation is sent or the match starts. (Requires presence/status tracking).	greyed out/hidden in the UI. Invitations to online and available friends succeed, and the invitation is sent or the match starts. (Requires presence/status tracking).	
19	Testing display of battle history details	Battles: Multiple completed battles	Each battle history item shows the opponent's username/identifier, outcome (Win, Loss, Draw), and likely scores. (Uses dummy data in the UI).	Each battle history item shows the opponent's username/identifier, outcome (Win, Loss, Draw), and likely scores. (Uses dummy data in the UI).	Pass
20	Testing report card metrics and opponent comparison	Battle: Completed battle	The Report Card displays metrics like Accuracy, Correct count, Avg Time, with a question-by-question breakdown including correct answers. It includes visual or tabular comparison data against the opponent(s) and/or past quizzes. (Uses dummy data).	The Report Card displays metrics like Accuracy, Correct count, Avg Time, with a question-by-question breakdown including correct answers. It includes visual or tabular comparison data against the opponent(s) and/or past quizzes. (Uses dummy data).	Pass
21	Testing competitor profile details visibility	Profile: Competitor from Leaderboard or Battle History	The competitor's profile screen displays information beyond username, such as performance metrics (score, accuracy) and possibly a subset of battle history, adhering to privacy considerations.	The competitor's profile screen displays information beyond username, such as performance metrics (score, accuracy) and possibly a subset of battle history, adhering to privacy considerations.	Pass

			(Requires mock profile data).	(Requires mock profile data).	
22	Testing display of enrolled classes for a student	Classes: Enrolled in Math, Science; Not enrolled in History	The list of enrolled classes shows only Math and Science. History is not displayed. (Requires mock user and class enrollment data).	The list of enrolled classes shows only Math and Science. History is not displayed. (Requires mock user and class enrollment data).	Pass
23	Testing class discussion access for enrolled students only	Users: Student enrolled in class; Student not enrolled in class	Enrolled students can access the class forum and post messages. Non-enrolled students cannot access the forum or are prevented from posting if they gain access. (Requires backend permission checks).	Enrolled students can access the class forum and post messages. Non-enrolled students cannot access the forum or are prevented from posting if they gain access. (Requires backend permission checks).	Pass
24	Testing access to teacher-uploaded class materials	Student: Enrolled in class	The list of materials contains only items uploaded or linked by the teacher(s) of that class. Materials from other classes or general system materials are not present. (Requires mock class materials data).	The list of materials contains only items uploaded or linked by the teacher(s) of that class. Materials from other classes or general system materials are not present. (Requires mock class materials data).	Pass
25	Testing access to teacher-created class quizzes	Student: Enrolled in class	The list of quizzes for the class contains only quizzes created by the teacher(s) of that class, excluding quizzes from other classes or general subject quizzes. (Requires mock class quiz data).	The list of quizzes for the class contains only quizzes created by the teacher(s) of that class, excluding quizzes from other classes or general subject quizzes. (Requires mock class quiz data).	Pass

26	Testing submission of teacher-assigned class assignments	Assignment: Teacher-created; Non-teacher-created	The list of assignments contains only those created by the teacher(s) of the class. Attempts to submit non-teacher-created assignments fail (e.g., no interface or validation error). (Requires mock class assignment data).	The list of assignments contains only those created by the teacher(s) of the class. Attempts to submit non-teacher-created assignments fail (e.g., no interface or validation error). (Requires mock class assignment data).	Pass
27	Testing streak increment for specific activities	Activities: Streak-contributing (e.g., daily quiz); Non-contributing (e.g., view profile)	Streak-contributing activities (e.g., daily quiz) increase or maintain the streak counter. Non-contributing activities (e.g., viewing profile) do not change the streak counter. (Requires mock/real streak tracking).	Streak-contributing activities (e.g., daily quiz) increase or maintain the streak counter. Non-contributing activities (e.g., viewing profile) do not change the streak counter. (Requires mock/real streak tracking).	Pass
28	Testing display of earned badges for a student	Badges: Earned and unearned	Earned badges are displayed as unlocked/colored. Unearned badges are displayed as locked/greyed out or not shown, depending on UI design.	Earned badges are displayed as unlocked/colored. Unearned badges are displayed as locked/greyed out or not shown, depending on UI design.	Pass
29	Testing coin awards for completed quests	Quest: Coin Quest, abandoned then completed	Abandoning a quest does not change the coin balance. Completing a quest increases the coin balance by the awarded amount. (Requires mock/real coin quest tracking and award system).	Abandoning a quest does not change the coin balance. Completing a quest increases the coin balance by the awarded amount. (Requires mock/real coin quest tracking and award system).	Pass

30	Testing accuracy and relevance of AI responses to student queries	Query: Related to curriculum or app features	The AI response is factually correct based on the curriculum/app functionality and directly addresses the query. (Qualitative test requiring manual review and subject matter expertise).	The AI response is factually correct based on the curriculum/app functionality and directly addresses the query. (Qualitative test requiring manual review and subject matter expertise).	Pass
31	Testing overall score calculation from completed activities	User: History of completed activities	The displayed score, grade, and percentage accurately reflect a weighted or summed calculation of all completed activities tracked for the user. (Requires backend calculation logic and frontend display).	The displayed score, grade, and percentage accurately reflect a weighted or summed calculation of all completed activities tracked for the user. (Requires backend calculation logic and frontend display).	Pass
32	Testing highlighting of overall weak performance areas	User: Low scores in Algebra topics	The "Areas to Improve" section lists topics or subjects where the user performed poorly, based on system analysis. (Requires backend performance analysis and frontend display).	The "Areas to Improve" section lists topics or subjects where the user performed poorly, based on system analysis. (Requires backend performance analysis and frontend display).	Pass
33	Testing score calculation for a specific book	User: Completed quizzes for Mathematics	The displayed score and percentage on the Mathematics course card reflect only quizzes/activities completed for Mathematics, excluding other subjects like Physics or Biology.	The displayed score and percentage on the Mathematics course card reflect only quizzes/activities completed for Mathematics, excluding other subjects like Physics or Biology.	Pass

			(Requires backend calculation logic and frontend display).	(Requires backend calculation logic and frontend display).	
34	Testing highlighting of weak areas within a specific book	User: Weak in Algebra, strong in Calculus within Mathematics	The Chapter Performance list highlights or shows lower percentages for areas within Mathematics (e.g., Algebra fundamentals) where the user performed poorly. (Requires backend performance analysis and frontend display).	The Chapter Performance list highlights or shows lower percentages for areas within Mathematics (e.g., Algebra fundamentals) where the user performed poorly. (Requires backend performance analysis and frontend display).	Pass
35	Testing timing of teacher registration confirmation email	Registration: Teacher account	The confirmation email is sent to the provided email address within 2 minutes of successful registration completion. (Requires backend email service integration and timing check).	The confirmation email is sent to the provided email address within 2 minutes of successful registration completion. (Requires backend email service integration and timing check).	Pass
36	Testing class name uniqueness for a teacher	Class Name: "Math 101" for two classes	The second attempt to create a class named "Math 101" fails with an error message indicating the class name is already used by this teacher.	The second attempt to create a class named "Math 101" fails with an error message indicating the class name is already used by this teacher.	Pass
37	Testing auto-generation of class codes	Class: Newly created	The system generates a 6-character alphanumeric code for the class. (Requires backend generation logic).	The system generates a 6-character alphanumeric code for the class. (Requires backend generation logic).	Pass

38	Testing email format validation when adding students	Email: Invalid format (e.g., "student@domain" , "student.com")	The system rejects invalid email formats with a validation error message displayed next to the email input field or upon attempting to send the invitation.	The system rejects invalid email formats with a validation error message displayed next to the email input field or upon attempting to send the invitation.	Pass
39	Testing expiration of student invitations after 7 days	Invitation: Sent, 7 days elapsed (simulated)	Attempting to accept the invitation after 7 days fails with a message indicating the invitation has expired.	Attempting to accept the invitation after 7 days fails with a message indicating the invitation has expired.	Pass
40	Testing real-time updates of student performance for teachers	Quiz: Submitted by student	The student's quiz submission data (e.g., score, completion status) updates on the Teacher's monitoring dashboard without requiring a page refresh. (Requires real-time communication like websockets or polling).	The student's quiz submission data (e.g., score, completion status) updates on the Teacher's monitoring dashboard without requiring a page refresh. (Requires real-time communication like websockets or polling).	Pass
41	Testing AI insights generation threshold for teacher monitoring	Assessments: Fewer than 3 completed; 3 or more completed	No AI insights are displayed for students with fewer than 3 completed assessments. AI insights (e.g., suggested areas for focus) are displayed for students with 3 or more completed assessments. (Requires backend logic to count assessments and trigger AI analysis).	No AI insights are displayed for students with fewer than 3 completed assessments. AI insights (e.g., suggested areas for focus) are displayed for students with 3 or more completed assessments. (Requires backend logic to count assessments and trigger AI analysis).	Pass

42	Testing mandatory fields for teacher assignment creation	Assignment Fields: No title; No deadline	Attempts to create or save an assignment without a Title or Deadline fail with validation errors indicating these fields are mandatory.	Attempts to create or save an assignment without a Title or Deadline fail with validation errors indicating these fields are mandatory.	Pass
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8.4 Integration Testing:

No.	Test Case/Test Script	Attribute and Value	Expected Result	Actual Result	Result
1	Testing successful registration flow from UI to backend	User Data: Valid, unique Full Name, Username/Email, Password	Registration succeeds, user navigates to Home screen.	Registration succeeds, user navigates to Home screen.	Pass
2	Testing duplicate email error during registration	Email: Already in DB	Error message displays: "Email already in use".	Error message displays: "Email already in use".	Pass
3	Testing invalid data error during registration	Email: Invalid format (e.g., "test@ ")	Error message displays: "Invalid email format".	Error message displays: "Invalid email format".	Pass
4	Testing successful login flow from UI to backend	Credentials: Valid for existing user	Login succeeds, user navigates to Home screen.	Login succeeds, user navigates to Home screen.	Pass
5	Testing invalid credentials error during login	Credentials: Valid Username/Email, incorrect Password	Error message displays: "Invalid credentials".	Error message displays: "Invalid credentials".	Pass
6	Testing user not found error during login	Credentials: Non-existent user	Error message displays: "User not found".	Error message displays: "User not found".	Pass

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7	Testing logout flow from UI to backend	Action: Click "Logout" on Menu screen	Logout succeeds, user navigates to Welcome/Login screen.	Logout succeeds, user navigates to Welcome/Login screen.	Pass
8	Testing forgot password flow for registered email	Email: Registered address	Success message displays: "Reset email sent".	Success message displays: "Reset email sent".	Pass
9	Testing forgot password flow for unregistered email	Email: Unregistered address	Error message displays: "Email not found".	Error message displays: "Email not found".	Pass
10	Testing social login flow for new user	Social Account: Not yet registered	Social login succeeds, new user navigates to Home screen.	Social login succeeds, new user navigates to Home screen.	Pass
11	Testing social login flow for existing linked user	Social Account: Already linked	Social login succeeds, user navigates to Home screen.	Social login succeeds, user navigates to Home screen.	Pass
12	Testing account deletion flow from UI to backend	Action: Click "Delete Account" on Menu screen	Account deletion succeeds, user navigates to Welcome/Login screen.	Account deletion succeeds, user navigates to Welcome/Login screen.	Pass
13	Testing home screen dummy data loading	Data: Dummy user details, scores, subjects, promotions	Home screen displays dummy user details, scores, subjects, and promotions.	Home screen displays dummy user details, scores, subjects, and promotions.	Pass
14	Testing overall score screen dummy data loading	Data: Dummy overall score, subject breakdowns, areas to improve	Overall Score screen displays dummy score, breakdowns, and areas to improve.	Overall Score screen displays dummy score, breakdowns, and areas to improve.	Pass
15	Testing subject performance screen	Data: Dummy subject metrics, chapter	Subject Performance screen displays dummy metrics,	Subject Performance screen displays dummy metrics,	Pass

	dummy data loading	performance, recommendations	chapter performance, and recommendations.	chapter performance, and recommendations.	
16	Testing leaderboard screen dummy data loading	Data: Dummy weekly and monthly rankings	Leaderboard screen displays dummy weekly and monthly rankings with tab switching.	Leaderboard screen displays dummy weekly and monthly rankings with tab switching.	Pass
17	Testing my classes screen dummy data loading	Data: Dummy list of classes	My Classes screen displays dummy list of class cards.	My Classes screen displays dummy list of class cards.	Pass
18	Testing class details screen dummy data loading	Data: Dummy class details, materials, assignments, forum	Class Details screen displays dummy class details, materials, assignments, and forum data.	Class Details screen displays dummy class details, materials, assignments, and forum data.	Pass
19	Testing battle history screen dummy data loading	Data: Dummy user info, banner, battle history items	Battle Lobby screen displays dummy user info, banner, and battle history items.	Battle Lobby screen displays dummy user info, banner, and battle history items.	Pass
20	Testing quiz report card screen dummy data loading	Data: Dummy quiz results, performance metrics, question analysis	Quiz Report Card screen displays dummy results, metrics, and question analysis.	Quiz Report Card screen displays dummy results, metrics, and question analysis.	Pass
21	Testing notifications screen dummy data loading	Data: Dummy list of notifications	Notifications screen displays dummy list of notifications.	Notifications screen displays dummy list of notifications.	Pass
22	Testing battle initiation flow with UI navigation	Action: Select "Whole Book" and "Random Match"	Navigation proceeds to Brain Battle screen, simulating quiz flow.	Navigation proceeds to Brain Battle screen, simulating quiz flow.	Pass
23	Testing chapter battle initiation	Action: Select "By Chapter" and specific chapters	Navigation proceeds to Brain Battle screen, simulating	Navigation proceeds to Brain Battle screen, simulating	Pass

	flow with UI navigation		chapter-specific quiz flow.	chapter-specific quiz flow.	
24	Testing practice initiation flow with UI navigation	Action: Click "Practice Now" or "Practice This Chapter"	Navigation proceeds to Practice screen with dummy chapter context.	Navigation proceeds to Practice screen with dummy chapter context.	Pass
25	Testing class action navigation flow	Action: Click "View Materials" on Class Details screen	Navigation proceeds to Materials screen with dummy class data.	Navigation proceeds to Materials screen with dummy class data.	Pass
26	Testing menu item navigation flow	Action: Click "Select Language" on Menu screen	Navigation proceeds to Language Selection screen with dummy content.	Navigation proceeds to Language Selection screen with dummy content.	Pass
27	Testing navigation stack management after login	Action: Use system back action on Home screen after login	System back action exits app or moves it to background.	System back action exits app or moves it to background.	Pass
28	Testing bottom navigation switching	Action: Tap "Class" then "Home" in bottom nav	Bottom nav switches to Class, then back to Home with active state updates.	Bottom nav switches to Class, then back to Home with active state updates.	Pass
29	Testing header back button navigation consistency	Action: Click header back button in navigation sequence	Header back button navigates to the previous screen in the stack.	Header back button navigates to the previous screen in the stack.	Pass
30	Testing chapter selection impact on button state	Action: Select a chapter checkbox on Select Chapters screen	"Start Match" button becomes enabled after chapter selection.	"Start Match" button becomes enabled after chapter selection.	Pass
31	Testing tab selection impact on content display	Action: Switch to "Areas to Improve" tab on Overall Score screen	"Areas to Improve" content displays, hiding "Subject Breakdown".	"Areas to Improve" content displays, hiding "Subject Breakdown".	Pass
32	Testing list component	Data: Dummy list for Battle	List component renders dummy	List component renders dummy	Pass

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	rendering with dummy data	Lobby or My Classes	list items correctly.	list items correctly.	
33	Testing item component state rendering with dummy data	Data: Dummy question results for Report Card	Question items display dummy results with correct visual indicators.	Question items display dummy results with correct visual indicators.	Pass
34	Testing modal interaction for display and closure	Action: Click Score widget, then close modal	Score & Goals modal displays, then closes on 'X' button click.	Score & Goals modal displays, then closes on 'X' button click.	Pass